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[Intervention Review]

Spinal manipulative therapy for adults with chronic low back pain

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ABSTRACT

Rationale

Many therapies exist for the treatment of chronic low back pain (LBP), including spinal manipulative therapy (SMT), which is a worldwide, extensively practised intervention. The effectiveness of SMT for chronic LBP is not without dispute. This Cochrane review is an update of a Cochrane systematic review published in 2011.

Objectives

To evaluate the benefits and harms of SMT compared to (1) sham SMT/placebo intervention, (2) no treatment, and (3) other conservative interventions in people with chronic LBP (18+ years old).

Search methods

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase, two other databases, and two trial registers up to 18 October 2024, unrestricted by language. We also screened the reference lists of all included studies and relevant systematic reviews, and approached content experts to identify potentially missing studies.

Eligibility criteria

We included randomised controlled trials (RCTs) that examined the effect of spinal manipulation or mobilisation in adults with chronic LBP compared to sham SMT/placebo, no treatment, and other conservative interventions. We placed no restrictions on the setting. We excluded studies that exclusively examined sciatica.

Outcomes

Our critical outcomes were pain, functional status, and adverse events. The primary time point was one month for pain and functional status. We evaluated adverse events at the end of the intervention.

Risk of bias

We assessed bias in the included studies using the original Cochrane risk of bias tool (RoB 1).

Synthesis methods

Where possible, we synthesised results using meta-analysis with a generic inverse-variance approach and random-effects models; otherwise, we used narrative synthesis. We assessed the certainty of evidence using the GRADE approach. Our main comparisons were SMT versus (1) sham SMT/placebo treatment, (2) no treatment, and (3) other conservative interventions at one month. We converted all pain scales to a 100-point scale.

Included studies

Seventy-six RCTs (11,866 participants) met our inclusion criteria, 50 (66%) of which were not included in the previous version of this review. Seventeen trials (2021 participants) compared SMT to sham SMT/placebo, and four trials (435 participants) compared SMT to no treatment. Most trials (43, including 8291 participants) examined the effect of SMT compared to other conservative interventions. The remaining trials examined other comparisons.

Treatment allocation was appropriately conducted in just four sham SMT/placebo-controlled trials (24%), while only six trials 'blinded' participants to the intervention (35%), indicating a high risk of selection and performance bias. Similarly, the no-treatment controlled trials were as susceptible to selection bias (50%) and performance bias (75%).

All trials were conducted in high-income ($n = 53$) or middle-income ($n = 23$) countries. In most studies, the population was middle-aged and included men and women.

Synthesis of results

SMT versus sham SMT/placebo

We found very low-certainty evidence (downgraded for inconsistency and study limitations) that SMT may result in a small reduction in pain compared to sham SMT/placebo at one month (mean difference (MD) -7.01, 95% confidence interval (CI) -12.48 to -1.53; $I^2 = 94\%$; 16 studies, 1570 participants) and very low-certainty evidence (downgraded for study limitations and inconsistency) that SMT may result in a medium improvement in functional status compared to sham SMT/placebo at one month (standardised mean difference (SMD) -0.41, 95% CI -0.69 to -0.13; $I^2 = 82\%$; 13 studies, 1416 participants), but the evidence is very uncertain.

SMT versus no treatment

We found very low-certainty evidence (downgraded for study limitations, inconsistency, and imprecision) that SMT may result in a medium reduction in pain compared to no treatment at one month (MD -13.99, 95% CI -27.33 to -0.66; $I^2 = 89\%$; 4 studies, 325 participants), but the evidence is very uncertain. We found low-certainty evidence (downgraded for study limitations and imprecision) that SMT may result in a large improvement in functional status compared to no treatment at one month (SMD -0.84, 95% CI -1.32 to -0.35; $I^2 = 71\%$; 4 studies, 312 participants).

SMT versus other conservative interventions

Low-certainty evidence (downgraded for inconsistency) indicated that SMT may result in little to no difference in pain (MD -4.72, 95% CI -8.26 to -1.17; $I^2 = 89\%$; 31 studies, 4109 participants) and may result in a small improvement in functional status (SMD -0.25, 95% CI -0.38 to -0.11; $I^2 = 73\%$; 28 studies, 3940 participants) compared to other conservative interventions at one month.

These effects, however, should be interpreted with caution due to the substantial statistical heterogeneity for which there is no clear explanation.

Less than half of the studies (47%) reported on adverse events, of which 12 studies reported these systematically. Adverse events in the SMT group were limited to muscle soreness, stiffness, and/or transient increase in pain. None of the studies registered any serious complications related to either the experimental or control group treatment. The evidence is very uncertain about the adverse effects of SMT.

Authors' conclusions

When SMT is compared to sham SMT/placebo, it may result in a small improvement in pain and medium improvement in functional status in adults with chronic low back pain. When compared to no treatment, SMT may result in a medium improvement in pain and a large improvement in functional status. When compared to other conservative interventions, SMT may result in little to no difference in pain and a small improvement in functional status. The evidence is of low to very low certainty, largely due to the fact that the effects of SMT were examined in trials conducted in different settings and populations, with different types of SMT technique, dosage, and frequency of treatment. Continuing to conduct RCTs in the same manner will neither strengthen the evidence nor our confidence in it.

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Registration

Protocol (2009): DOI: 10.1002/14651858.CD008112

Original review (2011): DOI: 10.1002/14651858.CD000447.pub2

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of spinal manipulative therapy for chronic low back pain?

Key messages

- Spinal manipulative therapy (SMT – a hands-on treatment where a therapist moves joints in the spine) may slightly reduce pain and moderately improve function compared to 'sham' (fake) SMT.
- SMT may moderately reduce pain and may substantially improve function compared to no treatment.
- Less than half of studies reported on adverse (unwanted or harmful) effects. While adverse effects, including muscle soreness and a temporary increase in low back pain, were common, no serious adverse effects related to SMT were observed.

What is non-specific chronic low back pain?

Low back pain is a common and disabling disorder, representing a great burden to society. It often results in reduced quality of life, time lost from work, and substantial medical expenses. Chronic low back pain is defined here as pain lasting longer than 12 weeks. We focused on people with pain predominantly located in the lower back and those with pain radiating into the buttocks and legs.

How is chronic low back pain treated?

Spinal manipulative therapy (SMT) is a common treatment for chronic low back pain and is practised worldwide by healthcare professionals, including chiropractors, manual therapists, and osteopaths. SMT is a 'hands-on' treatment of the spine, including both manipulation and mobilisation to reduce pain, improve function, and help people return to usual activities. SMT may include gentle movements and stretching (known as mobilisation), or quick, controlled pushes often accompanied by an audible 'pop' (known as manipulation) to help improve someone's range of motion and reduce their pain.

What did we find out?

We wanted to find out if SMT improved pain and function, or caused harm, in people with chronic low back pain. We did not include people with low back pain caused by a specific condition, such as infection, tumour, or fracture.

What did we do?

We searched for studies that compared SMT to:

- fake or 'sham' SMT;
- no treatment;
- other 'conservative' treatment, meaning simple, non-surgical care, such as exercise.

We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes. This is an update of a review, last published in 2011.

What did we find?

In this updated review, we identified 76 studies involving 11,866 people (published from 1978 to 2024) assessing the effects of SMT in people with chronic low back pain. The participants were a mix of women and men, and most were middle-aged. Most studies took place in high-income countries (53 studies, e.g. USA and UK), while 23 studies took place in middle-income countries (e.g. Brazil, India). None were in low-income countries.

Main results: benefits

A difference of 10 points on a 100-point scale is considered meaningful for patients—it can reflect a noticeable improvement in symptoms.

Comparison between SMT and sham SMT

Pain

People receiving SMT rated their pain on average **7.0** points better on a 0 to 100 scale one month after the start of SMT compared with those who had sham SMT.

Function

People receiving SMT rated their function on average **8.8** points better on a 0 to 100 scale one month after the start of SMT compared with people who had sham SMT.

Comparison between SMT and no treatment**Pain**

People receiving SMT rated their pain on average **14** points better on a 0 to 100 scale one month after the start of SMT compared with people who had no treatment.

Function

People receiving SMT rated their function on average **12.9** points better on a 0 to 100 scale one month after the start of SMT compared with people who had no treatment.

Comparison between SMT and other conservative treatment**Pain**

People receiving SMT rated their pain on average **4.7** points better on a 0 to 100 scale one month after the start of SMT compared with people who had another conservative treatment.

Function

People receiving SMT rated their function on average **4.9** points better on a 0 to 100 scale one month after the start of SMT compared with people who had another conservative treatment.

Main results: harms

Only a few studies reported on the adverse effects (i.e. unwanted or harmful effects) of SMT. These included muscle soreness and a temporary increase in low back pain. No serious adverse effects related to SMT were observed.

What are the limitations of the evidence?

We have little confidence in the evidence because the studies used a variety of SMT techniques, tested them in varied amounts, and the reported results greatly varied. Several studies included few participants, and some studies were poorly conducted. Therefore, the effects of SMT may be overestimated.

How current is this review?

The review is current to 18 October 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Spinal manipulative therapy compared to sham spinal manipulative therapy/placebo for people with chronic low back pain

Spinal manipulative therapy compared to sham spinal manipulative therapy/placebo for people with chronic low back pain

Patient or population: people with chronic low back pain

Setting: primary, secondary, and tertiary care

Intervention: spinal manipulative therapy

Comparison: sham spinal manipulative therapy/placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with sham spinal manipulative therapy/placebo	Risk with spinal manipulative therapy				
Pain at 1 month Scale from: 0 to 100 follow-up: 1 months	The mean pain at 1 month ranged from 27 to 59 points	MD 7.01 points lower (12.48 lower to 1.53 lower)	-	1570 (16 RCTs)	⊕⊕⊕⊕ Very low ^{a,b}	Spinal manipulative therapy may result in a small reduction in pain, but the evidence is very uncertain.
Functional status Scale from: 0 to 24 follow-up: 1 months	The mean functional status ranged from 6.02 to 10.95 (for Roland Morris Disability Questionnaire (RMDQ)) points	MD 2.1 points lower (3.54 lower to 0.67 lower)	-	1416 (13 RCTs)	⊕⊕⊕⊕ Very low ^{a,c}	Spinal manipulative therapy may result in a medium improvement in functional status, but the evidence is very uncertain. Standardised mean difference (SMD) -0.41 (95% CI -0.69 to -0.13). ^d
Adverse events	Six trials examined serious adverse events. Two trials reported serious adverse events, but none of these were judged to be related to the study intervention. Four trials examined minor adverse events. They were commonly reported, transient, and of mild to moderate severity.			1034 (6 RCTs)	⊕⊕⊕⊕ Very low ^{e,f}	The evidence is very uncertain about the adverse effects of spinal manipulative therapy.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_441376301065242038.

- ^a Downgraded due to the presence of performance bias (lack of participant blinding) and selection bias because most of the weight (> 50%) comes from trials with a high risk of bias.
- ^b Downgraded because $I^2 = 93\%$, and treatment effects were in different directions and could not be explained.
- ^c Downgraded because $I^2 = 83\%$, and treatment effects were in different directions and could not be explained.
- ^d Standardised mean difference (SMD) back-translated to typical scales by multiplying the SMD by the standard deviation (SD) at 4 weeks in the sham group as reported in Licciardone 2013: 5.13 (Roland Morris Disability Questionnaire (RMDQ), 0 to 24 scale).
- ^e Downgraded due to the presence of selection bias because most of the weight (> 50%) comes from trials with a high risk of bias.
- ^f Narrative synthesis was conducted, but due to the format in which the data were reported and the low number of events reported, it was not possible to combine them in a meaningful synthesis.

Summary of findings 2. Summary of findings table - Spinal manipulative therapy compared to no treatment for people with chronic low back pain

Spinal manipulative therapy compared to no treatment for people with chronic low back pain

Patient or population: people with chronic low back pain
Setting: primary, secondary, and tertiary care
Intervention: spinal manipulative therapy
Comparison: no treatment

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no treatment	Risk with spinal manipulative therapy				
Pain at 1 month Scale from: 0 to 100 follow-up: 1 months	The mean pain at 1 month ranged from 26 to 48 points	MD 13.99 points lower (27.33 lower to 0.66 lower)	-	325 (4 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	Spinal manipulative therapy may result in a medium reduction in pain, but the evidence is very uncertain.
Functional status Scale from: 0 to 24 follow-up: 1 months	The mean functional status ranged from 6.94 to 8.9 (for the Roland Morris Disability Questionnaire (RMDQ)) points	MD 3.1 points lower (4.88 lower to 1.3 lower)	-	312 (4 RCTs)	⊕⊕⊕⊖ Low ^{a,c}	Spinal manipulative therapy may result in a large improvement in functional status. Standardised mean difference (SMD) -0.84 (95% CI -1.32 to -0.35) ^{d,e}

Adverse events	Only one trial reported no serious adverse events with spinal manipulative therapy. In all other trials, adverse events were not reported.	192 (1 RCT)	⊕○○○ Very low ^{f,g,h}	The evidence is very uncertain about the adverse effects of spinal manipulative therapy.
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***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_441263001007623984.

^a Downgraded due to the presence of performance bias (lack of participant blinding) in all trials. We did not downgrade for the other risk of bias domains because most of the weight (> 50%) comes from trials with a low risk of bias.

^b Downgraded because $I^2 > 50\%$, and treatment effects were in different directions and could not be explained.

^c Downgraded because sample < 400 participants.

^d Control group risk was estimated from the no-treatment group value at follow-up for functional status in Xia 2016.

^e Standardised mean difference (SMD) back-translated to typical scales by multiplying the SMD by the standard deviation (SD) at 3 weeks in the no-treatment group as reported in Xia 2016: mean (SD) functional status (Roland Morris Disability Questionnaire (RMDQ), 0 to 24 scale): 9.7 (3.7 both arms combined).

^f Downgraded due to the presence of performance bias (lack of participant blinding).

^g Downgraded because just one (small) trial provided data for this outcome.

^h Downgraded because just one trial provided a narrative description of adverse events.

Summary of findings 3. Summary of findings table - Spinal manipulative therapy compared to other conservative therapies for people with chronic low back pain

Spinal manipulative therapy compared to other conservative therapies for people with chronic low back pain

Patient or population: people with chronic low back pain

Setting: primary, secondary, and tertiary care

Intervention: spinal manipulative therapy

Comparison: other conservative therapies

Outcomes	Anticipated absolute effects* (95% CI)	Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
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	Risk with other conservative therapies	Risk with spinal manipulative therapy				
Pain at 1 month Scale from: 0 to 100 follow-up: 1 months	The mean pain at 1 month ranged from 10 to 49.6 points.	MD 4.72 lower (8.26 lower to 1.17 lower)	-	4109 (31 RCTs)	⊕⊕⊕⊕ Low ^{a,b}	Spinal manipulative therapy may result in little to no difference in reduction in pain.
Functional status at 1 month Scale from: 0 to 24 follow-up: 1 months	The mean functional status at 1 month ranged from 4.26 to 20.8 points (for the Roland Morris Disability Questionnaire (RMDQ))	MD 1.18 lower (1.79 lower to 0.52 lower)	-	3940 (28 RCTs)	⊕⊕⊕⊕ Low ^{a,c}	Spinal manipulative therapy may result in a small improvement in functional status. Standardised mean difference (SMD) -0.25 (95% CI -0.38 to -0.11). ^d
Adverse events	Fourteen trials reported adverse events with spinal manipulative therapy (SMT). No serious adverse events related to SMT were reported. Minor adverse events were reported by nine trials. The reported minor adverse events were transient (flare-up; muscle soreness and stiffness), required little or no change to activity levels, and were considered non-serious in all these trials.			3137 (14 RCTs)	⊕⊕⊕⊕ Low ^{a,e}	The evidence is very uncertain about the adverse effects of spinal manipulative therapy.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_441574849642265779.

^a Not downgraded for study limitations, because sensitivity analyses indicated that effect estimates from low-risk-of-bias studies were consistent with the overall findings.

^b Downgraded because $I^2 = 90\%$, and treatment effects were in different directions and could not be explained.

^c Downgraded because $I^2 = 73\%$, and treatment effects were in different directions and could not be explained.

^d Standardised mean difference (SMD) back-translated to typical scales by multiplying the SMD by the standard deviation (SD) at 4 weeks in the home exercise and advice group as reported in Bronfort 2011: 4.7 (Roland Morris Disability Questionnaire (RMDQ), 0 to 24 scale).

^e Downgraded due to adverse events being reported in a narrative and limited manner, which restricted the ability to fully assess and synthesise the data.

BACKGROUND

Low back pain (LBP) is a common and disabling disorder that affects many people worldwide, causes distress to individuals, and poses a great financial burden to society [1]. Therefore, adequate treatment of LBP is an important issue for patients, clinicians, and healthcare policymakers. Spinal manipulative therapy (SMT) is a widely used therapy for acute and chronic LBP and has been examined in numerous randomised controlled trials (RCTs). These trials have been summarised in systematic reviews [2, 3, 4], which form the basis for recommendations in clinical guidelines [5, 6, 7, 8]. These guidelines have historically offered mixed recommendations for SMT for chronic LBP. The current guidelines are more consistent in their recommendations.

Description of the condition

LBP is defined as pain and discomfort, localised below the costal margin and above the inferior gluteal folds, with or without referred leg pain. Chronic LBP is typically defined as pain persisting for more than 12 weeks [9]. Non-specific LBP is further defined as LBP not attributed to a recognisable, specific pathology (e.g. infection, tumour, fracture, or radicular syndrome).

Description of the intervention and how it might work

In this review, SMT is considered as any 'hands-on' treatment of the spine, including both manipulation and mobilisation [10, 11]. Mobilisations use low-grade velocity, small or large amplitude passive movement techniques within the patient's range of motion and control. Manipulation, on the other hand, uses a high-velocity impulse or thrust applied to a synovial joint over a short amplitude at or near the end of the passive or physiologic range of motion, often accompanied by an audible 'crack' [12]. The cracking sound is caused by cavitation of the joint, which is a term used to describe the formation and activity of bubbles within the fluid [13, 14]. Various practitioners – including chiropractors, manual therapists (physiotherapists trained in manipulative techniques), orthomaneual therapists (medical doctors trained in manipulative techniques), and osteopaths – use this intervention in their practices. While many SMT techniques are shared across these professions, some are unique to specific disciplines. The diagnostic approaches and philosophy of the various professions differ [15]. However, these professional distinctions are not expected to influence clinical outcomes in the context of this review. Therefore, our analysis focuses on SMT in general rather than the difference in effect between the professions.

Many hypotheses exist regarding the specific mechanism of action for spinal manipulation and mobilisation [16, 17, 18]. Some have postulated that, given their theoretically different mechanisms, mobilisation and manipulation should be assessed as separate entities [13, 19]. The specific modes of action might be divided into biomechanical and neurophysiological. The mechanistic (biomechanical) approach suggests that SMT acts on a manipulable lesion (often called the functional spinal lesion or subluxation) designed to reduce internal mechanical stresses resulting in a reduction of symptoms [20]. However, given the complexity of chronic LBP, which may involve nociceptive, neuropathic, and nociplastic mechanisms, a purely mechanistic theory alone cannot explain clinical improvement [18, 21, 22]. The neurophysiological approach suggests that SMT impacts the primary afferent neurones from paraspinal tissues, the motor control system, and pain

processing [21, 22, 23, 24, 25, 26]. In several recent systematic reviews summarising the mechanism of action of SMT, there is no clear consensus on how SMT induces a neurophysiological effect or supports a mechanistic approach, and the evidence is insufficient and of poor-to-moderate quality [27, 28, 29, 30].

Furthermore, it is important to understand shared/non-specific effects related to SMT [31]. These include factors like the patient-practitioner relationship (e.g. verbal and non-verbal communication) and the patient's beliefs (e.g. expectations of the treatment), which are not investigated in this review. Although research has recently started to focus on contextual factors [32, 33], research on how these treatment mechanisms work in the clinical context is scarce [34]. Therefore, our understanding of the mechanisms influencing the effects of SMT remains limited.

Why it is important to do this review

Recent guidelines recommend using SMT for chronic LBP as part of a treatment package [5, 6, 8]. However, the effectiveness of SMT for chronic LBP is not without dispute. Most notably, an earlier version of this Cochrane review concluded, amongst other things, that there is evidence suggesting a small, short-term effect on pain and functional status of SMT compared to other interventions [2]. However, the review was last updated 15 years ago. Since then, many new studies have been published, warranting an updated synthesis. In addition, evidence for the other comparisons, such as SMT versus no treatment and versus sham SMT/ placebo, was more limited, raising concerns about the certainty of the evidence. Additionally, data were particularly sparse for outcomes besides pain and function, such as recovery, return to work, and health-related quality of life; therefore, the effect of SMT on these outcomes is uncertain.

This review updates the Cochrane review published in 2011 [2]. Since then, many large, seemingly well-conducted studies have been published, and Cochrane methodology has evolved, warranting an updated synthesis.

This review aims to update the previous Cochrane review in alignment with the most recent guidelines developed by Cochrane in general [35] and by the Cochrane Musculoskeletal Group in particular.

OBJECTIVES

To evaluate the benefits and harms of SMT compared to (1) sham SMT/ placebo intervention, (2) no treatment, and (3) other conservative interventions in people with chronic LBP (18+ years old).

METHODS

Since the publication of the review protocol in 2009 [36], the background and the following methods (types of participants, types of interventions, types of outcomes, risk of bias assessment, and investigation of heterogeneity (subgroup and sensitivity analyses)) have been updated (see [Supplementary material 12](#) for justification). The main comparisons are SMT versus sham SMT/placebo and no treatment, as advocated by the Cochrane Musculoskeletal Group.

We followed the Methodological Expectations for Cochrane Intervention Reviews (MECIR) when conducting the review [37], and PRISMA 2020 for the reporting [38].

Criteria for considering studies for this review

Types of studies

We included only RCTs published as full texts. Cross-over and cluster-randomised trials were considered. We excluded trials using an inadequate randomisation procedure (e.g. alternate allocation, allocation based upon birthdate).

Types of participants

Inclusion criteria

Participants

- Adults (≥ 18 years of age)
- With a median duration of the current episode of LBP greater than 12 weeks for the study population, meaning that more than 50% of the study population had pain lasting longer than three months

Studies

- In which the unique contribution of SMT to the observed effects could be implied; that is, studies in which the effects of SMT could be isolated. This included studies in which SMT was delivered as part of a package of care; for example, studies comparing SMT plus exercise with exercise alone would be included, whereas studies comparing SMT plus exercise with SMT alone would not.
- Studies with a mixed population, including participants with and without radiculopathy, were only included when more than 75% of the study population had no sign of radiculopathy.

Exclusion criteria

Participants

- Postpartum LBP or pelvic pain due to pregnancy
- Pain unrelated to the low back, e.g. coccydynia
- Postoperative studies or participants with 'failed-back syndrome'
- Serious pathologies, such as tumour, fracture, or infection

Studies

- Examined 'maintenance care' or prevention
- Were designed to test the immediate post-intervention effect of a single treatment only, with no additional follow-up
- Exclusively examined back-related pathologies, e.g. sciatica (defined here as radiating pain following the sciatic distribution and exhibiting signs of radiculopathy)

Types of interventions

Experimental intervention

This review focusses on the effects of both spinal manipulation (i.e. high-velocity, low-amplitude techniques (HVLA)) and mobilisation (i.e. low-velocity, low-amplitude (LVLA)) techniques in participants with chronic LBP. This includes techniques such as flexion-distraction but excludes those that focus on axial traction alone.

Additionally, the techniques considered are those encompassing a form of 'hands-on' treatment but exclude those techniques which focus on 'adjusting instruments', such as Activator.

Types of comparison

Primary analyses

The effect of SMT was examined in the following main clusters:

- SMT versus no treatment;
- SMT versus sham SMT/placebo control (i.e. participants were 'blinded' to the intervention).

In the control arm, 'no treatment' participants received no treatment or were on a waiting list. The group described as a sham/placebo control is judged to be intended by the trial authors to be a sham SMT or placebo (e.g. detuned ultrasound). Participants unknowingly participated in a placebo or sham treatment that was feasibly effective. Sham SMT was used to closely mimic the procedure of SMT without delivering its therapeutic thrust, whereas placebo interventions served as inert controls that did not replicate the SMT technique, allowing us to isolate the specific effects of manipulation.

Secondary analyses

- SMT versus other conservative interventions
- SMT as adjuvant therapy to any other therapy
- SMT versus another kind of SMT; for example, HVLA SMT versus LVLA SMT (i.e. manipulation versus mobilisation), localised SMT versus generalised SMT

We categorised other conservative interventions according to the specific treatment type; for example, as advice or education, exercise therapy, electrotherapy, psychological therapy, non-exercise physical therapy (multimodal), back school, relaxation, anti-inflammatories ([Supplementary material 8](#)).

Outcome measures

Our selection of major outcome domains and measurements is based upon recommendations for the core outcome set for LBP clinical trials [39]. We evaluated only those patient-reported, self-perceived outcome measures listed below for treatment effects at one, three, six, and 12 months. We did not include physiological measures, such as spinal flexibility or degrees achieved with a straight leg raise test.

Critical outcomes

- Pain (e.g. visual analogue scale (VAS), numerical rating scale (NRS), typically measured on a 0- to 100-point scale)
- Functional status (as expressed on a back pain-specific scale (e.g. Roland-Morris Disability Questionnaire (RMDQ), Oswestry Disability Index (ODI))
- Total adverse events, including (1) minor adverse events, defined as minor short-term reactions, such as muscle soreness, tiredness, and increased pain, and are self-limiting, and (2) serious adverse events, defined as a life-threatening event or an event requiring hospitalisation. Our approach to adverse events is exploratory, meaning that we captured any adverse events that happen to be reported, recording the measurement approach in the trial, and recording adverse events as serious

or minor, when reported. These events are summarised narratively.

Important outcomes

- Global improvement or self-perceived recovery (defined as the number of participants reported to be recovered or nearly recovered when presented on a Likert-scale)
- Health-related quality of life (e.g. 36-item Short Form Health Survey (SF-36; as measured by the General Health subscale), EuroQol, general health (e.g. as measured on a VAS scale), or a similarly validated index)
- Return to work (defined as the proportion of the number of participants who were 'sick' at baseline due to their LBP and subsequently returned to work during the course of follow-up)

Search methods for identification of studies

We present all search methods and search strategies in [Supplementary material 1](#). This update includes searches up to 18 October 2024.

Electronic searches

We identified RCTs by electronically searching the following databases to 18 October 2024:

- Cochrane Central Register of Controlled Trials (CENTRAL, which includes the Back and Neck Group trials register) in the Cochrane Library, Issue 4, on 18 October 2024 ([Supplementary material 1](#));
- MEDLINE OvidSP, 1946 to 18 October 2024 ([Supplementary material 1](#));
- MEDLINE In-Process & Other Non-Indexed Citations, OvidSP, on 18 October 2024 ([Supplementary material 1](#));
- Embase OvidSP, 1980 to 18 October 2024 ([Supplementary material 1](#));
- CINAHL (Cumulative Index to Nursing and Allied Health Literature) EBSCO, 1981 to 18 October 2024 ([Supplementary material 1](#));
- Physiotherapy Evidence Database (PEDro) to 18 October 2024 ([Supplementary material 1](#));
- Index to Chiropractic Literature (ICL), to 18 October 2024 ([Supplementary material 1](#));
- PubMed ([Supplementary material 1](#)).

Searches were updated in 2014, 2016, 2018, 2020, 2021, and 2024 ([Supplementary material 1](#)). In 2014, we added MEDLINE In-Process & Other Non-Indexed Citations to the strategy, as well as PubMed, to identify studies not in MEDLINE, using the strategy recommended by Duffy and colleagues [40]. An experienced information specialist conducted the searches according to the recommendations in the *Cochrane Handbook for Systematic Reviews of Interventions* [41], and guidelines recommended by the Musculoskeletal Group. We used the Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE: sensitivity-maximising version (2008 revision), a filter, as recommended in the *Cochrane Handbook* [41].

Please refer to the previous versions of this review for details about the searches conducted for those versions [2, 11].

Searching other resources

In addition to the comprehensive electronic search described above, we also: (1) screened the reference lists of all included studies and systematic reviews pertinent to this topic; (2) searched trial registers (18 October 2024); specifically, ClinicalTrials.gov and the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) for ongoing or registered trials (see [Supplementary material 1](#) for search strategies); and (3) sent our selection of studies to trial authors and authorities in the field of SMT to identify trials missed in any of the aforementioned steps. During the selection and full-text review process, we paid attention to any post-publication concerns, errata, and retractions of potential studies by checking PubMed.

Data collection and analysis

Selection of studies

Citations were managed using EndNote X8 software [42].

In this update, two review authors (of AdZ, MJP, SMR, TI) performed the following steps independently of one another: title and abstract screening, risk of bias evaluation, data extraction, and assessment of the certainty of the evidence (GRADE). Consensus was reached during meetings; an arbiter was not necessary.

We obtained potentially relevant studies as full-text publications and independently assessed them for inclusion. Only full papers were evaluated. We imposed no language restrictions. We contacted study authors when inclusion criteria remained unclear.

Data extraction and management

We used a standardised form to extract the following data.

- Study design (RCT, sample size calculation)
- Study characteristics (e.g. country where the study was conducted, patient population source or setting, methods and period of recruitment and randomisation, inclusion and exclusion criteria)
- Participant characteristics (e.g. number of participants, age, gender, duration of LBP, presence of comorbidity)
- Intervention and control characteristics (e.g. description of the therapies, including frequency and duration of care, non-SMT component, treatment provider)
- Outcome measures and follow-up intervals
- Other (e.g. source of funding, declaration of interests by the authors)
- Risk of bias (i.e. selection, performance, attrition, detection, selective outcome reporting, and publication bias)

Working independently, two review authors (of AdZ, MJP, SMR, TI) performed the data extraction.

We assessed data relating to the primary and secondary outcomes for inclusion in the meta-analyses, and extracted final value scores (means and standard deviations (SD)). We converted change scores into a mean value for the respective follow-up measurement. We used WebPlotDigitizer to extract data from graphs or figures if means and measures of variance were not reported in the text of the included studies [43]. We summarised key findings in a narrative format if we were unable to extract effect data.

The studies measured quality of life using different instruments and scales (e.g. SF-36, 0- to 100-point generic scale). We extracted data for the General Health subscale score from the SF-36 when studies used this instrument. If a study did not report that score, we extracted the Physical Component Summary score instead. When studies reported multiple results for a given construct – for example, the RMDQ and ODI (Castro-Sanchez 2016 [44]; Krekousias 2017 [45, 46]; Vismara 2012 [47, 48]) – we extracted data for the RMDQ only because most studies used this instrument.

For the extraction of the names of the intervention and control arms, we used the original terminology reported by the authors, even when sham SMT was described as "placebo SMT".

Risk of bias assessment in included studies

Working independently, two review authors (of AdZ, MJP, SMR, TI) assessed the studies' risk of bias using the 13 criteria recommended by Cochrane [49, 50], and the Cochrane Back and Neck Review Group [51]. (Supplementary material 9 describes the 13 domains, and Supplementary material 10 describes how we applied the criteria.) We evaluated the following criteria: selection bias, performance bias, detection bias, attrition bias, selective outcome reporting bias, and any other forms of bias, such as conflicts of interest (industry-sponsored trials). We classified each of these sources of bias as 'low risk', 'high risk', or 'unclear risk', with justification for our judgement in the risk of bias table. We excluded studies with a 'fatal flaw' from the meta-analyses, defined as those studies with: (1) a large dropout rate at any of the follow-up measurements; or (2) statistically and clinically relevant important baseline differences for one or more primary outcomes (i.e. pain, functional status) if this appeared not to be due to a small sample size, suggesting possibly improper randomisation or selective exclusion of data.

Review authors were not blinded to the authors of the individual studies, the institution, or the journal. When necessary, we attempted to contact study authors to clarify methodological issues when information was unclear. We sent the risk of bias evaluation for each newly included study to the respective authors to give them an opportunity to provide input, as was done in the 2011 version of the review [2].

Measures of treatment effect

We calculated a treatment effect measure if we considered the pooled studies to be sufficiently clinically homogenous, defined a priori by setting, population, and comparison group.

Continuous outcomes

We expressed continuous outcomes as mean differences (MDs) with 95% confidence intervals (CIs) when trials essentially used the same instrument to measure outcomes; otherwise, we used a standardised mean difference (SMD) (i.e. functional status, quality of life). We converted all pain scales to a 100-point scale. Re-scaling is accepted and common in the back pain field [52], and facilitates comparison and interpretability of the syntheses.

A negative effect size indicates that SMT is more beneficial than the comparison therapy, meaning participants have less pain, better functional status, or better quality of life. In order to enhance the interpretability of continuous outcomes, we back-transformed the pooled SMD values for functional status to a common metric

by multiplying the SMD and 95% CIs by a representative pooled SD for one of the included trials. This determination was based upon studies with a similar population, risk of bias, and type of intervention. These results are presented in the summary of findings tables (SoF) only.

Dichotomous outcomes

We expressed dichotomous outcomes as risk ratios (RRs) and the event was defined as the number of participants recovered or returned to work. A positive RR indicates that SMT results in a greater chance of recovery or return to work. In order to enhance the interpretability of dichotomous outcomes, we calculated the number needed to treat for an additional beneficial outcome (NNTB) or number needed to treat for an additional harmful outcome (NNTH) for the primary outcome only.

When we were unable to extract quantitative data from individual studies for the meta-analyses, we described these data, as well as the data for adverse events, narratively.

Time interval

We assessed outcomes at one, three, six, and 12 months post-randomisation, and analysed data according to the time closest to these intervals. We defined the assessment time points as very short-term (one month; main time interval), short-term (three months), intermediate (six months), and long-term (12 months).

Minimal importance difference

For the comparisons of SMT to sham SMT/placebo and no treatment, we based our approach to defining minimal important differences (for the outcomes of pain and functional status) upon a similar approach used in the American College of Physicians 2017 guidelines for LBP [5], as follows.

- Little to no difference: MD < 5% of the scale (e.g. < 5 mm on a 100-mm VAS); SMD difference < 0.2.
- Small: MD 5% to 10% of the scale (e.g. 5 to 10 mm on a 100-mm VAS); SMD difference 0.2 to 0.5; relative risk < 1.25 or > 0.8.
- Medium: MD 10% to 20% of the scale; SMD 0.5 to 0.8; relative risk 1.25 to 2.0 or 0.5 to 0.8.
- Large: MD > 20% of the scale; SMD ≥ 0.8; relative risk > 2.0 or < 0.5.

For the comparison of SMT to other conservative interventions, even a smaller difference in outcomes may be considered important, depending on the CI. If the CI is narrow and does not cross the threshold of clinical significance or the line of no effect, even a small difference in outcomes can be meaningful and impactful in clinical practice.

Unit of analysis issues

The unit of analysis was each participant recruited to the included trials. We identified no eligible cluster-RCTs. We combined data from trials with multiple comparisons when we thought that similar contrasts were used and the outcomes were clinically similar [53, section 16.5.4]. In practice, we took this approach for just one study, Ferreira 2007 [54, 55], which included two similar forms of exercise (general exercise and motor control exercise) as the comparator. In all other cases, when multiple contrasts from the same trial were examined in the same comparison (e.g. SMT versus physiotherapy and SMT versus standard medical care, which are both included in the comparison of SMT to other conservative interventions),

we halved the number of participants in the shared comparison (SMT) group. This step corrects for error introduced by 'double-counting' of participants for the 'shared comparison' in the same meta-analysis. We conducted a sensitivity analysis to determine if combining similar contrasts resulted in a different effect.

We included two studies with a cross-over design, and used only the data prior to the cross-over in the analysis (Evans 1978 [56]; Teixeira 2016 [57]).

Dealing with missing data

When extraction of meta-data from a publication was not possible, we used the individual patient data from the original study, if available. The research team has received data from most studies published between 2000 and 2018 for an individual patient data meta-analysis that we conducted. In all other cases, if data were missing, we attempted to contact the study authors. If we did not receive a response, and data were reported as a median and interquartile range (IQR), we assumed that the median was equivalent to the mean and the width of the IQR equivalent to 1.35 times the SD according to the methods recommended in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* [58]. If a range was presented instead of an IQR, the SD was estimated to be one-quarter of the range [58]. If a standard deviation (SD) was presented at baseline, but not follow-up, we used the baseline SD. If no measure of variation was reported anywhere in the text, we estimated the SD based upon other studies with a similar population and risk of bias, in accordance with methods described in Chapter 10 of the *Cochrane Handbook* [59].

Reporting bias assessment

We created and examined funnel plots only for the outcomes of pain and functional status for comparisons 1 (SMT versus sham/placebo treatment) and 3 (SMT versus other conservative interventions) because all other comparisons included too few studies.

We examined and noted the funding source(s) and the authors' disclosures of interest in the description of the studies.

Synthesis methods

We conducted all analyses in Review Manager (RevMan) [60]. Our main comparisons were SMT versus sham/placebo treatment and SMT versus no treatment. Other comparisons were: SMT versus other conservative interventions, SMT as an adjunct intervention, and SMT versus other SMT. Our main time interval of interest was one month. Other time intervals were three, six, and 12 months.

We undertook meta-analyses only when this was meaningful; that is, if the treatments, participants, and underlying clinical question were similar enough for pooling to make sense.

We used a random-effects model for all analyses because we expected some between-study variation in terms of population, type of intervention, and comparator [61]. The random-effects model incorporates an assumption that the different studies are estimating different, yet related, intervention effects [59]. This method is based on the inverse-variance approach, making an adjustment to the study weights according to the extent of variation or heterogeneity among the varying intervention effects. We used the Mantel-Haenszel method for dichotomous data. For each treatment comparison, we calculated an effect size and a 95% confidence interval (CI).

We assessed the clinical homogeneity of the studies, looking for potential differences in population, and stratified the analyses by treatment comparison, outcome (e.g. pain, functional status), and timing of the outcome. We defined statistical heterogeneity in two ways: (1) informally, by visually inspecting the forest plot ('eyeball test'), and (2) formally, by using the Q-test (Chi²) and I² statistic. Where possible, we attempted to explain cases of considerable heterogeneity (defined as an I² statistic value of 75% or greater, or based upon the review authors' judgement), by examining the characteristics of the individual studies [59]. Furthermore, we examined methodological heterogeneity in the various sensitivity analyses. Further details are described in [Investigation of heterogeneity and subgroup analysis](#) below.

Investigation of heterogeneity and subgroup analysis

We planned and conducted the following subgroup analyses, based upon the aggregate data from the individual trials, to explore possible sources of heterogeneity (I² > 75% and > 10 trials).

- Duration of LBP (trials which included subacute and chronic LBP versus exclusively chronic LBP)
- Type of LBP (i.e. non-specific LBP, including participants with radiating pain to the knee versus those participants with pain below the knee or those with clear neurological deficit)
- Trials where the effect of SMT was examined alone as opposed to trials which examined the effect of SMT in a larger, 'multimodal' context (e.g. when combined with advice and reassurance and/or home exercises or even when offered as a package of care, such as 'chiropractic care' or 'osteopathic care')
- Type of technique (high-velocity, low-amplitude manipulation or low-velocity, low-amplitude mobilisation)
- Country income status (high- versus middle- and low-income countries)
- For the comparison of SMT to other conservative interventions, we explored:
 - the type of contrast, specifically, the effect of SMT versus exercise therapy;
 - active versus passive treatments; and
 - recommended versus non-recommended conservative interventions. We classified an intervention as recommended or non-recommended if it was labelled as such in two or more recent international LBP guidelines [5, 6, 9].

We performed meta-regression for those comparisons, outcomes, and time intervals that contained sufficient data when statistical heterogeneity appeared substantial and was likely to influence the overall pooled effect. Before conducting these analyses, we first examined the discriminative ability of the variables defined in the subgroup analyses by examining the distribution of the data, and excluded those variables when this was insufficient. In the subsequent step, we modelled just three variables:

- duration of LBP (subacute/chronic versus chronic only);
- 'multimodal' SMT (SMT delivered alone versus SMT delivered in conjunction with other modalities thought to have limited or no clinical effect); and
- country income status (high-income countries versus middle- and low-income countries).

We constructed the final model with variables demonstrating the strongest effect in univariate analyses. We report the analyses for the comparisons of SMT versus sham SMT/placebo and other conservative interventions in [Table 1](#) and [Table 2](#). All analyses were conducted in STATA, version 17.0.

Equity-related assessment

We evaluated the effectiveness of SMT for different age groups (young (< 65 years) versus old (65+)) and income level of the country (high versus middle and low).

Sensitivity analysis

Our single, pre-planned sensitivity analysis involved restricting the analysis to studies we deemed to have an adequate allocation procedure, to check if the review's findings were robust to the decisions made in the process of obtaining them [\[59\]](#).

Certainty of the evidence assessment

We presented summary of findings tables for the comparisons of SMT to: sham SMT/placebo, no treatment, and other conservative interventions. These tables include the critical outcomes of pain, functional status, and adverse events at one-month follow-up.

We evaluated the overall certainty of the evidence and strength of recommendations for each outcome using GRADE, as recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [\[62\]](#). The certainty of the evidence is based upon the following five domains: limitations of design (risk of bias), inconsistency of results (statistical heterogeneity), indirectness (generalisability), imprecision (few events or participants leading to wide estimates of effect), and other factors (e.g. publication bias). Single studies ($N < 400$ for continuous outcomes and < 300 events for dichotomous outcomes) were considered inconsistent and imprecise and provided 'low-certainty evidence', which could be further downgraded to 'very low' if there were limitations in design or indirectness. We only considered RCTs; therefore, the evidence was thought to be of high certainty and subsequently reduced by one (or two) levels for each domain that was not met. The operational definitions are described in the appendix ([Supplementary material 7](#)).

In cases where sensitivity analyses indicated that effect estimates from low-risk-of-bias studies were consistent with the overall findings, we did not downgrade the certainty of evidence for study limitations (risk of bias). This applied to the comparison of SMT versus other conservative interventions for the outcomes pain and functional status, even though performance bias was present (due to lack of participant blinding) and more than 50% of the total weight was derived from trials assessed as having a high risk of bias. This was a post hoc decision.

The levels of certainty of the evidence are as follows.

- High certainty: further research is very unlikely to change our confidence in the estimate of effect. There are sufficient data with narrow confidence intervals. There are no known or suspected reporting biases.
- Moderate certainty: further research is likely to have an important impact on confidence in the estimate of effect and may change the estimate; one of the domains is not met.
- Low certainty: further research is very likely to have an important impact on confidence in the estimate of effect and is likely to change the estimate; two of the domains are not met.
- Very low certainty: great uncertainty about the estimate; three of the domains are not met.
- No evidence: no evidence from RCTs.

Consumer involvement

Consumers were not involved in this review due to limited resources. However, we used core outcome sets for the review's outcomes, which were developed with consumer involvement [\[39\]](#).

RESULTS

Description of studies

Results of the search

We updated the search for the current review on 18 October 2024, searching for trials published since June 2009 – the date of the search in the previous version of the review [\[2\]](#).

The results of the search are presented in [Figure 1](#). The updated search strategy identified 8545 new citations through database searching and 388 records from other sources (5335 after de-duplication). Of these, we assessed 228 records in full text and included 50 new trials (72 articles) (Ali 2019 [\[63\]](#); Balthazard 2012 [\[64\]](#); Bialosky 2014 [\[65\]](#); Blanco-Giménez 2024 [\[66, 67, 68\]](#); Bond 2020 [\[69, 70\]](#); Bronfort 2011 [\[71, 72\]](#); Bronfort 2014 [\[73\]](#); Carrasco-Martínez 2019 [\[74\]](#); Castro-Sanchez 2016; Castro-Sanchez 2021 [\[75\]](#); Cecchi 2010 [\[76, 77, 78\]](#); Cook 2013 [\[79, 80, 81\]](#); De Oliveira 2020a [\[82\]](#); De Oliveira 2020b [\[83\]](#); Didehdar 2020 [\[84\]](#); Dogan 2021 [\[85\]](#); Dougherty 2014a [\[86\]](#); Dougherty 2014b [\[87\]](#); Fatima 2021 [\[88\]](#); Gevers-Montoro 2024 [\[89, 90\]](#); Grande-Alonso 2019 [\[91\]](#); Griswold 2019 [\[92\]](#); Haas 2014 [\[93, 94, 95\]](#); Ibrahim 2023 [\[96\]](#); Kamali 2019 [\[97\]](#); Krekoukias 2017; Licciardone 2013 [\[98, 99, 100, 101, 102, 103\]](#); Lizis 2021 [\[104, 105\]](#); McCarthy 2019 [\[106\]](#); Nambi 2018 [\[107\]](#); Nejati 2019 [\[108\]](#); Nim 2020 [\[109, 110, 111\]](#); Özcan 2022 [\[112\]](#); Petersen 2011 [\[113\]](#); Samir 2016 [\[114\]](#); Saracoglu 2022 [\[115\]](#); Sarker 2017 [\[116\]](#); Schulz 2019 [\[117\]](#); Senna 2011 [\[118\]](#); Sturion 2020 [\[119\]](#); Tavares 2017 [\[120\]](#); Teixeira 2016; Thomas 2020 [\[121\]](#); Ulger 2017 [\[122\]](#); Verma 2013 [\[123\]](#); Vismara 2012; Walker 2013 [\[124, 125\]](#); Wang 2019 [\[126\]](#); Xia 2016 [\[127\]](#); Zaidi 2020 [\[128\]](#)).

Figure 1. PRISMA flow diagram of included studies

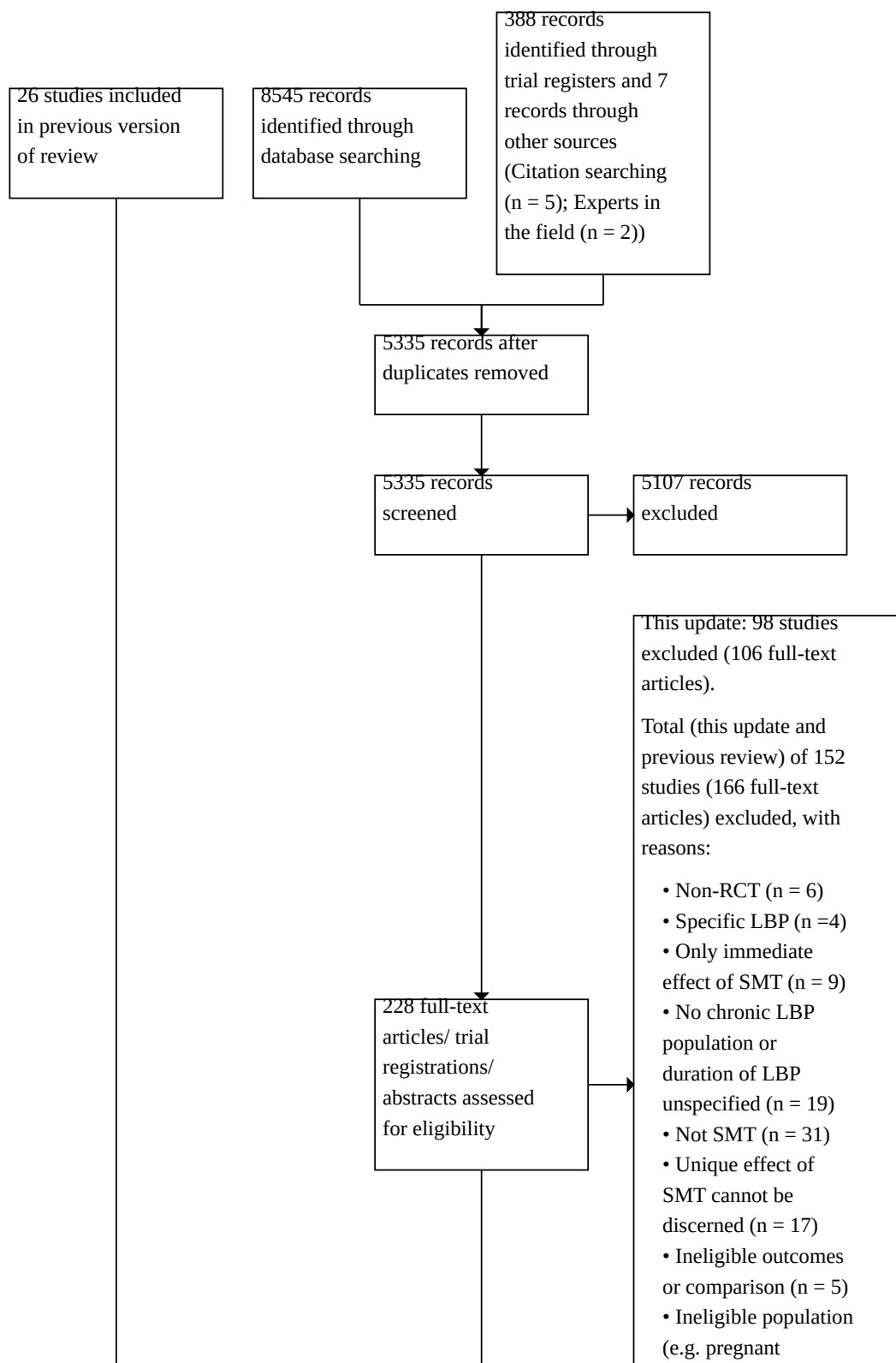
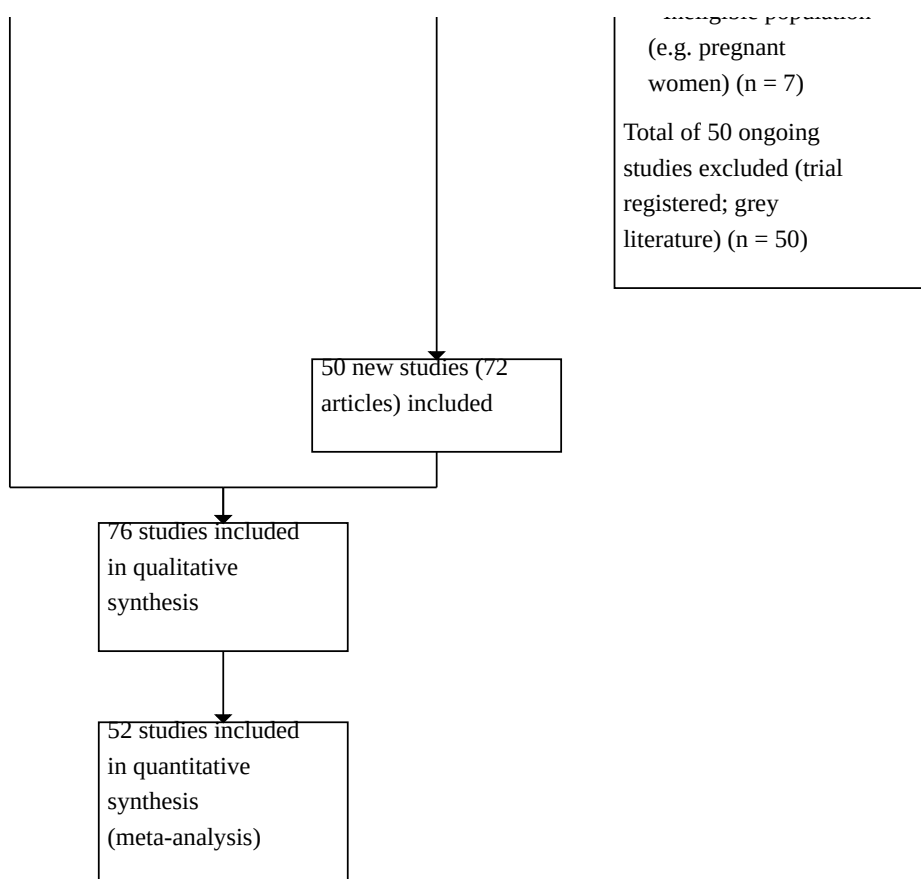


Figure 1. (Continued)



We included 26 trials from the previous version of the review (Brønfort 1996 [129]; Chown 2008 [130]; Evans 1978; Ferreira 2007; Ghroubi 2007 [131]; Gibson 1985 [132]; Goldby 2006 [133]; Gudavalli 2006 [134, 135, 136]; Hemmila 2002 [137, 138]; Hondras 2009 [139, 140]; Hsieh 2002 [141]; Hurwitz 2002 [142, 143, 144, 145, 146, 147, 148, 149, 150]; Koes 1992 [151, 152, 153, 154]; Licciardone 2003 [155]; Mohseni-Bandpei 2006 [156]; Muller 2005 [157, 158, 159]; Paatelma 2008 [160]; Pope 1994 [161, 162]; Postacchini 1988 [163]; Rasmussen 2008 [164]; Rasmussen-Barr 2003 [165]; Skillgate 2007 [166, 167, 168]; UK BEAM Trial 2004 [169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181]; Waagen 1986 [182]; Wilkey 2008 [183]; Zaproudina 2009 [184, 185]).

The 2011 review identified 10 ongoing or completed studies. Since then, four studies have reported results and are included in this review update (Bronfort 2011; Haas 2014; Licciardone 2013; Schulz 2019). Three trials were not published and thus remain listed as ongoing studies (ISRCTN47636118 2000 [186]; ISRCTN61808774 2000; NCT00410397 2006). We excluded the remaining three studies (Bronfort 2022 [187]: not SMT; Goertz 2013 [188]: acute LBP; NCT00269503 2006 [189]: subacute sciatica).

We contacted four authors to clarify the duration of LBP in their studies. Of these, we included two RCTs as they involved participants with chronic LBP. We excluded the other two: one due to its focus on acute LBP, and the other after multiple unsuccessful attempts to contact the author.

There are no studies awaiting assessment.

Thus, in this update, we have included a total of 76 trials.

Included studies

A complete record of all included studies and analyses for this review is available in [Supplementary material 2](#) and [Supplementary material 5](#), and a full data package is available in [Supplementary material 6](#) and [Supplementary material 11](#).

[Table 3](#) is a summary of the included studies and syntheses. [Table 4](#) is a summary of specific clinical and treatment characteristics important for interpreting the syntheses.

Study design and setting

Forty-four trials had two intervention arms; 25 trials had three intervention arms; five trials had four arms; and two trials had six arms (see [Table 3](#)).

The countries in which the trials were conducted varied. In total, 20 trials were conducted in the USA (Bialosky 2014; Bond 2020; Bronfort 2011; Bronfort 2014; ; Cook 2013; Dougherty 2014a; Dougherty 2014b; Gudavalli 2006; Haas 2014; Hondras 2009; Hsieh 2002; Hurwitz 2002; Licciardone 2003; Licciardone 2013; Pope 1994; Schulz 2019; Thomas 2020; Waagen 1986; Xia 2016); eight trials in the UK (Chown 2008; Evans 1978; Gibson 1985; Goldby 2006;

McCarthy 2019; Mohseni-Bandpei 2006; UK BEAM Trial 2004; Wilkey 2008); six trials in Spain (Blanco-Giménez 2024; Carrasco-Martinez 2019; Castro-Sanchez 2016; Castro-Sanchez 2021; Gevers-Montoro 2024; Grande-Alonso 2019); five trials in India (Ali 2019; Fatima 2021; Nambi 2018; Sarker 2017; Verma 2013) and Brazil (De Oliveira 2020a; De Oliveira 2020b; Sturion 2020; Tavares 2017; Teixeira 2016); four trials in Turkey (Dogan 2021; Özcan 2022; Saracoglu 2022; Ulger 2017); three in Finland (Hemmila 2002; Paatelma 2008; Zaproudina 2009), Australia (Ferreira 2007; Muller 2005; Walker 2013), Iran (Didehdar 2020; Kamali 2019; Nejati 2019), Denmark (Petersen 2011; Nim 2020; Rasmussen 2008), Italy (Postacchini 1988; Cecchi 2010; Vismara 2012), and Egypt (Ibrahim 2023; Samir 2016; Senna 2011); two in Sweden (Rasmussen-Barr 2003; Skillgate 2007) and Switzerland (Balthazard 2012; Griswold 2019), and one each in the Netherlands (Koes 1992), Greece (Krekoukias 2017), Pakistan (Zaidi 2020), Poland (Lizis 2021), China (Wang 2019), and Tunisia (Ghroubi 2007). All trials were published in English except the trial conducted in Tunisia, which was published in French.

Participants

In total, 11,866 participants were included; study sample sizes ranged from 10 to 1334 (median (IQR) 152 (55 to 207)).

Most trials included middle-aged participants (mean age ranged from 22 to 77) with or without radiating pain. The study populations included an average of 53% women (range from 1% to 100%). Two trials included exclusively older adults (Hondras 2009; Schulz 2019).

Eleven trials included participants without radiating pain (Bond 2020; Carrasco-Martinez 2019; Chown 2008; Dougherty 2014a; Dougherty 2014b; Ghroubi 2007; Grande-Alonso 2019; Griswold 2019; Muller 2005; Pope 1994; Wang 2019). However, in 34 trials, it was unclear if participants with radiating pain were included (Ali 2019; Bialosky 2014; Blanco-Giménez 2024; Castro-Sanchez 2016; Castro-Sanchez 2021; Cecchi 2010; Cook 2013; De Oliveira 2020a; De Oliveira 2020b; Didehdar 2020; Dogan 2021; Fatima 2021; Gibson 1985; Goldby 2006; Ibrahim 2023; Krekoukias 2017; Licciardone 2013; Lizis 2021; McCarthy 2019; Mohseni-Bandpei 2006; Nim 2020; Özcan 2022; Samir 2016; Saracoglu 2022; Sarker 2017; Senna 2011; Skillgate 2007; Sturion 2020; Tavares 2017; Teixeira 2016; Thomas 2020; Ulger 2017; Verma 2013; Vismara 2012; Walker 2013).

Forty-nine trials examined participants with exclusively chronic LBP (that is, had an inclusion criterion specifying that the symptoms must have been present for three months or longer) (Ali 2019; Balthazard 2012; Blanco-Giménez 2024; Bond 2020; Carrasco-Martinez 2019; Castro-Sanchez 2016; Castro-Sanchez 2021; Cecchi 2010; Chown 2008; De Oliveira 2020a; De Oliveira 2020b; Didehdar 2020; Dogan 2021; Dougherty 2014a; Dougherty 2014b; Fatima 2021; Ferreira 2007; Gevers-Montoro 2024; Ghroubi 2007; Goldby 2006; Grande-Alonso 2019; Gudavalli 2006; Haas 2014; Ibrahim 2023; Krekoukias 2017; Licciardone 2003; Licciardone 2013; Lizis 2021; Mohseni-Bandpei 2006; Muller 2005; Nambi 2018; Nejati 2019; Nim 2020; Özcan 2022; Rasmussen 2008; Samir 2016; Saracoglu 2022; Sarker 2017; Senna 2011; Sturion 2020; Tavares 2017; Teixeira 2016; Thomas 2020; Ulger 2017; Verma 2013; Vismara 2012; Wang 2019; Wilkey 2008; Zaproudina 2009). Most trials indicated that participants had a current episode of LBP consisting of months to years.

Interventions

Interventions are summarised in [Table 3](#) and [Table 4](#). More details are provided in [Supplementary material 11](#) and [Supplementary material 2](#).

SMT versus sham SMT/placebo

Seventeen trials including 2021 participants examined the effect of SMT versus sham SMT/placebo (Balthazard 2012; Bialosky 2014; Bond 2020; Didehdar 2020; Dougherty 2014b; Gevers-Montoro 2024; Ghroubi 2007; Gibson 1985; Koes 1992; Krekoukias 2017; Licciardone 2003; Licciardone 2013; Senna 2011; Tavares 2017; Thomas 2020; Waagen 1986; Walker 2013). The effects of SMT were compared to sham SMT in 11 trials and to placebo treatment in six trials (see [Table 3](#)).

SMT versus no treatment

Four trials including 453 participants examined the effect of SMT versus no treatment (Bialosky 2014; Licciardone 2003; Tavares 2017; Xia 2016) (see [Table 3](#)).

SMT versus other conservative interventions

In total, 43 trials with a total of 8291 participants compared the effects of SMT to other conservative interventions. Most commonly, the other conservative intervention was exercise (26 trials) (Ali 2019; Bronfort 2011; Blanco-Giménez 2024; Brønfort 1996; Castro-Sanchez 2021; Cecchi 2010; Chown 2008; De Oliveira 2020a; Dougherty 2014a; Fatima 2021; Ferreira 2007; Gibson 1985; Goldby 2006; Griswold 2019; Gudavalli 2006; Haas 2014; Hemmila 2002; Hondras 2009; Hsieh 2002; Hurwitz 2002; Kamali 2019; Koes 1992; Krekoukias 2017; Mohseni-Bandpei 2006; Muller 2005; Nejati 2019; Özcan 2022; Paatelma 2008; Petersen 2011; Pope 1994; Postacchini 1988; Rasmussen-Barr 2003; Samir 2016; Sarker 2017; Schulz 2019; Skillgate 2007; Sturion 2020; Teixeira 2016; UK BEAM Trial 2004; Ulger 2017; Wilkey 2008; Zaidi 2020; Zaproudina 2009) (see [Table 3](#)).

Effect of SMT as adjuvant therapy

Seventeen trials (2963 participants) examined the adjuvant effects of SMT when combined with analgesic therapy (Evans 1978), exercise (Blanco-Giménez 2024; Dogan 2021; Ibrahim 2023; Nejati 2019; Rasmussen 2008; Saracoglu 2022; Schulz 2019; Verma 2013; Vismara 2012; Wang 2019), exercise plus laser therapy (Nambi 2018), exercise plus education (Bronfort 2014), myofascial therapy (Hsieh 2002), standard medical care in combination with exercise (UK BEAM Trial 2004), usual care (Licciardone 2003), or education (Grande-Alonso 2019).

SMT versus another type of SMT

In total, nine trials examined the effect of head-to-head comparisons of different types of SMT (1227 participants) (Carrasco-Martinez 2019; Castro-Sanchez 2016; Cook 2013; De Oliveira 2020b; Hondras 2009; Lizis 2021; McCarthy 2019; Nim 2020; Xia 2016).

Number and duration of treatments

It is unclear how many treatments the participants received, on average, because trials did not typically report this. The maximum number of treatments allowed by the protocol was, on average, eight (SD 4.3; data from 67 trials range from one to 24). In other trials, this was at the discretion of the therapist/physician and terminated sooner if the patient recovered ([Table 4](#)). Similarly, the

treatment period was also varied. The duration of the treatment was protocolised for, on average, six weeks (SD 3.7; data from 74 trials range from one week to five months).

Outcome measures: types, timing

Critical outcomes

- **Pain.** All but one trial reported on pain (Chown 2008). All trials measured this construct via a VAS or NRS, except four which used the pain subscale from the modified von Korff scale (Haas 2014; Petersen 2011; Skillgate 2007; UK BEAM Trial 2004), two which used a 0 to 4 scale (Evans 1978; Postacchini 1988), one which used the West Haven-Yale Multidimensional Pain Inventory (Koes 1992), and one which used the Roland Morris Pain Rating Scale (Cecchi 2010).
- **Functional status.** Most trials reported back pain-specific functional status (except Dogan 2021; Evans 1978; Gibson 1985; Nim 2020; Pope 1994; Rasmussen 2008; Samir 2016; Sarker 2017; Verma 2013; Waagen 1986). This consisted of either the RMDQ (29 RCTs) (Bronfort 2011; Bronfort 2014; Brønfort 1996; Castro-Sanchez 2016; Castro-Sanchez 2021; Cecchi 2010; De Oliveira 2020b; Ferreira 2007; Grande-Alonso 2019; Gudavalli 2006; Hondras 2009; Hsieh 2002; Hurwitz 2002; Krekorkias 2017; Licciardone 2013; Licciardone 2003; McCarthy 2019; Nambi 2018; Özcan 2022; Paatelma 2008; Petersen 2011; Schulz 2019; Sturion 2020; Teixeira 2016; Thomas 2020; UK BEAM Trial 2004; Vismara 2012; Wilkey 2008; Xia 2016) or the ODI (35 RCTs) (Ali 2019; Balthazard 2012; Bialosky 2014; Blanco-Giménez 2024; Bond 2020; Carrasco-Martinez 2019; Castro-Sanchez 2016; Chown 2008; Cook 2013; De Oliveira 2020a; Didehdar 2020; Dougherty 2014a; Dougherty 2014b; Fatima 2021; Gevers-Montoro 2024; Griswold 2019; Ghroubi 2007; Goldby 2006; Hemmila 2002; Ibrahim 2023; Kamali 2019; Krekorkias 2017; Lizis 2021; Mohseni-Bandpei 2006; Muller 2005; Nejati 2019; Rasmussen-Barr 2003; Saracoglu 2022; Senna 2011; Tavares 2017; Ulger 2017; Vismara 2012; Wang 2019; Zaidi 2020; Zaproudina 2009). However, other scales were also used, such as the modified von Korff scale (Haas 2014; Skillgate 2007; UK BEAM Trial 2004) (disability data presented separately), Disability Rating Index (Rasmussen-Barr 2003), Functional Rating Index (Walker 2013), Sickness Impact Profile (Koes 1992), back performance scale (Saracoglu 2022), and a four-point non-validated scale (Postacchini 1988).
- **Adverse events.** Less than half of the trials (36/76) reported on adverse events. In only 13 trials, the adverse events were actively monitored. Adverse events in the SMT group were limited to muscle soreness, stiffness, and/or transient increase in pain. None of the trials registered any serious complications related to either the experimental or control group. A detailed summary of the adverse events is given in [Table 5](#).

Important outcomes

Relatively few trials reported on the review's important outcomes, such as return to work or aspects thereof, such as perceived recovery (Bronfort 2011; Brønfort 1996; Cook 2013; De Oliveira 2020b; Evans 1978; Ferreira 2007; Gibson 1985; Grande-Alonso

2019; Griswold 2019; Gudavalli 2006; Haas 2014; Hondras 2009; Hurwitz 2002; Koes 1992; Petersen 2011; Senna 2011; Skillgate 2007; Walker 2013; Zaproudina 2009), return to work, e.g. number of sick leave days (Cecchi 2010; Gibson 1985; Hemmila 2002; Hsieh 2002; Licciardone 2003; Licciardone 2013; Petersen 2011; Schulz 2019), or health-related quality of life, such as via the SF-36 (Bronfort 2011; Bronfort 2014; Castro-Sanchez 2016; Castro-Sanchez 2021; Dougherty 2014a; Dougherty 2014b; Gudavalli 2006; Hondras 2009; Hsieh 2002; Licciardone 2003; Licciardone 2013; Muller 2005; Petersen 2011; Schulz 2019; Senna 2011; UK BEAM Trial 2004; Ulger 2017), SF-12 (Haas 2014), EuroQoL (Bronfort 2014; Chown 2008; Haas 2014; Sarker 2017; UK BEAM Trial 2004), HRQoL - 15D questionnaire (Zaproudina 2009), CDC HRQoL-4 (Nambi 2018), Nottingham Health Profile (Goldby 2006), general health status (expressed on a 10-cm VAS scale) (Rasmussen-Barr 2003), and other instruments (Dartmouth Primary Care Cooperative Information Project chart system) (Brønfort 1996). In addition, when the SF-36 was measured, data were not always available for the general health subscale, as some trials reported either an overall score (Hondras 2009; Hsieh 2002; Licciardone 2003), or presented other subscales (Dougherty 2014a; Dougherty 2014b; UK BEAM Trial 2004; Xia 2016). Three trials examined a mixed population (neck and low back) (Koes 1992; Skillgate 2007; Walker 2013), and data are presented for the low back only.

Timing of the outcome measures ranged from two weeks to two years post-randomisation. Only two trials examined follow-up beyond 12 months (Goldby 2006; Skillgate 2007). The majority reported short- and intermediate-term outcomes.

Regarding the reporting of SMT interventions, 82 SMT arms (78%) had a TIDieR (Template for Intervention Description and Replication) score of 7 or higher (including the response reported and not applicable) [190], while the remaining 22 SMT arms (22%) fell below this threshold. While this may initially suggest a relatively high percentage of trials reporting their interventions satisfactorily, it underscores the substantial absence of important items regarding SMT interventions in the trial reports.

Excluded studies

We excluded 152 studies during full-text screening and documented the reasons for their exclusion in the 'Characteristics of excluded studies' ([Supplementary material 3](#)). The most common reasons for exclusion were: no chronic LBP population, unspecified duration of LBP, non-RCT, specific LBP, and the unique effect of SMT could not be discerned. For a detailed description of the excluded studies, see [Supplementary material 11](#) and [Supplementary material 3](#).

We added 37 new registered trials or grey literature items to the 'ongoing studies' category (n = 50) ([Supplementary material 4](#)).

Risk of bias in included studies

A summary of our judgements on the risk of bias items for each included trial is presented in [Figure 2](#), and presented as percentages across all trials in [Figure 3](#).

Figure 2. Risk of bias summary: summary of authors' judgements on risk of bias domains for each included study

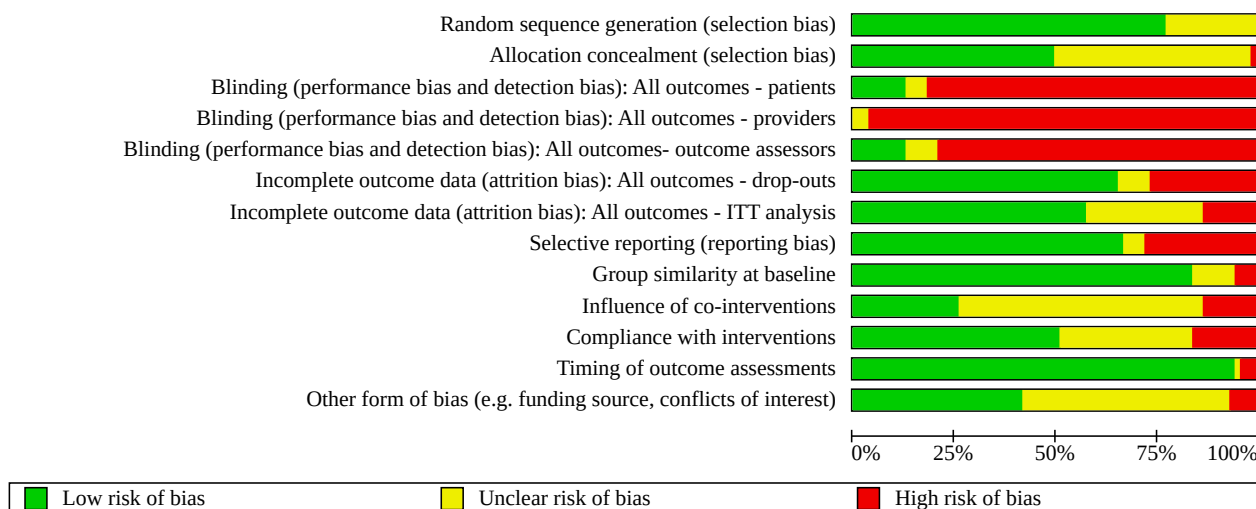
	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding (performance bias and detection bias): All outcomes - patients	Blinding (performance bias and detection bias): All outcomes - providers	Blinding (performance bias and detection bias): All outcomes- outcome assessors	Incomplete outcome data (attrition bias): All outcomes - drop-outs	Incomplete outcome data (attrition bias): All outcomes - ITT analysis	Selective reporting (reporting bias)	Group similarity at baseline	Influence of co-interventions	Compliance with interventions	Timing of outcome assessments	Other form of bias (e.g. funding source, conflicts of interest)
Ali 2019	+	?	-	-	-	-	+	+	+	+	+	+	+
Balthazard 2012	+	+	-	-	-	+	-	-	+	?	+	+	+
Bialosky 2014	+	?	-	-	-	+	?	+	+	?	+	+	+
Blanco-Giménez 2024	?	?	-	-	?	?	+	+	?	?	+	+	+
Bond 2020	+	+	+	-	+	+	+	+	+	?	+	+	+
Bronfort 2011	+	+	-	-	-	+	+	+	+	+	+	+	+
Bronfort 2014	+	+	-	-	-	+	+	-	+	+	+	+	+
Brønfort 1996	+	+	-	-	-	-	+	-	+	+	+	+	?
Carrasco-Martinez 2019	+	+	-	-	-	-	-	+	+	?	-	+	?
Castro-Sanchez 2016	+	+	-	-	-	+	+	+	+	?	+	+	-
Castro-Sanchez 2021	+	+	-	-	-	+	+	+	+	?	?	+	?
Cecchi 2010	+	+	-	-	-	+	+	+	+	-	+	+	+
Chown 2008	+	?	-	-	-	-	?	-	+	?	-	+	?
Cook 2013	+	?	-	-	-	+	+	-	+	-	+	+	?
De Oliveira 2020a	+	+	-	-	-	+	+	+	+	+	+	+	?

Figure 2. (Continued)

	1	2	3	4	5	6	7	8	9	10	11	12
De Oliveira 2020a	+	+	-	-	-	+	+	+	+	+	+	?
De Oliveira 2020b	+	+	+	-	+	+	+	+	+	?	+	+
Didehdar 2020	?	?	+	-	+	+	+	+	+	+	+	?
Dogan 2021	?	?	-	-	-	+	?	?	?	?	+	+
Dougherty 2014a	+	+	-	-	-	+	+	-	+	?	-	+
Dougherty 2014b	+	?	-	-	-	+	+	+	+	+	?	+
Evans 1978	?	?	-	-	-	+	?	-	+	+	?	?
Fatima 2021	+	+	-	-	-	+	+	+	+	+	?	?
Ferreira 2007	+	+	-	-	-	+	+	+	+	?	+	?
Gevers-Montoro 2024	+	?	+	-	+	+	+	+	+	?	+	+
Ghroubi 2007	+	?	?	-	?	+	?	+	+	?	+	?
Gibson 1985	?	?	-	-	-	+	?	-	-	?	+	?
Goldby 2006	+	?	-	-	-	+	-	+	+	?	-	?
Grande-Alonso 2019	+	?	-	-	-	+	+	+	+	+	+	+
Griswold 2019	+	+	-	-	-	+	+	+	+	+	+	?
Gudavalli 2006	+	+	-	-	-	-	-	+	+	?	?	?
Haas 2014	+	+	-	-	-	+	+	+	+	+	+	+
Hemmila 2002	+	+	-	-	-	+	?	-	+	-	+	?
Hondras 2009	+	+	-	-	-	-	+	+	+	?	-	?
Hsieh 2002	+	?	-	-	-	+	+	+	+	+	-	?
Hurwitz 2002	+	+	-	-	-	+	+	+	+	-	-	?
Ibrahim 2023	+	?	-	-	-	+	+	+	+	?	+	?
Kamali 2019	+	?	-	-	-	-	-	+	?	?	+	+
Koes 1992	+	+	-	-	-	+	+	-	+	-	?	?
Krekoukias 2017	+	?	-	-	-	+	+	+	+	+	+	-
Licciardone 2003	+	?	+	-	+	-	?	-	+	+	?	?
Licciardone 2013	+	?	+	-	+	+	+	+	+	+	+	?
Lizis 2021	+	+	+	-	+	+	+	+	+	+	+	?
McCarthy 2019	?	?	?	-	?	-	?	+	+	?	?	?
Mohseni-Bandpei 2006	?	?	-	-	-	-	?	+	+	?	?	?
Muller 2005	+	?	-	-	-	-	-	+	+	?	-	?
Nambi 2018	+	+	-	-	-	+	+	+	+	+	?	+
Nejati 2019	+	+	-	-	-	+	+	+	-	+	+	+
Nim 2020	+	-	+	-	+	+	+	+	+	?	+	+
Özcan 2022	?	?	?	?	?	?	?	+	+	?	?	-
Parkkinen 2000	+	?	-	-	-	-	-	+	+	?	?	?

Figure 2. (Continued)

Özcan 2022	?	?	?	?	?	?	?	+	+	?	?	+	-
Paatelma 2008	+	?	-	-	-	-	+	+	+	?	?	+	?
Petersen 2011	+	+	-	-	-	-	+	+	+	?	+	+	+
Pope 1994	?	?	-	-	-	+	?	-	+	?	+	+	?
Postacchini 1988	?	?	-	-	-	?	-	-	?	-	?	+	?
Rasmussen 2008	?	?	-	-	-	-	+	-	+	?	+	+	?
Rasmussen-Barr 2003	+	?	-	-	-	-	?	+	+	?	?	+	?
Samir 2016	?	?	?	?	?	?	?	-	+	-	-	+	?
Saracoglu 2022	+	+	-	?	?	+	+	+	?	?	?	+	?
Sarker 2017	?	?	-	-	-	+	?	-	+	-	-	+	-
Schulz 2019	+	+	-	-	-	+	+	+	+	+	+	+	+
Senna 2011	?	?	+	-	+	-	?	+	+	?	?	+	+
Skillgate 2007	+	+	-	-	-	+	+	+	+	?	?	+	?
Sturion 2020	+	+	-	-	-	+	+	-	?	?	+	+	+
Tavares 2017	+	+	-	-	-	-	-	+	-	?	+	+	+
Teixiera 2016	+	+	-	-	-	+	+	-	?	?	?	-	+
Thomas 2020	+	?	-	-	-	+	+	+	+	-	+	+	+
UK BEAM Trial 2004	+	+	-	-	-	-	+	+	+	?	?	+	+
Ulger 2017	+	?	-	-	-	-	-	+	+	?	-	+	+
Verma 2013	?	+	-	-	-	?	?	?	+	?	?	+	?
Vismara 2012	+	+	-	-	-	+	?	+	-	?	+	+	?
Waagen 1986	?	?	+	-	+	-	?	-	+	?	?	+	?
Walker 2013	+	+	-	-	-	+	+	-	+	?	+	+	+
Wang 2019	?	?	-	-	-	+	?	-	?	-	?	+	?
Wilkey 2008	+	+	-	-	-	+	?	?	+	?	-	+	?
Xia 2016	+	+	-	-	-	+	+	?	+	?	+	+	-
Zaidi 2020	+	?	-	-	-	?	?	+	+	?	?	+	+
Zaproudina 2009	+	+	-	-	-	+	-	+	+	?	?	-	+

Figure 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies

We identified three trials with a 'fatal flaw' and excluded them from the meta-analyses: two trials demonstrated more than 50% dropout at the first follow-up measurement (Chown 2008; Muller 2005), and one trial was found to have clinically relevant baseline differences between the intervention groups for one or more primary outcomes (Goldby 2006). Upon closer examination, it was clear that only a per-protocol analysis had been conducted. We sent our risk of bias assessment to all authors of the newly included studies in this update. Thirteen authors responded (Ali 2019; Carrasco-Martinez 2019; Castro-Sanchez 2016; De Oliveira 2020a; Dougherty 2014a; Gevers-Montoro 2024; Haas 2014; Licciardone 2013; McCarthy 2019; Nim 2020; Özcan 2022; Sturion 2020; Vismara 2012), but this did not lead to any major changes in our assessment.

Selection bias

Slightly less than half of the trials (49%; 37/76) used both an adequate sequence generation and allocation procedure and were judged to be at low risk of selection bias. In sixteen trials, both randomisation and allocation were unclear. Twenty-three trials had either an adequate sequence generation or allocation procedure and a high or unclear risk of bias for either sequence generation or allocation procedure.

Additionally, for group similarities at baseline, we judged four trials to have a high risk of bias.

Performance bias, blinding

In trials of hands-on therapies such as SMT, prevention of performance bias poses a challenge, as blinding of the participants is difficult and blinding of the treatment providers is not possible. Performance bias will play a major role in trials comparing SMT to sham SMT, as the placebo effect in each arm will be different. However, trials in a pragmatic setting see a lesser impact of performance bias as the placebo effect in each arm will be quite similar. We assigned a low risk of bias if the participants had similar perceptions about the effectiveness of the interventions.

In total, 10 trials (13%) attempted to blind participants and outcome assessors to the assigned intervention and were judged to be at low risk of performance and detection bias. The reporting of performance and detection bias was unclear in four trials.

In all trials, the healthcare provider administering the intervention was not blinded, except for three trials in which the blinding was judged unclear.

Attrition bias

We judged 36 of 76 (47%) trials to have a low risk of attrition bias, because they provided an adequate overview of withdrawals or dropouts, kept these to a minimum for the subsequent follow-up measurements, and conducted intention-to-treat (ITT) analysis. We classified 11 trials as having an unclear risk in this domain because of insufficient information about missing data, or authors did not report an ITT analysis, or both.

We judged the remaining 24 trials to be at high risk of bias, as there was a difference in dropouts between the groups, the study did not perform an ITT analysis, or both.

In nine trials, there was a difference in the dropout rate between groups. Eleven trials reported a large dropout, but the dropout rate was equal between the groups.

Selective reporting bias

Almost half of the trials (45%; 34/76) published or registered their protocol. In 32 trials, the reported outcomes were consistent with the protocol, while in five trials the reported outcomes were inconsistent with the protocol, and one trial did not report outcomes in their registration.

Additionally, we identified 20 trials that had no published or registered protocol, but reported all the expected primary outcomes.

In total, we judged 51 of 76 trials (67%) to have a low risk of selective reporting bias.

Synthesis of results

See [Summary of findings 1](#) (SMT versus sham SMT/placebo), [Summary of findings 2](#) (SMT versus no treatment), and [Summary of findings 3](#) (SMT versus other conservative interventions).

We converted all pain scales to a 100-point scale.

The minimal important difference for all outcomes was categorised as follows.

- Little to no difference: MD < 5% of the scale (e.g. < 5 mm on a 100 mm VAS); SMD difference < 0.2.
- Small: MD 5% to 10% of the scale (e.g. 5 to 10 mm on a 100 mm VAS); SMD difference 0.2 to 0.5; relative risk < 1.25 or > 0.8.
- Medium: MD 10% to 20% of the scale; SMD 0.5 to 0.8; relative risk 1.25 to 2.0 or 0.5 to 0.8.
- Large: MD > 20% of the scale; SMD ≥ 0.8; relative risk > 2.0 or < 0.5.

Comparison 1. SMT versus sham SMT/placebo

Critical outcomes

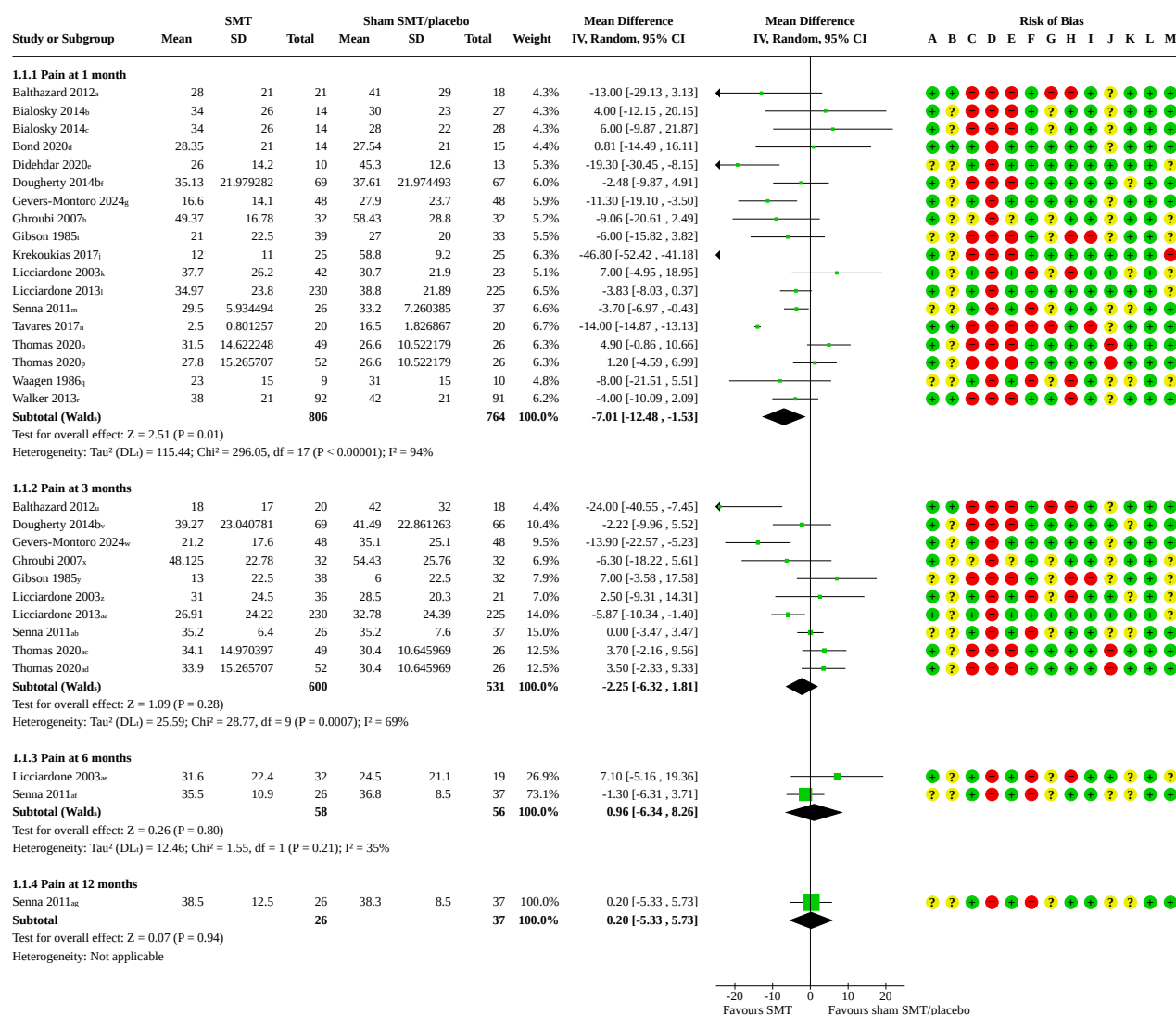
Pain

Sixteen trials examined the effect of SMT on pain (1570 participants) (Balthazard 2012; Bialosky 2014; Bond 2020; Didehdar 2020;

Dougherty 2014b; Gevers-Montoro 2024; Ghroubi 2007; Gibson 1985; Krekoulas 2017; Licciardone 2003; Licciardone 2013; Senna 2011; Tavares 2017; Thomas 2020; Waagen 1986; Walker 2013).

There is very low-certainty evidence (downgraded for inconsistency by one level and for study limitations by two levels) that SMT may result in a small reduction in pain compared to sham SMT/placebo at one month (MD -7.01, 95% CI -12.48 to -1.53; $I^2 = 94\%$; 16 studies, 1570 participants), but the evidence is very uncertain. There is very low-certainty evidence (downgraded for study limitations by two levels and for inconsistency by one level) that SMT may result in little to no difference in pain compared to sham SMT/placebo at three months (MD -2.25, 95% CI -6.32 to 1.81; $I^2 = 69\%$; 9 studies, 1131 participants), but the evidence is very uncertain. Additionally, there is very low-certainty evidence (downgraded for study limitations by two levels, and for inconsistency, indirectness, and imprecision by one level) that SMT may result in little to no difference in pain compared to sham SMT/placebo at six months (MD 0.96, 95% CI -6.34 to 8.26; $I^2 = 35\%$; 2 studies, 114 participants) and 12 months (MD 0.20, 95% CI -5.33 to 5.73; 1 study, 63 participants), but the evidence is very uncertain ([Figure 4](#)).

Figure 4. Comparison 1: SMT compared with sham SMT/placebo for chronic LBP; Outcome 1: Pain (0-100)



Footnotes

- ^aSMT with active exercise vs detuned US with active exercise; VAS 0-100; data extracted from table 2; 1st follow up is after discharge (4-8 weeks)
- ^bSMT vs standard sham SMT, described as placebo SMT by trialists; NRS 0-100; data based upon figures presented in Fig.4; change scores presented and SDs used from baseline; data at 2 weeks
- ^cSMT vs enhanced sham SMT, described as placebo SMT by trialists; NRS 0-100; data based upon figures presented in Fig.4; change scores presented and SDs used from baseline; data at 2 weeks
- ^dSMT versus sham SMT; NPRS 0-10; converted to 100 scale; change score presented and SD based upon Balthazard 2012; data at 3 weeks
- ^eSMT vs sham SMT; NRS 0-10 converted to 0-100; data extracted from table 2; data at 5 weeks
- ^fSMT versus sham SMT; VAS 0-100; data extracted from table 2; data at 5 weeks
- ^gSMT vs sham SMT, described as placebo SMT by trialists; NPRS 0-100; data extracted from table 4; data (average pain) at 4 weeks
- ^hUnclear SMT vs. sham SMT; VAS 0-10 converted to 0-100; data extracted from table 3; data at 1 month
- ⁱSMT vs detuned diathermy; median (range) presented in the study and converted; data extracted from table 2; data (daytime pain) at 4 weeks
- ^jSMT vs sham SMT; NRS 0-10; converted to 0-100 scale; data extracted from table 2; data at 5 weeks
- ^kOsteopathic MT vs. sham SMT; VAS 0-10 converted to 0-100; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 1 month
- ^lOsteopathic MT vs sham MT; VAS 0-10; converted to 0-100; data received from the author; data at 1 month
- ^mSMT ('non-maintained') vs sham SMT ('control group'); VAS 0-100; SE converted into SD; data extracted from table 3; data at 1 month
- ⁿSMT vs sham SMT; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 5 weeks
- ^oManipulation vs sham SMT; NPRS 0-10; converted to 0-100; data based upon figures presented in Fig 2 using WebPlot Digitizer; data at 4 weeks
- ^pMobilisation vs sham SMT; NPRS 0-10; converted to 0-100; data based upon figures presented in Fig 2 using WebPlot Digitizer; data at 4 weeks
- ^qChiropractic/HVLA SMT vs. sham SMT; VAS 0-10; data converted to 0-100; SD's presented here are approximated from similar populations; data at 2 weeks?
- ^rChiropractic care vs detuned ultrasound and sham activator; NRS 0-10; converted to 0-100; data extracted from table 4; data at 2 weeks
- ^sCI calculated by Wald-type method.
- ^tTau² calculated by DerSimonian and Laird method.
- ^uSMT with active exercise vs detuned US with active exercise; VAS 0-100; data extracted from table 2; 1st Follow up is after discharge (4-8 weeks), 2nd follow up is 3 months after discharge
- ^vSMT versus sham SMT; VAS 0-100; data extracted from table 2; data at 12 weeks
- ^wMT vs sham SMT, described as placebo SMT by trialists; NPRS 0-100; data extracted from table 4; data at 8 weeks
- ^xUnclear SMT vs. sham SMT; VAS 0-10; converted to 0-100; data extracted from table 3; data at 2 months
- ^ySMT vs detuned diathermy; median (range) presented in the study and converted; data extracted from table 2; data (daytime pain) at 12 weeks
- ^zOsteopathic SMT vs. sham SMT; VAS 0-10 converted to 0-10; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 3 months
- ^{aa}Osteopathic MT vs sham MT; VAS 0-10; converted to 0-100; data received from the author; data at 12 weeks
- ^{ab}SMT ('non-maintained') vs sham SMT ('control group'); VAS 0-100; SE converted into SD; data extracted from table 3; data at 4 months

Figure 4. (Continued)

⌚Osteopathic SMT vs. sham SMT; VAS 0-10 converted to 0-100; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 3 months
⌚Osteopathic MT vs sham MT; VAS 0-10; converted to 0-100; data received from the author; data at 12 weeks
⌚SMT ('non-maintained') vs sham SMT ('control group'); VAS 0-100; SE converted into SD; data extracted from table 3; data at 4 months
⌚Manipulation vs sham SMT; NPRS 0-10; converted to 0-100; data based upon figures presented in Fig 2 using WebPlot Digitizer; data at 7 weeks
⌚Mobilisation vs sham SMT; NPRS 0-10; converted to 0-100; data based upon figures presented in Fig 2 using WebPlot Digitizer; data at 7 weeks
⌚Osteopathic SMT vs. sham SMT; VAS 0-10; converted to 0-100; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 6 months
⌚SMT ('non-maintained') vs sham SMT ('control group'); VAS 0-100; SE converted into SD; data extracted from table 3; data at 7 months
⌚SMT ('non-maintained') vs sham SMT ('control group'); VAS 0-100; SE converted into SD; data extracted from table 3; data at 10 months

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding (performance bias and detection bias): All outcomes - patients
- (D) Blinding (performance bias and detection bias): All outcomes - providers
- (E) Blinding (performance bias and detection bias): All outcomes- outcome assessors
- (F) Incomplete outcome data (attrition bias): All outcomes - drop-outs
- (G) Incomplete outcome data (attrition bias): All outcomes - ITT analysis
- (H) Selective reporting (reporting bias)
- (I) Group similarity at baseline
- (J) Influence of co-interventions
- (K) Compliance with interventions
- (L) Timing of outcome assessments
- (M) Other form of bias (e.g. funding source, conflicts of interest)

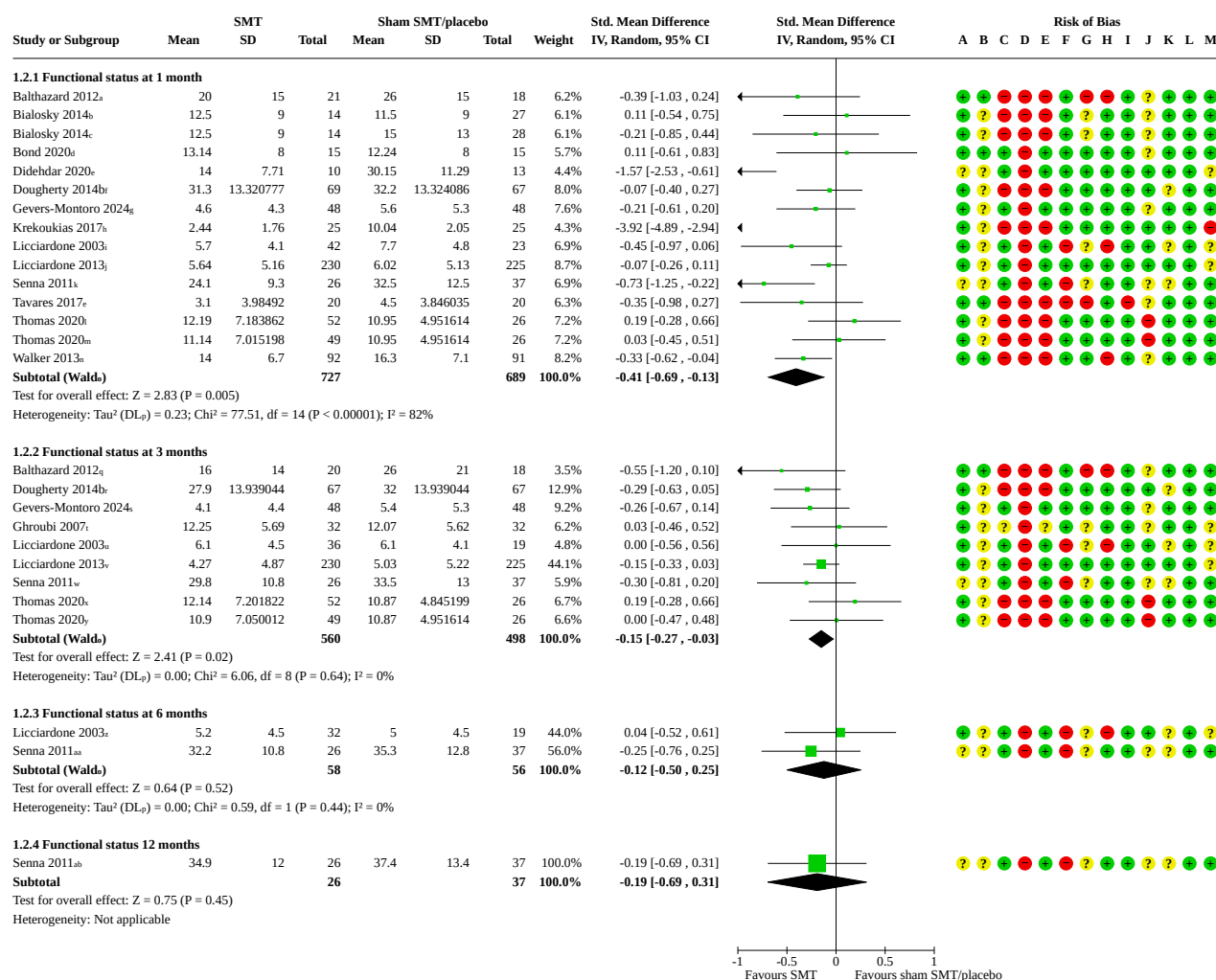
Functional status

Thirteen trials examined the effect of SMT on functional status (1416 participants) (Balthazard 2012; Bialosky 2014; Bond 2020; Didehdar 2020; Dougherty 2014b; Gevers-Montoro 2024; Krekoulakis 2017; Licciardone 2003; Licciardone 2013; Senna 2011; Tavares 2017; Thomas 2020; Walker 2013). There is very low-certainty evidence (downgraded for study limitations by two levels and inconsistency by one level) that SMT may result in a medium improvement in functional status compared to sham SMT/placebo at one month (SMD -0.41, 95% CI -0.69 to -0.13; $I^2 = 82\%$; 13 studies, 1416 participants), but the evidence is very uncertain. There is low-certainty evidence (downgraded for study limitations by two levels) that SMT may result in little to no difference in functional status compared to sham SMT/placebo at three months (SMD -0.15, 95% CI -0.27 to -0.03; $I^2 = 0\%$; 8 studies, 1058 participants).

Additionally, there is very low-certainty evidence (downgraded for study limitations and imprecision by two levels, and inconsistency

and indirectness by one level) that SMT may result in little to no difference in functional status compared to sham SMT/placebo at six months (SMD -0.12, 95% CI -0.50 to 0.25; $I^2 = 0\%$; 2 studies, 114 participants) and 12 months (SMD -0.19, 95% CI -0.69 to 0.31; 1 study, 63 participants), but the evidence is very uncertain (Figure 5). Converted to an MD on the Roland Morris Disability Questionnaire of 0 to 24, these effects amount to an MD of -2.10, 95% CI -3.54 to -0.67 at 1-month follow-up, MD of -0.78, 95% CI -1.41 to -0.016 at 3-month follow-up, MD of -0.54, 95% CI -2.25 to 1.13 at 6-month follow-up (based on an SD of 5.13 at 4 weeks, SD of 5.22 at 3 months, SD of 4.5 at six months from Licciardone 2013, as this is a large, good-quality study that used a common functional status scale (RDMQ)). Because only one study reported functional status at 12 months (Senna 2011), we present its MD of -2.5 (95% CI -8.82 to 3.82) on the Oswestry Disability Index at 12-month follow-up, rather than back-translating the SMD.

Figure 5. Comparison 1: SMT compared with sham SMT/placebo for chronic LBP; Outcome 2: functional status



Footnotes

- ^aSMT with active exercise vs detuned US with active exercise; ODI 0-100; data extracted from table 2; 1st Follow up is after discharge (4-8 weeks)
- ^bSMT vs standard sham SMT, described as placebo SMT by trialists; ODI 0-100; data based upon figures presented in Fig. 4; change scores presented and SDs used from baseline; data at 2 weeks
- ^cSMT vs enhanced sham SMT, described as placebo SMT by trialists; ODI 0-100; data based upon figures presented in Fig. 4; change scores presented and SDs used from baseline; data at 2 weeks
- ^dSMT vs sham SMT; ODI 0-100; Change score presented and SD based upon Didehar 2020; data at 3 weeks
- ^eSMT vs sham SMT; ODI 0-100; data extracted from table 2; data at 5 weeks
- ^fSMT vs sham SMT; ODI 0-100; data extracted from table 2 (mean and CI); data at 5 weeks
- ^gSMT vs sham SMT, described as placebo SMT by trialists; ODI 0-100; data extracted from table 4; data at 4 weeks
- ^hSMT vs sham SMT; RMDQ 0-24; data extracted from table 2; data at 5 weeks
- ⁱOsteopathic SMT vs sham; RMDQ 0-24; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 1 month
- ^jOsteopathic MT vs sham MT; RMDQ 0-24; data received from the author; data at 1 month
- ^kSMT ('non-maintained') vs sham SMT ('control group'); ODI 0-100; SE converted into SD; data extracted from table 3; data at 1 month
- ^lMobilisation vs sham SMT; RMDQ 0-24; data based upon figures presented in Fig. 2 using WebPlot Digitizer; data at 4 weeks
- ^mManipulation vs sham SMT; RMDQ 0-24; data based upon figures presented in Fig. 2 using WebPlot Digitizer; data at 4 weeks
- ⁿChiropractic care vs detuned ultrasound and sham activator; Functional Rating Index 0-100; used data extracted from table 4; data at 2 weeks
- ^oCI calculated by Wald-type method.
- ^pTau² calculated by DerSimonian and Laird method.
- ^qSMT with active exercise vs detuned US with active exercise; ODI 0-100; data extracted from table 2; 1st Follow up is after discharge (4-8 weeks); 2nd follow up is 3 months after discharge
- ^rSMT vs sham SMT; ODI 0-100; data extracted from table 2 (mean and CI); data at 12 weeks
- ^sSMT vs sham SMT, described as placebo SMT by trialists; ODI 0-100; data extracted from table 4; data at 8 weeks
- ^tUnclear SMT vs sham SMT; ODI 0-100; data extracted from table 3 and 4; data at 2 months
- ^uOsteopathic SMT vs. sham SMT; RMDQ 0-24; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 3 months
- ^vOsteopathic MT vs sham MT; RMDQ 0-24; data received from the author; data at 12 weeks
- ^wSMT ('non-maintained') vs sham SMT ('control group'); ODI; SE converted into SD; data extracted from table 3; data at 4 months
- ^xMobilisation vs sham SMT; RMDQ 0-24; data based upon figures presented in Fig. 2 using WebPlot Digitizer; data at 7 weeks
- ^yManipulation vs sham SMT; RMDQ 0-24; data based upon figures presented in Fig. 2 using WebPlot Digitizer; data at 7 weeks
- ^zOsteopathic SMT vs sham SMT; RMDQ; data based upon figures presented in Fig. 3 (mean+ SE); SE converted to SD; data at 6 months
- ^{aa}SMT ('non-maintained') vs sham SMT ('control group'); ODI; SE converted into SD; data extracted from table 3; data at 7 months
- ^{ab}SMT ('non-maintained') vs sham SMT ('control group'); ODI; SE converted into SD; data extracted from table 3; data at 10 months

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)

Figure 5. (Continued)
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding (performance bias and detection bias): All outcomes - patients
- (D) Blinding (performance bias and detection bias): All outcomes - providers
- (E) Blinding (performance bias and detection bias): All outcomes - outcome assessors
- (F) Incomplete outcome data (attrition bias): All outcomes - drop-outs
- (G) Incomplete outcome data (attrition bias): All outcomes - ITT analysis
- (H) Selective reporting (reporting bias)
- (I) Group similarity at baseline
- (J) Influence of co-interventions
- (K) Compliance with interventions
- (L) Timing of outcome assessments
- (M) Other form of bias (e.g. funding source, conflicts of interest)

Data could not be extracted from one study (Koes 1992, N = 76), which reported a statistically significant difference in functional status and global perceived effect ($P < 0.05$) between SMT and detuned physiotherapy modalities at six weeks.

Adverse events

Six trials examined adverse events (Bond 2020; Dougherty 2014b; Licciardone 2013; Senna 2011; Thomas 2020; Walker 2013). In all other trials, adverse events were not reported. The evidence is very uncertain about the adverse effects of SMT (downgraded for study limitations by one level and imprecision by two levels).

Serious adverse events

Six trials examined serious adverse events (Bond 2020; Dougherty 2014b; Licciardone 2013; Senna 2011; Thomas 2020; Walker 2013). Of these, two trials reported the occurrence of serious adverse events, but none were judged to be related to the study intervention (Dougherty 2014b; Licciardone 2013) (see details [Table 5](#)).

Minor adverse events

Four trials reported on minor adverse events (Bond 2020; Dougherty 2014b; Senna 2011; Walker 2013). Most adverse events that were commonly reported (30% to 50% of participants in the SMT group) were transient, began within 24 hours of treatment, and were of mild to moderate severity (see details [Table 5](#)).

Important outcomes
Recovery

Three trials examined the effect of SMT on perceived recovery (250 events) (Gibson 1985; Senna 2011; Walker 2013). There is low-certainty evidence (downgraded for study limitations and imprecision by one level) that SMT may result in a small increase in recovery compared to sham SMT/placebo at one month (RR 1.72, 95% CI 1.06 to 2.78; $I^2 = 0\%$; 3 studies, 250 participants; Analysis 1.3). The absolute difference was 12.3% (95% CI 10% more to 30.3% more) more participants who experienced recovery in the SMT group. The percentage of participants who experienced recovery was 29.3% in the intervention group and 17% in the control group. There is very low-certainty evidence (downgraded for study limitations by two levels and inconsistency, indirectness, and imprecision by one level) that SMT may result in little to no difference in recovery compared to sham SMT/placebo at three months (RR 0.96, 95% CI 0.50 to 1.86; 1 study, 51 participants; number needed to treat for an additional harmful outcome (NNTH) = 59), but the evidence is very uncertain. The absolute difference was 1.8% (95% CI 21.9% fewer to 37.6% more) fewer participants

who experienced recovery in the SMT group. The percentage of participants who experienced recovery was 42% in the intervention group and 43.8% in the control group. There are no data for six and 12 months.

Health-related quality of life

One study examined the effect of SMT versus sham SMT/placebo on quality of life (63 participants) (Senna 2011). There is very low-certainty evidence (downgraded for study limitations by two levels and inconsistency, indirectness, and imprecision by one level) that SMT may result in little to no difference in health-related quality of life compared to sham SMT/placebo at one month (MD 4.50, 95% CI 0.46 to 8.54; 1 study, 63 participants), three months (MD 2.80, 95% CI -1.24 to 6.84; 1 study, 63 participants), six months (MD 1.70, 95% CI -2.34 to 5.74; 1 study, 63 participants), and 12 months (MD 1.70, 95% CI -2.34 to 5.74; 1 study, 63 participants) (Analysis 1.4), but the evidence is very uncertain.

Return to work

There are no data for return to work.

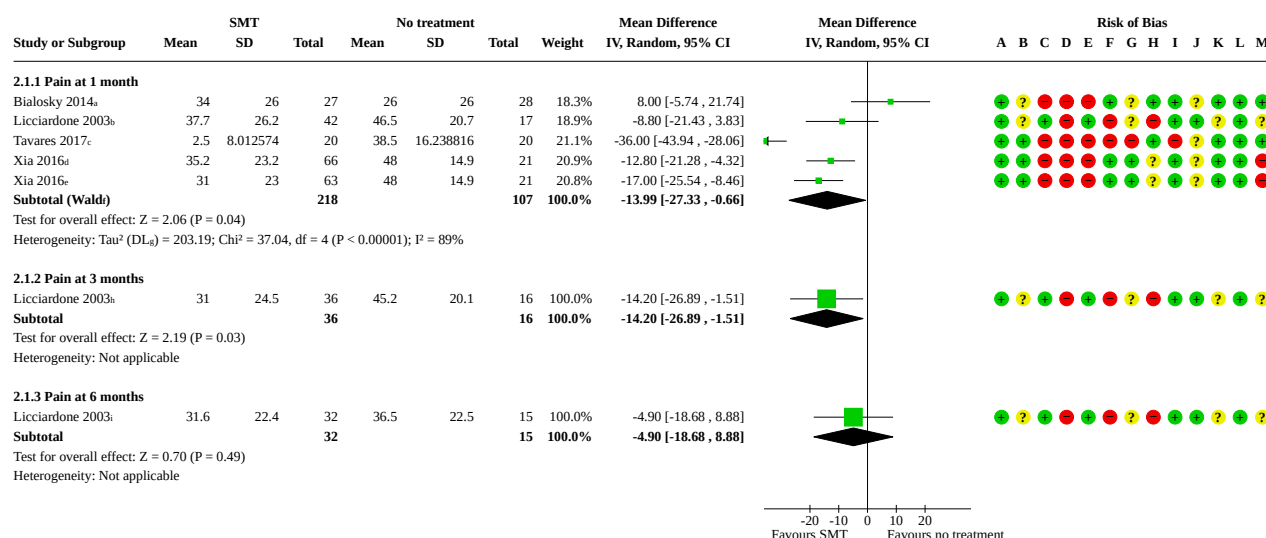
Comparison 2. SMT versus no treatment

Four trials examined the effect of SMT versus no treatment (Bialosky 2014; Licciardone 2003; Tavares 2017; Xia 2016) (see [Table 3](#)).

Critical outcomes
Pain

All four trials examined the effect of SMT on pain (315 participants) (Bialosky 2014; Licciardone 2003; Tavares 2017; Xia 2016). There is very low-certainty evidence (downgraded for study limitations, inconsistency, and imprecision, all by one level) that SMT results in a medium reduction in pain compared to no treatment at one month (MD -13.99, 95% CI -27.33 to -0.66; $I^2 = 89\%$; 4 studies, 325 participants; [Figure 6](#)), but the evidence is very uncertain. There is very low-certainty evidence (downgraded for study limitations by two levels, inconsistency, indirectness, and imprecision by one level) that SMT may result in a medium reduction in pain compared to no treatment at three months (MD -14.20, 95% CI -26.89 to -1.51; 1 study, 52 participants), but the evidence is very uncertain. Additionally, there is very low-certainty evidence (downgraded for study limitations and imprecision by two levels, inconsistency, and indirectness by one level) that SMT may result in little to no difference in pain compared to no treatment at six months (MD -4.90, 95% CI -18.68 to 8.88; 1 study, 47 participants), but the evidence is very uncertain. There are no data for 12 months.

Figure 6. Comparison 2: SMT compared with no treatment for chronic LBP; Outcome 1: pain (0-100)



Footnotes

^aSMT vs. no treatment; NRS 0-100; data based upon figures presented in Fig.4; change scores presented and SD's used from baseline; data at 2 weeks^bOsteopathic SMT vs no treatment; VAS 0-100; data extracted from figure 3; SD calculated; data at 1 month^cSMT vs no treatment; NPRS 0-10; data converted to 0-100; data extracted from table 2; data at 5 weeks^dLVVA ('non-thrust') SMT vs waiting list control; VAS 0-100; adjusted mean estimates; CI in the article, SD calculated; data at 3 weeks^eHVLA ('thrust') SMT vs waiting list control; VAS 0-100; adjusted mean estimates; CI in the article, SD calculated; data at 3 weeks^fCI calculated by Wald-type method.^g τ^2 calculated by DerSimonian and Laird method.^hOsteopathic SMT vs no treatment; VAS 0-100; data extracted from figure 3 SD calculated; data at 3 monthsⁱOsteopathic SMT vs no treatment; VAS 0-100; data extracted from figure 3 SD calculated; data at 6 months

Risk of bias legend

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

(C) Blinding (performance bias and detection bias): All outcomes - patients

(D) Blinding (performance bias and detection bias): All outcomes - providers

(E) Blinding (performance bias and detection bias): All outcomes- outcome assessors

(F) Incomplete outcome data (attrition bias): All outcomes - drop-outs

(G) Incomplete outcome data (attrition bias): All outcomes - ITT analysis

(H) Selective reporting (reporting bias)

(I) Group similarity at baseline

(J) Influence of co-interventions

(K) Compliance with interventions

(L) Timing of outcome assessments

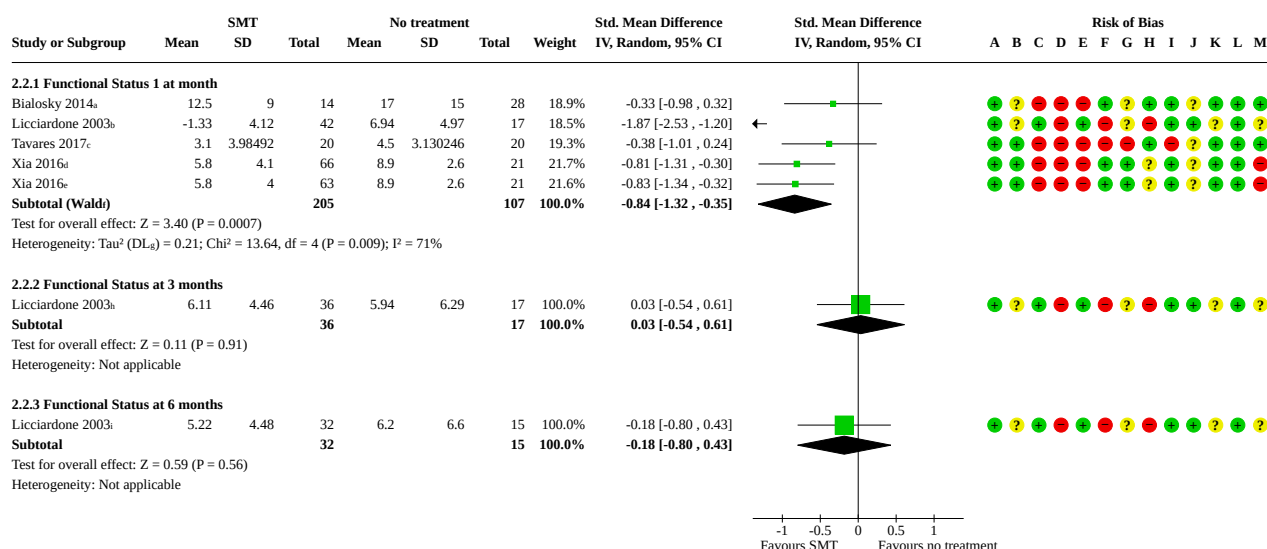
(M) Other form of bias (e.g. funding source, conflicts of interest)

Functional status

All four trials examined the effect of SMT on functional status (315 participants) (Bialosky 2014; Licciardone 2003; Tavares 2017; Xia 2016). There is low-certainty evidence (downgraded for study limitations and imprecision, each by one level) that SMT may result in a large improvement in functional status compared to no treatment at one month (SMD -0.84, 95% CI -1.32 to -0.35; $I^2 = 71\%$; 4 studies, 312 participants). There is very low-certainty evidence (downgraded for study limitations and imprecision by two levels, indirectness and imprecision by one level) that SMT may result in little to no difference in functional status compared to no treatment at three months (SMD 0.03, 95% CI -0.54 to 0.61; 1 study,

53 participants) and six months (SMD -0.18, 95% CI -0.80 to 0.43; 1 study, 47 participants), but the evidence is very uncertain (Figure 7). There are no data for 12 months. Converted to an MD on the Roland Morris Disability Questionnaire of 0 to 24, these effects amount to an MD of -3.10, 95% CI -4.88 to -1.30 at 1-month follow-up (based on an SD at 3 weeks (3.7 both arms combined) from Xia 2016, as this is a large, good-quality study that uses a common functional status scale. Because only one study reported functional status at three and six months (Licciardone 2003), we present its MD 0.17 (95% CI -3.16 to 3.50) at three months and MD -0.98 (95% CI -4.66 to 2.70) at six months on the Roland Morris Disability Questionnaire, rather than back-translating the SMD.

Figure 7. Comparison 2: SMT compared with no treatment for chronic LBP; Outcome 2: functional status



Footnotes

- ^aSMT vs. no treatment; ODI 0-100; data based upon figures presented in Fig.4; change scores presented and SD's used from baseline; data at 2 weeks
- ^bOsteopathic SMT vs no treatment; RMDQ 0-24; data extracted from figure 3 SD calculated; data at 1 month
- ^cSMT vs no treatment; ODI 0-100; data extracted from tabel 2; data at 5 weeks
- ^dLVVA (non-thrust) SMT vs waiting list control; RMDQ 0-24; adjusted mean estimates; CI in the article, SD calculated; data at 3 weeks
- ^eHVLA ('thrust') SMT vs waiting list control; RMDQ 0-24; adjusted mean estimates; CI in the article, SD calculated; data at 3 weeks
- ^fCI calculated by Wald-type method.
- ^gTau² calculated by DerSimonian and Laird method.
- ^hOsteopathic SMT vs no treatment; RMDQ 0-24; data extracted from figure 3; SD calculated; data from 3 month
- ⁱOsteopathic SMT vs no treatment; RMDQ0-24 data extracted from figure 3 SD calculated; data from 6 month

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding (performance bias and detection bias): All outcomes - patients
- (D) Blinding (performance bias and detection bias): All outcomes - providers
- (E) Blinding (performance bias and detection bias): All outcomes - outcome assessors
- (F) Incomplete outcome data (attrition bias): All outcomes - drop-outs
- (G) Incomplete outcome data (attrition bias): All outcomes - ITT analysis
- (H) Selective reporting (reporting bias)
- (I) Group similarity at baseline
- (J) Influence of co-interventions
- (K) Compliance with interventions
- (L) Timing of outcome assessments
- (M) Other form of bias (e.g. funding source, conflicts of interest)

Adverse events

One trial reported no serious adverse events with SMT (Xia 2016). In all other trials, adverse events were not reported. The evidence is very uncertain about the adverse effects of SMT (downgraded for study limitations and inconsistency by one level, and imprecision by two levels).

Important outcomes

Recovery

There are no data for perceived recovery.

Health-related quality of life

One study examined the effect of SMT on health-related quality of life (171 participants) (Xia 2016). There is very low-certainty evidence (downgraded for study limitations, inconsistency, and imprecision by one level) that SMT may have little to no effect on health-related quality of life compared to no treatment at one month (MD 4.95, 95% CI 3.20 to 6.71; 171 participants; Analysis 2.3), but the evidence is very uncertain.

Return to work

There are no data for return to work.

Comparison 3. SMT versus other conservative interventions

We identified 42 trials, with 8291 participants, that compared the effects of SMT versus other conservative interventions (Ali 2019; Bronfort 2011; Brønfort 1996; Castro-Sanchez 2014; Cecchi 2010; Chown 2008; De Oliveira 2020a; Dougherty 2014a; Fatima 2021; Ferreira 2007; Gibson 1985; Goldby 2006; Griswold 2019; Gudavalli 2006; Haas 2014; Hemmila 2002; Hondras 2009; Hsieh 2002; Hurwitz 2002; Kamali 2019; Koes 1992; Krekoukias 2017; Mohseni-Bandpei 2006; Muller 2005; Nejati 2019; Özcan 2022; Paatelma 2008; Petersen 2011; Pope 1994; Postacchini 1988; Rasmussen-Barr 2003; Samir 2016; Sarker 2017; Skillgate 2007; Sturion 2020; Teixeira 2016; UK BEAM Trial 2004; Ulger 2017; Wilkey 2008; Zaidi 2020; Zaproudina 2009). The most common other conservative treatment was exercise (25 trials) (for details see Table 3).

Of the aforementioned trials, 34 provided data for the meta-analyses. We identified three trials with a 'fatal flaw' and excluded them from the meta-analyses: two trials demonstrated more than

50% dropout at the first follow-up measurement (Chown 2008; Muller 2005), and one study was found to have clinically relevant baseline differences between the intervention groups for one or more primary outcomes (Goldby 2006).

We could not extract data from four trials (Koes 1992; Pope 1994; Postacchini 1988; Teixeira 2016). Koes 1992 (N = 76) reported a statistically significant difference in functional status and global perceived effect between SMT and detuned physiotherapy modalities at six weeks. Pope 1994 (N = 127) demonstrated no statistically significant difference in pain between SMT and use of a corset or transcutaneous muscle stimulation. In Postacchini 1988 (N = 95), it is unclear whether there was a statistically significant difference in improvement between SMT and the control group at three weeks or six months. Teixeira 2016 was a cross-over study. Authors did not provide data before the cross-over, even though they reported that the relief of pain and improvement of functional status were more significant in the manipulative group.

Critical outcomes

Pain

Thirty-four trials examined the effects of SMT versus other conservative interventions for pain (6385 participants) (Ali 2019; Bronfort 2011; Brønfort 1996; Castro-Sanchez 2021; Cecchi 2010; De Oliveira 2020a; Dougherty 2014a; Fatima 2021; Ferreira 2007; Gibson 1985; Griswold 2019; Gudavalli 2006; Haas 2014; Hemmila 2002; Hondras 2009; Hsieh 2002; Hurwitz 2002; Kamali 2019; Krekoulas 2017; Mohseni-Bandpei 2006; Nejati 2019; Özcan 2022; Paatelma 2008; Petersen 2011; Rasmussen-Barr 2003; Samir 2016; Sarker 2017; Schulz 2019; Skillgate 2007; Sturion 2020; UK BEAM Trial 2004; Ulger 2017; Wilkey 2008; Zaidi 2020; Zaproudina 2009).

There is low-certainty evidence (downgraded for inconsistency by two levels) that SMT may result in little to no difference in pain compared to other conservative interventions at one month (MD -4.72, 95% CI -8.26 to -1.17; $I^2 = 89\%$; 31 studies, 4109 participants) and three months (MD -2.32, 95% CI -4.83 to 0.19; $I^2 = 79\%$; 19 studies, 3044 participants), but the evidence is very uncertain (Figure 8).

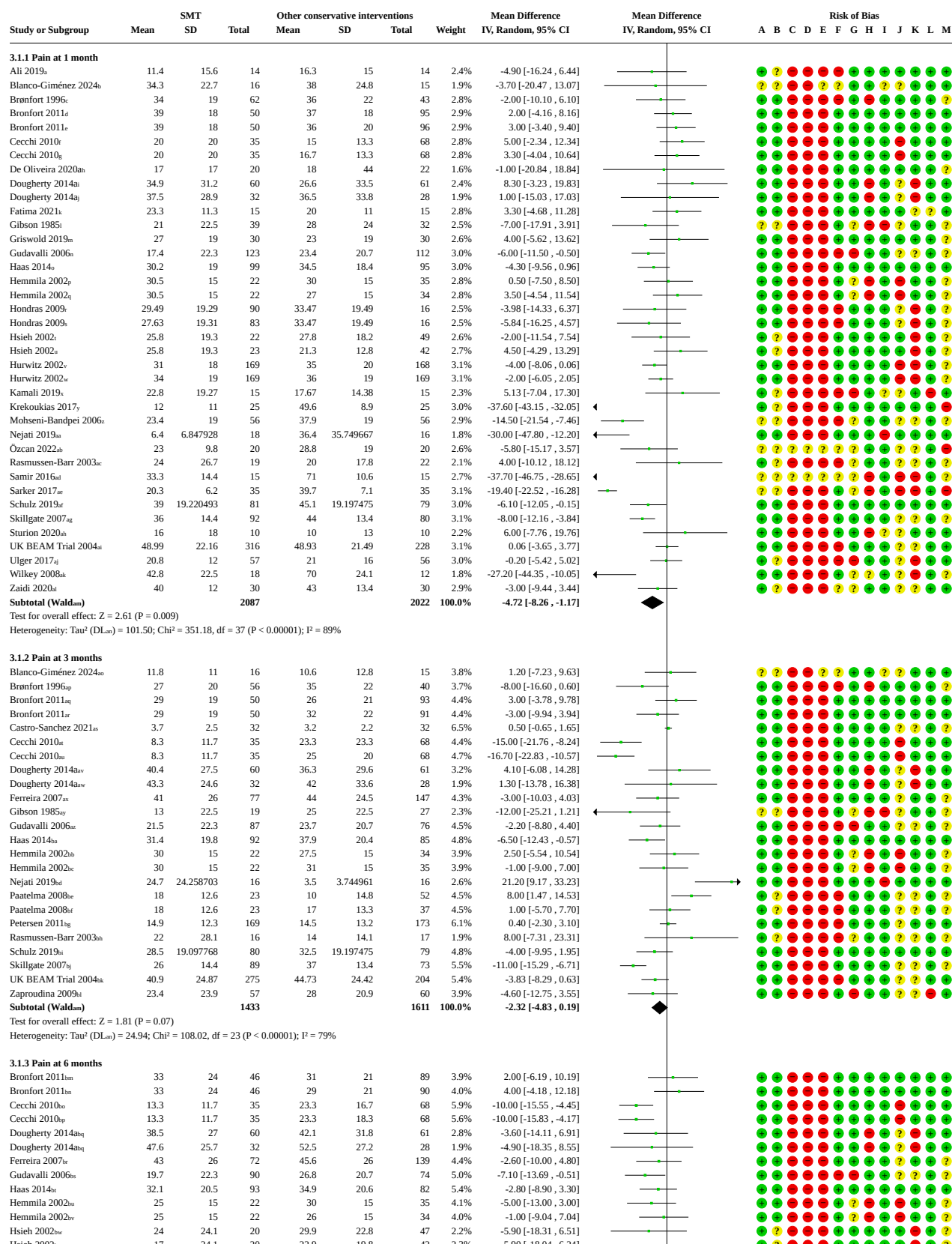
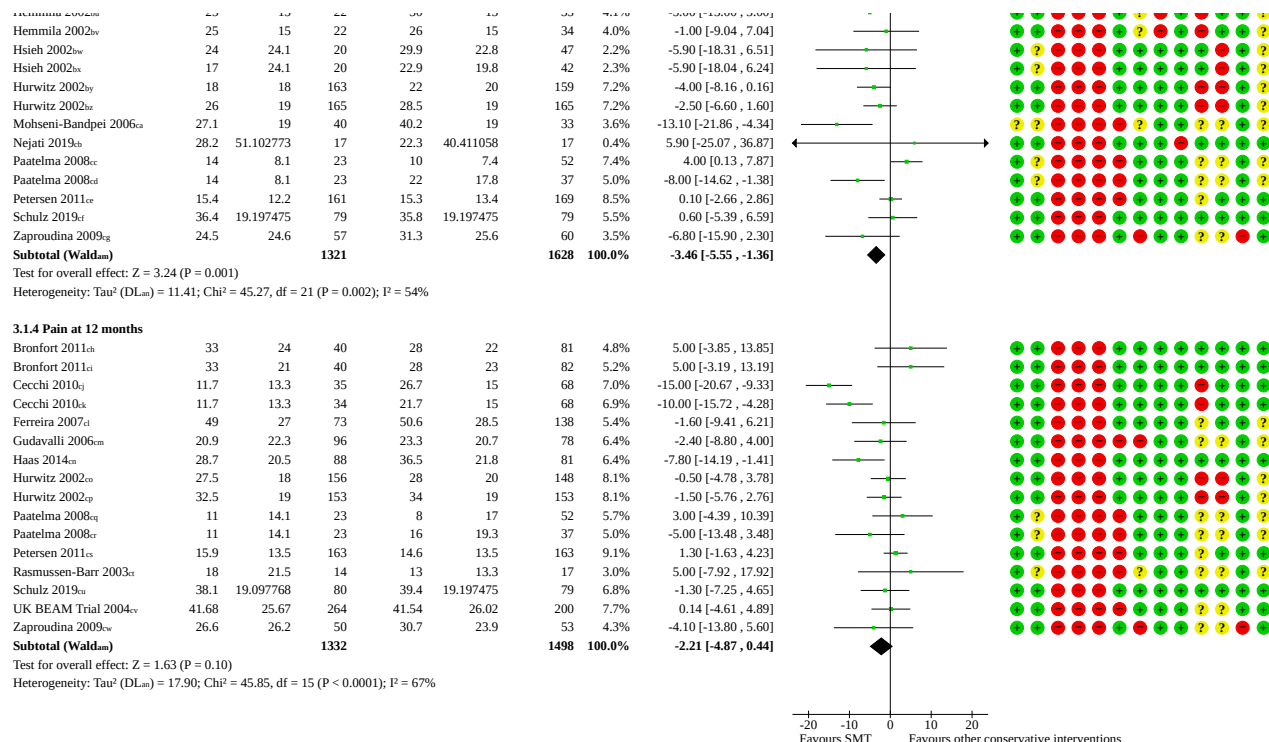
Figure 8. Comparison 3: SMT compared with other conservative interventions for chronic LBP; Outcome 1: pain (0-100)

Figure 8. (Continued)



Footnotes

- ^aMaitland Mobilisation vs Mulligan Mobilisation; NPRS 0-10 converted to 0-100; data extracted from table 2; data at 4 weeks
- ^aSMT + exercise vs kinesiotaping + exercise; VAS (0-10); data extracted from table 2; data at 3 weeks
- ^aHVLA-SMT + strength exercises vs. NSAID + strength exercises; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 5 weeks
- ^aSMT vs supervised exercise therapy; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 4 weeks
- ^aSMT vs home exercise and advice; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 4 weeks
- ^aSMT vs individualized physiotherapy; RMPRS 0-6; data converted to 0-100; data extracted from table 2; First follow-up is at discharge (4-6 weeks)
- ^aSMT vs back school; RMPRS 0-6; data converted to 0-100; data extracted from table 2; First follow up is at discharge (4-6 weeks)
- ^aOsteopathic manipulation vs active control group (exercise); VAS 0-100; data extracted from table 2; data at 5 weeks
- ^aSMT vs active exercise therapy + negative on clinical prediction rule; VAS 0-100; data extracted from table 4; data at 5 weeks
- ^aSMT vs active exercise therapy + positive on clinical prediction rule; VAS 0-100; data extracted from table 4; data at 5 weeks
- ^aHVLA SMT vs muscle energy technique; NPRS 0-10 converted to 0-100; data extracted from table 2; data presumably at 2 weeks
- ^aSMT vs. diathermy; median (range) presented in the study and converted; data extracted from table 2; data (daytime pain) at 4 weeks
- ^aNon-thrust manipulation vs dry needling; NPRS 0-10 converted to 0-100; data extracted from table 2; data at 3 weeks
- ^aSMT (flexion-distraction) vs. exercise; VAS 0-100; change scores presented - used SD from baseline; data at 4 weeks
- ^aSMT vs light massage; Von Korf pain scale; data extracted from table 2; data at 6 weeks
- ^aSMT vs. exercise; VAS 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 6 weeks
- ^aSMT vs. physiotherapy; mean data estimated from figure (6 wks) - no SD presented in the publication and est. from other studies; VAS, average pain.
- ^aHVLA-SMT vs medical care; VAS 0-100; adjusted scores; IPD data provided by author; data at 3 weeks
- ^aLVA-SMT (flexion-distraction) vs. medical care; VAS 0-100; adjusted scores; IPD data provided by author; data at 3 weeks
- ^aSMT vs. myofascial therapy; VAS 0-10; converted to 0-100; data extracted from table 3; data at 3 weeks
- ^aSMT vs. back school; VAS 0-10; converted to 0-100; data extracted from table 3; data at 3 weeks
- ^aChiropractic care + physical modalities (DCpm) vs. medical care + physical therapy (MDpt); NPRS 0-10; converted to 0-100; data estimated from graphs; SD used from baseline; data at 6 weeks
- ^aChiropractic care only vs. medical care only; NPRS 0-10; converted to 0-100; data estimated from graphs; SD used from baseline; data at 6 weeks
- ^aMT vs stabilizing exercise; VAS 0-100; data extracted from table 2; data at 2-weeks follow up for MT and 4 weeks for stabilizing exercise
- ^aSMT vs. conventional physiotherapy (stretching, TENS, massage); NPRS 0-10; converted to 0-100; data extracted from table 2; data at 5 weeks
- ^aSMT + exercise vs. ultrasound + exercise; VAS 0-100; data extracted from table 2; data at 6 weeks
- ^aManipulation vs exercise therapy; VAS 0-10; converted to 0-100; data extracted from table 1; SDs are imputed from similar studies; data at 6 weeks
- ^aChiropractic manipulation vs Mulligan mobilisation; NPRS 0-10; converted to a 0-100 scale; data extracted from table 2; data at 4 weeks
- ^aSMT vs. stabilizing training; VAS 0-10; data converted to 0-100; IPD data; data at 6 weeks
- ^aMaitland mobilisation vs Mulligan mobilisation; VAS 0-10; converted to 0-100 data extracted from table 2; data at 4 weeks
- ^aHVLA SMT vs. core stability exercises; NPRS 0-10; converted to 0-100; data extracted from text; data at 2 weeks
- ^aSMT + home exercise program vs Supervised exercise and home exercise program; NPRS 0-10 converted to 0-100; data extracted from table 3; data at 4 weeks
- ^aSMT (naprapathy) vs. standard medical care; Modified von Korff - pain subscale only 0-100; IPD data provided by author; data at 3 weeks
- ^aHLVA SMT vs muscle energy technique; NPRS 0-10; converted to 0-100; data extracted from table 1; data at 15-days
- ^aBest care + SMT vs. best care + exercise; Modified von Korff - pain subscale only 0-100; IPD data provided by the authors; data at 1 month
- ^aSMT vs spinal stabilization exercise; VAS 0-10; converted to a 0-100; data extracted from table 3; data at 6 weeks
- ^aSMT ('chiropractic management') vs. outpatient hospital pain clinic; NPRS 0-10; converted to 0-100; IPD provided by the author; data at 4 weeks
- ^aMaitland mobilisation vs muscle energy technique; VAS 0-10; converted to 0-100; data extracted from table 2 and 3; data at 4 weeks
- ^aCI calculated by Wald-type method.
- ^aTau² calculated by DerSimonian and Laird method.
- ^aSMT + exercise vs kinesio taping + exercise; VAS (0-10); data extracted from table 2; data at 12 weeks
- ^aHVLA-SMT + strength exercises vs. NSAID + strength exercises; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 11 weeks
- ^aSMT vs supervised exercise therapy; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 12 weeks
- ^aSMT vs home exercise and advice; NPRS 0-10; converted to 0-100; data extracted from table 2; data at 12 weeks
- ^aSacral manipulative therapy vs myofascial release; NPRS 0-10 converted to 0-100; data extracted from table; data at 3 months
- ^aSMT vs back school; RMPRS 0-6; data converted to 0-100; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
- ^aSMT vs individualized physiotherapy; data converted to 0-100; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
- ^aSMT vs active exercise therapy + negative on clinical prediction rule; SMT vs active exercise therapy + positive on clinical prediction rule; VAS 0-100; data extracted from table 4; data at 12 weeks

Figure 8. (Continued)

25SMT vs back school; RMPRS 0-6; data converted to 0-100; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
 26SMT vs individualized physiotherapy; data converted to 0-100; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
 27SMT vs active exercise therapy + negative on clinical prediction rule; SMT vs active exercise therapy + positive on clinical prediction rule; VAS 0-100; data extracted from table 4; data at 12 weeks
 28SMT vs active exercise therapy + positive on clinical prediction rule; VAS 0-100; data extracted from table 4; data at 12 weeks
 29SMT vs. general and motor exercise therapy (combined); VAS 0-10; data converted to 0-100; data extracted from table 2; data at 8 weeks
 30SMT vs. diathermy; median (range) presented in the study and converted; data extracted from table 2; data (daytime pain) at 12 weeks
 31SMT (flexion distraction) vs. exercise; VAS 0-100; change scores presented - used SD from baseline; data at 3 months
 32SMT vs light massage; Von Korf pain scale; data extracted from table 2; data at 12 weeks
 33SMT vs. physiotherapy; VAS 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 3 months
 34SMT vs. exercise; VAS 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 3 months
 35Manipulation vs exercise therapy; VAS 0-10; converted to 0-100; mean and p values presented from table 2; data at 12-weeks
 36SMT vs. McKenzie exercise; VAS 0-100; median (IQR) - converted to mean (SD); data extracted from table 2; data at 3 months
 37SMT vs. advice-group (counseling session + educational back booklet); VAS 0-100; median (IQR) - converted to mean (SD); data extracted from table 2; data at 3 months
 38SMT vs McKenzie exercise; Modified von Korff - pain subscale only 0-10; converted to 0-100; McKenzie 67.5; IPD data provided by the authors; follow up post treatment (treatment period max. 12 weeks; mean days: SMT 60.5 days)
 39SMT vs. stabilizing training; VAS 0-10; data converted to 0-100; IPD data; data at 3 months
 40SMT + home exercise program vs Supervised exercise and home exercise program; NPRS 0-10 converted to 0-100; data extracted from table 3; data at 12 weeks
 41SMT (naprapathy) vs. standard medical care; Modified von Korff - pain subscale only 0-10; converted to 0-100; IPD data provided by author; data from 12 weeks
 42Best care + SMT vs. best care + exercise; Modified von Korff - pain subscale only 0-10; converted to 0-100; IPD data provided by the authors; data at 3 months
 43SMT vs. physiotherapy (massage, stretching and exercise); VAS 0-100; IPD data provided by authors; data at 3 months (1 month post treatment)
 44SMT vs home exercise and advice; NPRS 0-10; converted 0-100; data extracted from table 2; data at 26 weeks
 45SMT vs supervise exercise therapy; NPRS 0-10; converted 0-100; data extracted from table 2; data at 26 weeks
 46SMT vs back school; RMPRS 0-6; data converted to 0-100; data extracted from table 2; data at 6 months after discharge (4-6 weeks)
 47SMT vs Individualized physiotherapy; RMPRS 0-6; data converted to 0-100; data extracted from table 2; data at 6 months after discharge (4-6 weeks)
 48SMT vs active exercise therapy + negative on clinical prediction rule; VAS 0-100; data extracted from table 4; data at 24 weeks
 49SMT vs. general and motor exercise therapy (combined); VAS 0-10; data converted to 0-100; data extracted from table 2; data at 6 months
 50SMT vs. exercise; VAS 0-10; data converted to 0-100; change scores presented - used SD from baseline; data at 6 months
 51SMT vs light massage; Von Korf pain scale; data extracted from table 2; data at 24 weeks
 52SMT vs. exercise; VAS 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 6 months
 53SMT vs. physiotherapy; VAS 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 6 months
 54SMT vs. myofascial therapy; VAS 0-10; converted to 0-100; data extracted from table 3; data at 6 months
 55SMT vs. back school; VAS 0-10; converted to 0-100; data extracted from table 3; data at 6 months
 56Chiropractic care + physical modalities (DCPm) vs. medical care + physical therapy (MDpt); NPRS 0-10; converted to 0-100; data estimated from graphs; SD used from baseline; data at 6 months
 57Chiropractic care only vs. medical care only; NPRS 0-10; converted to 0-100; data estimated from graphs; SD used from baseline; data at 6 months
 58SMT + exercise vs. ultrasound + exercise; VAS 0-100; data extracted from table 2; data at 6 months
 59Manipulation vs exercise therapy; VAS 0-10; converted to 0-100; mean and p values presented from table 2; data at 6 months
 60SMT vs. McKenzie exercise; VAS 0-100; median (IQR) - converted to mean (SD); data extracted from table 2; data at 6 months
 61SMT vs. advice-group (counseling session + educational back booklet); VAS 0-100; median (IQR) - converted to mean (SD); data extracted from table 2; data at 6 months
 62SMT vs McKenzie; Modified von Korff - pain subscale only 0-10; converted to 0-100; IPD data provided by the authors; data at 2 months post treatment (treatment period max 12 weeks; mean days: SMT 60.5 days; McKenzie 67.5)
 63SMT + home exercise program vs Supervised exercise and home exercise program; NPRS 0-10 converted to 0-100; data extracted from table 3; data at 26 weeks
 64SMT vs. physiotherapy (massage, stretching and exercise); VAS 0-100; IPD data provided by authors; data at 9 months (6 months post treatment)
 65SMT vs home exercise and advice; NPRS 0-10; converted 0-100; data extracted from table 2; data at 52 weeks
 66SMT vs supervise exercise therapy; NPRS 0-10; converted 0-100; data extracted from table 2; data at 52 weeks
 67SMT vs Individualized physiotherapy; RMPRS 0-6; data converted to 0-100; data extracted from table 2; data at 12 months after discharge (4-6 weeks)
 68SMT vs back school; RMPRS 0-6; data converted to 0-100; data extracted from table 2; data at 12 months after discharge (4-6 weeks)
 69SMT vs. general and motor exercise therapy (combined); VAS 0-10; data converted to 0-100; data extracted from table 2; data at 52 weeks
 70SMT (flexion-distraction) vs. exercise; VAS 0-100; change scores presented - used SD from baseline; data at 12 months
 71SMT vs light massage; Von Korf pain scale; data extracted from table 2; data from 52 weeks
 72Chiropractic care + physical modalities (DCPm) vs. medical care + physical therapy (MDpt); NPRS 0-10; converted to 0-100; data estimated from graphs; SD used from baseline; data at 12 months
 73Chiropractic care only vs. medical care only; NPRS 0-10; converted to 0-100; data estimated from graphs; SD used from baseline; data at 12 months
 74SMT vs. McKenzie exercise; VAS 0-100; median (IQR) - converted to mean (SD); data extracted from table 2; data at 12 months
 75SMT vs. advice-group (counseling session + educational back booklet); VAS 0-100; median (IQR) - converted to mean (SD); data extracted from table 2; data at 12 months
 76SMT vs McKenzie; Modified von Korff - pain subscale only 0-10; converted to 0-100; IPD data provided by the authors; data at 12 months follow up post treatment (treatment period max 12 weeks; mean days: SMT 60.5 days; McKenzie 67.5)
 77MT vs. stabilizing training; VAS 0-10; data converted to 0-100; IPD data; data at 12 months
 78SMT + home exercise program vs Supervised exercise and home exercise program; NPRS 0-10 converted to 0-100; data extracted from table 3; data at 52 weeks
 79Best care + SMT vs. best care + exercise; Modified von Korff - pain subscale only 0-10; converted to 0-100; IPD data provided by the authors; data at 12 months
 80SMT vs. physiotherapy (massage, stretching and exercise); VAS 0-100; IPD data provided by authors; 12 months post-tx. = 15 months

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding (performance bias and detection bias): All outcomes - patients
- (D) Blinding (performance bias and detection bias): All outcomes - providers
- (E) Blinding (performance bias and detection bias): All outcomes - outcome assessors
- (F) Incomplete outcome data (attrition bias): All outcomes - drop-outs
- (G) Incomplete outcome data (attrition bias): All outcomes - ITT analysis
- (H) Selective reporting (reporting bias)
- (I) Group similarity at baseline
- (J) Influence of co-interventions
- (K) Compliance with interventions
- (L) Timing of outcome assessments
- (M) Other form of bias (e.g. funding source, conflicts of interest)

There is low-certainty evidence (downgraded for inconsistency by two levels) that SMT may result in little to no difference in pain compared to other conservative interventions at six months (MD -3.46, 95% CI -5.55 to -1.36; $I^2 = 54\%$; 15 studies, 2949 participants) and 12 months (MD -2.21, 95% CI -4.87 to 0.44; $I^2 = 67\%$; 12 studies, 2830 participants).

Functional status

Thirty-two trials examined the effects of SMT versus other conservative interventions for functional status (6116 participants) (Ali 2019; Bronfort 2011; Brønfort 1996; Castro-Sanchez 2021;

Cecchi 2010; De Oliveira 2020a; Dougherty 2014a; Fatima 2021; Ferreira 2007; Griswold 2019; Gudavalli 2006; Haas 2014; Hemmilla 2002; Hondras 2009; Hsieh 2002; Hurwitz 2002; Kamali 2019; Krekoulas 2017; Mohseni-Bandpei 2006; Nejati 2019; Özcan 2022; Paatelma 2008; Petersen 2011; Rasmussen-Barr 2003; Schulz 2019; Skillgate 2007; Sturion 2020; UK BEAM Trial 2004; Ulger 2017; Wilkey 2008; Zaidi 2020; Zaproudina 2009).

There is low-certainty evidence (downgraded for inconsistency by two levels) that SMT may result in a small improvement in functional status compared to other conservative interventions at one month (SMD -0.25, 95% CI -0.38 to -0.11; $I^2 = 73\%$; 28 studies,

3940 participants), and little to no difference in functional status at three months (SMD -0.06, 95% CI -0.20 to 0.08; $I^2 = 70\%$; 19 studies, 3237 participants) and 12 months (SMD -0.11, 95% CI -0.23 to 0.02; $I^2 = 64\%$; 13 studies, 2963 participants). Additionally, there

is low-certainty evidence (downgraded for study limitations and inconsistency, both by one level) that SMT results in little to no difference in functional status at six months (SMD -0.09, 95% CI -0.19 to 0.01; $I^2 = 42\%$; 15 studies, 3084 participants; [Figure 9](#)).

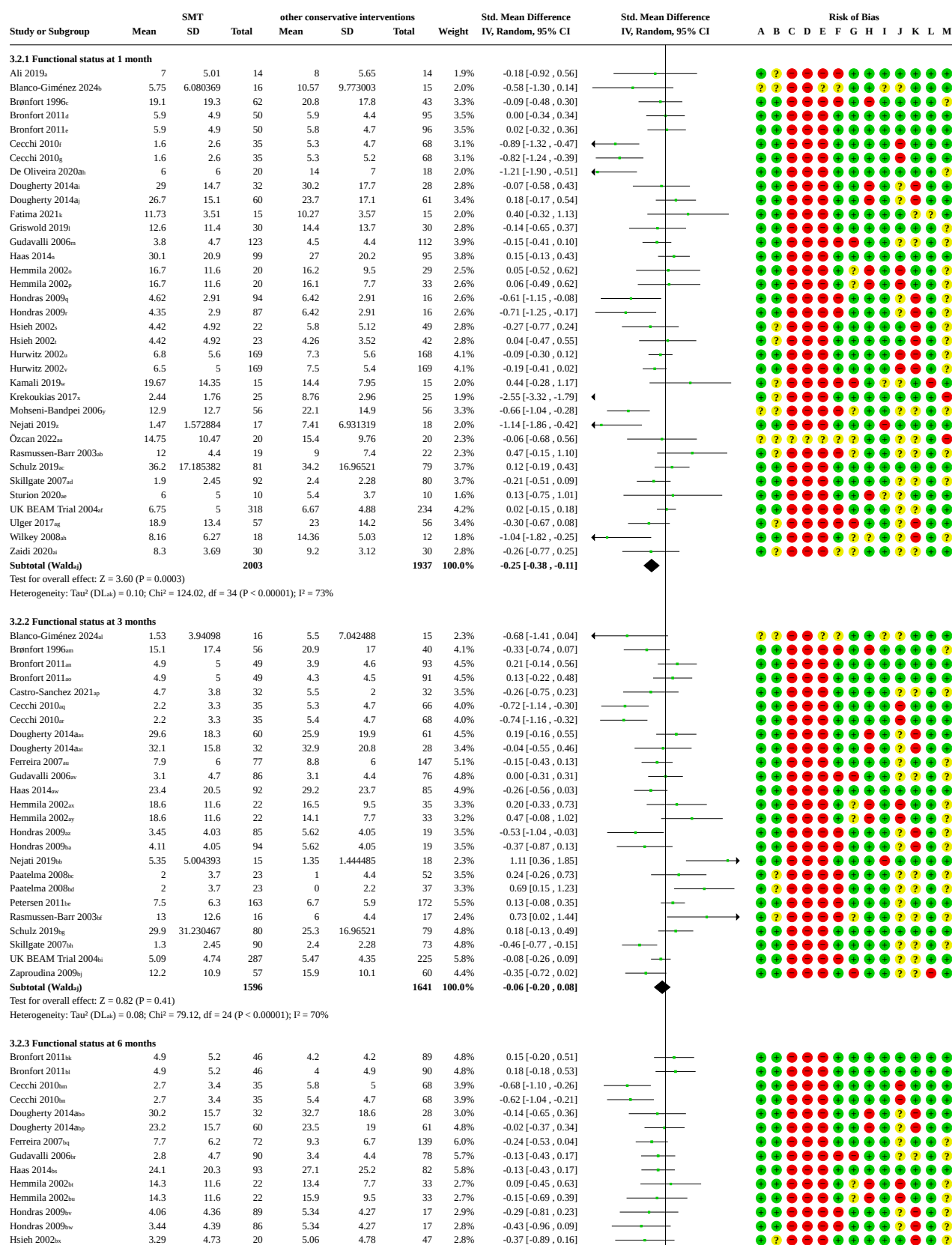
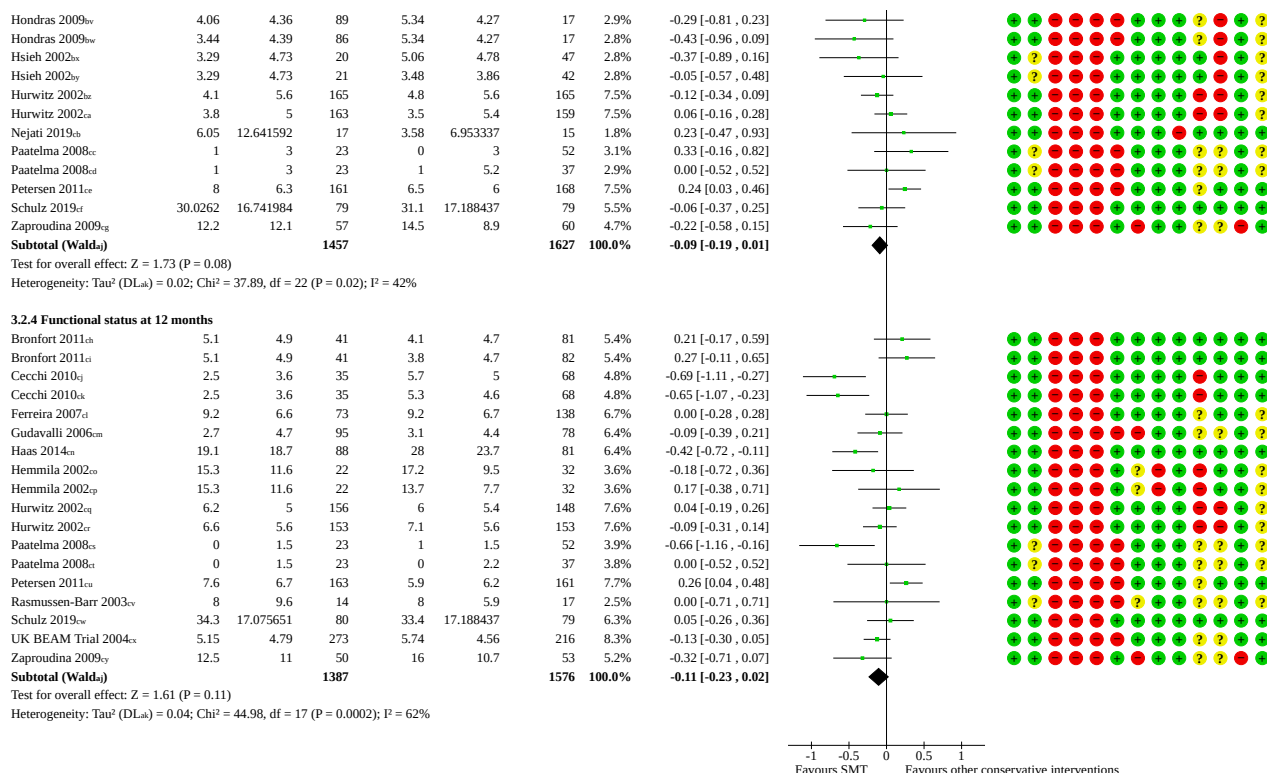
Figure 9. Comparison 3: SMT compared with other conservative interventions for chronic LBP; Outcome 2: functional status

Figure 9. (Continued)



Footnotes

- ^aMaitland Mobilisation vs Mulligan Mobilisation; ODI 0-100; data extracted from table 2; data at 4 weeks
- ^aSMT + exercise vs kinesiotaping + exercise; ODI 0-100; data extracted from figure 4 with webplotdigitizer; data at 3 weeks
- ^aHVLA-SMT + strength exercises vs. NSAID + strength exercises; RMDQ 0-24; data extracted from table 2; data at 5 weeks
- ^aSMT vs supervised exercise therapy; RMDQ 0-24; data extracted from table 2; data at 4 weeks
- ^aSMT vs home exercise and advice; RMDQ 0-24; data extracted from table 2; data at 4 weeks
- ^aSMT vs back school; RMDQ 0-24; data extracted from table 2; First follow-up is at discharge (4-6 weeks)
- ^aSMT vs individualized physiotherapy; RMDQ 0-24; data extracted from table 2; First follow-up is at discharge (4-6 weeks)
- ^aOsteopathic manipulation vs active control group (exercise); ODI 0-100; data extracted from table 2; data at 5 weeks
- ^aSMT vs active exercise therapy + negative on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 5 weeks
- ^aSMT vs active exercise therapy + positive on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 5 weeks
- ^aHVLA SMT vs muscle energy technique; ODI 0-100; data extracted from table 2; data presumably at 2 weeks
- ^aNon-thrust manipulation vs dry needling; ODI 0-100; data extracted from table 2; data at 3 weeks
- ^aSMT (flexion-distraction) vs. exercise; RMDQ 0-24; change scores presented - used SD from baseline; data at 4 weeks
- ^aSMT vs light massage; Von Korf disability scale; data extracted from table 2; data at 6 weeks
- ^aSMT vs. exercise; ODI 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 6 weeks
- ^aSMT vs. physiotherapy; ODI 0-100; mean data estimated from figure; no SD presented estimated from other studies; data at 6 weeks
- ^aHVLA-SMT vs medical care; RMDQ 0-24; adjusted scores; IPD data provided by author; data at 3 weeks
- ^aLVA-SMT (flexion-distraction) vs. medical care; RMDQ 0-24; adjusted scores; IPD data provided by author; data at 3 weeks
- ^aHVLA-SMT vs. Myofascial therapy; RMDQ 0-24; data extracted from table 3; data at 3 weeks
- ^aHVLA-SMT vs. back school; RMDQ 0-24; data extracted from table 3; data at 3 weeks
- ^aChiropractic care + physical modalities (DCPm) vs. medical care + physical therapy (MDpt); RMDQ 0-24; data estimated from graphs; SD used from baseline score; data at 6 weeks
- ^aChiropractic care only vs. medical care only; RMDQ 0-24; data estimated from graphs; SD used from baseline score; data at 6 weeks
- ^aMT vs stabilizing exercise; ODI 0-100; data extracted from table 2; data at 2 weeks for MT and 4 weeks for stabilizing exercise
- ^aSMT vs conventional physiotherapy; RMDQ 0-24; data extracted from table 2; data at 5 weeks
- ^aSMT (Maitland) + exercise vs. ultrasound + exercise; ODI 0-100; data extracted from table 2; data at 6 weeks
- ^aManipulation vs exercise therapy; RMDQ 0-24; data extracted from table 2; SDs are imputed from similar studies; data at 6 weeks
- ^aChiropractic manipulation vs Mulligan mobilisation; RMDQ (0-24); data extracted from table 2; data at 4 weeks
- ^aManual therapy vs. stabilization training; ODI 0-100; median (IQR) converted to mean (SD); IPD data provided by author; data at 3 weeks
- ^aSMT + home exercise program vs Supervised exercise and home exercise program; RMDQ 0-100; data extracted from table 3; data at 4 weeks
- ^aSMT (Naprapathy) vs. std. medical care; von Korf disability scale; IPD data provided by author; data at 3 weeks
- ^aHVLA SMT vs muscle energy technique; RMDQ 0-24; data extracted from table 1; data at 15 days
- ^aBest care + SMT vs. best care + exercise; RMDQ 0-24; IPD data provided by the authors; data at 1 month
- ^aSMT vs spinal stabilization exercise; ODI 0-100; data extracted from table 3; data at 6 weeks
- ^aSMT ('chiropractic management') vs. outpatient hospital pain clinic; RMDQ 0-24; IPD data provided by the author data at 4 weeks
- ^aMaitland mobilisation vs muscle energy technique; ODI 0-100; data extracted from table 2 and 3; data at 4 weeks
- ^aCI calculated by Wald-type method.
- ^a τ^2 calculated by DerSimonian and Laird method.
- ^aSMT + exercise vs kinesiotaping + exercise; ODI 0-100; data extracted from figure 4 with webplotdigitizer; data at 12 weeks
- ^aHVLA-SMT + strength exercises vs. NSAID + strength exercises; RMDQ 0-24; data extracted from table 2; data at 11 weeks
- ^aSMT vs supervised exercise therapy; RMDQ 0-24; data extracted from table 2; data at 12 weeks
- ^aSMT vs home exercise and advice; RMDQ 0-24; data extracted from table 2; data at 12 weeks RMDQ 0-24; data extracted from table 2; data at 12 weeks
- ^aSacral manipulative therapy vs myofascial release; RMDQ 0-24; data extracted from table 1 and 2; data at 3 months
- ^aSMT vs back school; RMDQ 0-24; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
- ^aSMT vs individualized physiotherapy; RMDQ 0-24; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
- ^aSMT vs active exercise therapy + positive on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 12 weeks
- ^aSMT vs active exercise therapy + negative on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 12 weeks

Figure 9. (Continued)

- ^aSMT vs individualized physiotherapy; RMDQ 0-24; data extracted from table 2; data from 3 months after discharge (4-6 weeks)
- ^aSMT vs active exercise therapy + positive on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 12 weeks
- ^aSMT vs active exercise therapy + negative on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 12 weeks
- ^aSMT vs. general and motor exercise therapy (combined); RMDQ 0-24; data extracted from table 2; data at 8 weeks
- ^aSMT (flexion-distraction) vs. exercise; RMDQ 0-24; change scores presented - used SD from baseline; data at 3 months
- ^aSMT vs light massage; Von Korff disability scale 0-10; data extracted from table 2; data at 12 weeks
- ^aSMT vs. exercise; ODI 0-100; mean data estimated from figure - no SD presented in the publication and estimated from other studies; data at 3 months
- ^aSMT vs. physiotherapy; ODI 0-100; mean data estimated from figure - no SD presented in the publication and estimated from other studies; data at 3 months
- ^aLVVA-SMT (flexion-distraction) vs. medical care; RMDQ 0-24; adjusted scores; IPD data provided by author; data at 12 weeks
- ^bHVLA-SMT vs medical care; RMDQ 0-24; adjusted scores; IPD data provided by author; data at 12 weeks
- ^bManipulation vs exercise therapy; RMDQ 0-24; mean and p values presented from table 2; data measured at 12 weeks
- ^bSMT vs. McKenzie exercise; RMDQ 0-24; median (IQR) - converted to mean (SD); data extracted from table 2; data at 3 months
- ^bSMT vs. advice-group (counseling session + educational back booklet); RMDQ 0-24; median (IQR) - converted to mean (SD); data extracted from table 2; data at 3 months
- ^bSMT vs McKenzie; RMDQ 0-24; IPD data provided by the authors; post treatment (treatment period max 12 weeks; mean days: SMT 60.5 days;McKenzie 67.5)
- ^bManual therapy vs. stabilization training; median (IQR) converted to mean (SD); ODI 0-100; IPD data provided by the author; data at 3 months
- ^bSMT + home exercise program vs Supervised exercise and home exercise program; RMDQ 0-100; data extracted from table 3; data at 12 weeks
- ^bSMT (naprapathy) vs. standard medical care; von Korff disability scale 0-10; IPD data provided by the authors; data at 12 weeks
- ^bBest care + SMT vs. best care + exercise; RMDQ 0-24; IPD data provided by the authors; data at 3 months
- ^bSMT vs. physiotherapy (massage, stretching and exercise); ODI 0-100; IPD data provided by author; data at 3 months (1 month post-treatment)
- ^bSMT vs home exercise and advice; RMDQ 0-24; data extracted from table 2; data at 26 weeks
- ^bSMT vs supervised exercise therapy; RMDQ0-24; data extracted from table 2; data at 26 weeks
- ^bSMT vs individualized physiotherapy; data from 6 months after discharge (4-6 weeks); RMDQ 0-24; data extracted from table 2; data at 6 months after discharge (4-6 weeks)
- ^bSMT vs back school; data from 6 months after discharge (4-6 weeks); RMDQ 0-24; data extracted from table 2; data at 6 months after discharge (4-6 weeks)
- ^bSMT vs active exercise therapy + negative on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 24 weeks
- ^bSMT vs active exercise therapy + positive on clinical prediction rule; ODI 0-100; data extracted from table 4; data at 24 weeks
- ^bSMT vs. general and motor exercise therapy (combined); RMDQ 0-24; data extracted from table 2; data at 6 months
- ^bSMT (flexion-distraction) vs. exercise; RMDQ 0-24; change scores presented - used SD from baseline; data at 6 months
- ^bSMT vs light massage; Von Korff disability scale 0-10; data extracted from table 2; data at 24 weeks
- ^bSMT vs. physiotherapy; ODI 0-100; mean data estimated from figure - no SD presented, estimated from other studies; data at 6 months
- ^bSMT vs. exercise; ODI 0-100; mean data estimated from figure; no SD presented, estimated from other studies; data at 6 months
- ^bHVLA-SMT vs medical care; RMDQ 0-24; adjusted scores; IPD data provided by author; data at 26 weeks
- ^bLVVA-SMT (flexion-distraction) vs. medical care;RMDQ 0-24; adjusted scores; IPD data provided by author; data at 26 weeks
- ^bHVLA-SMT vs. Myofascial therapy; RMDQ 0-24; data extracted from table 3; data at 6 months
- ^bHVLA-SMT vs. back school; RMDQ 0-24; data extracted from table 3; data at 6 months
- ^bChiropractic care only vs. medical care only; RMDQ 0-24; data estimated from graphs; SD used from baseline score; data at 6 months
- ^cChiropractic care + physical modalities (DCPm) vs. medical care + physical therapy (MDpt); RMDQ 0-24; data estimated from graphs; SD used from baseline score; data at 6 months
- ^cManipulation vs exercise therapy; RMDQ 0-24; mean and p values presented from table 2; data at 6 months
- ^cSMT vs. McKenzie exercise; RMDQ 0-24; median (IQR) - converted to mean (SD); data extracted from table 2; data at 6 months
- ^cSMT vs. advice-group (counseling session + educational back booklet)RMDQ 0-24; median (IQR) - converted to mean (SD); data extracted from table 2; data at 6 months
- ^cSMT vs McKenzie; RMDQ 0-24; IPD data provided by the authors; data at 2 months post treatment (treatment period max 12 weeks; mean days: SMT 60.5 days; McKenzie 67.5)
- ^cSMT + home exercise program vs Supervised exercise and home exercise program; RMDQ 0-100; data extracted from table 3; data at 26 weeks
- ^cSMT vs. physiotherapy (massage, stretching and exercise); 6 months post-tx. = 6 months; ODI 0-100; IPD data provided by authors; data at 9 months (6 months post treatment)
- ^cSMT vs home exercise and advice;RMDQ 0-24; data extracted from table 2; data at 52 weeks
- ^cSMT vs supervised exercise therapy; RMDQ 0-24; data extracted from table 2; data at 52 weeks
- ^cSMT vs individualized physiotherapy; RMDQ 0-24; data extracted from table 2; data at 12 months after discharge (4-6 weeks)
- ^cSMT vs back School; RMDQ 0-24; data extracted from table 2; data at 12 months after discharge (4-6 weeks)
- ^cSMT vs. general and motor exercise therapy (combined); RMDQ 0-24; data extracted from table 2; data at 52 weeks
- ^cSMT (flexion-distraction) vs. exercise; RMDQ 0-24; change scores presented - used SD from baseline; data at 12 months
- ^cSMT vs light massage; Von Korff disability scale 0-10; data extracted from table 2; data at 52 weeks
- ^cSMT vs. exercise; ODI 0-100; mean data estimated from figure; no SD presented, estimated from other studies; data at 52 weeks
- ^cSMT vs. physiotherapy; ODI 0-100; mean data estimated from figure; no SD presented, estimated from other studies; data at 52 weeks
- ^cChiropractic care + physical modalities (DCPm) vs. medical care + physical therapy (MDpt);RMDQ 0-24; data estimated from graphs; SD used from baseline; data at 12 months
- ^cSMT (chiropractic care) only vs. medical care only; RMDQ 0-24; data estimated from graphs; SD used from baseline; data at 12 months
- ^cSMT vs. McKenzie exercise; RMDQ 0-24; median (IQR) - converted to mean (SD); data extracted from table 2; data at 12 months
- ^cSMT vs. advice-group (counseling session + educational back booklet); RMDQ 0-24; median (IQR) - converted to mean (SD); data extracted from table 2; data at 12 months
- ^cSMT vs McKenzie; RMDQ 0-12; IPD data provided by the authors; data at 12 months follow up post treatment (treatment period max 12 weeks; mean days: SMT 60.5 days;McKenzie 67.5)
- ^cMT vs. stabilization training; ODI 0-100; median (IQR) converted to mean (SD); IPD data; data at 12 months
- ^cSMT + home exercise program vs Supervised exercise and home exercise program; RMDQ 0-100; data extracted from table 3; data at 52 weeks
- ^cBest care + SMT vs. best care + exercise; RMDQ 0-24; IPD data provided by the authors; data at 12 months
- ^cSMT vs. physiotherapy (massage, stretching and exercise); ODI 0-100; IPD data provided by authors; 12 months post-treatment is 15 months

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding (performance bias and detection bias): All outcomes - patients
- (D) Blinding (performance bias and detection bias): All outcomes - providers
- (E) Blinding (performance bias and detection bias): All outcomes- outcome assessors
- (F) Incomplete outcome data (attrition bias): All outcomes - drop-outs
- (G) Incomplete outcome data (attrition bias): All outcomes - ITT analysis
- (H) Selective reporting (reporting bias)
- (I) Group similarity at baseline
- (J) Influence of co-interventions
- (K) Compliance with interventions
- (L) Timing of outcome assessments
- (M) Other form of bias (e.g. funding source, conflicts of interest)

Converted to an MD on the Roland Morris Disability Questionnaire of 0 to 24, these effects amount to an MD of -1.18, 95% CI -1.79 to -0.52 at 1-month follow-up, MD of -0.27, 95% CI -0.9 to 0.36 at 3-month follow-up, MD of -0.44, 95% CI -0.93 to 0.05 at 6-month follow-up, and MD of -0.52, 95% CI -1.08 to 0.09 at 12-month follow-up (based on the SD of 4.7 at 1 month, SD of 4.5 at 3 months, SD of 4.9 at 6 months, and SD of 4.7 at 12 months from Bronfort 2011 (home exercise and advice arm) as this is a large, good-quality study that used a common functional status scale).

Adverse events

Fifteen trials examined adverse events (Bronfort 2011; Brønfort 1996; Castro-Sanchez 2021; Cecchi 2010; Dougherty 2014a; Gudavalli 2006; Haas 2014; Hondras 2009; Hsieh 2002; Kamali 2019; Muller 2005; Schulz 2019; Skillgate 2007; UK BEAM Trial 2004; Zaproudina 2009) (Table 5). The evidence is very uncertain about the adverse effects of SMT (downgraded for study limitations by one level and imprecision by two levels).

Four trials reported no adverse events with SMT (Castro-Sanchez 2021; Cecchi 2010; Gudavalli 2006; Kamali 2019), while the other trials specified the occurrence and nature of adverse events.

Serious adverse events

Five trials reported no serious adverse events with SMT (Bronfort 2011; Brønfort 1996; Skillgate 2007; UK BEAM Trial 2004; Zaproudina 2009). Brønfort 1996 reported one adverse event that was unrelated to the control group. Schulz 2019 reported six serious adverse events that were unrelated to the study interventions.

Minor adverse events

Nine trials examined adverse events (Bronfort 2011; Brønfort 1996; Dougherty 2014a; Haas 2014; Hondras 2009; Hsieh 2002; Muller 2005; Skillgate 2007; Zaproudina 2009). The reported minor adverse events were transient (flare-ups; muscle soreness and stiffness), required little or no change to activity levels, and were considered non-serious in all these trials.

Important outcomes

Recovery

Six trials examined the effects of SMT versus other conservative interventions for perceived recovery (1258 participants) (Bronfort 2011; Gibson 1985; Gudavalli 2006; Hemmila 2002; Petersen 2011; Zaproudina 2009) (Analysis 3.3). There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT likely results in little to no difference in recovery compared to other conservative interventions at one month (RR 1.22, 95% CI 1.07 to 1.40; $I^2 = 0\%$; 4 studies, 652 participants). There is very low-certainty evidence (downgraded for study limitations and imprecision by one level and inconsistency by two levels) that SMT may result in little to no difference in recovery compared to other conservative interventions at three months (RR 1.11, 95% CI 0.77 to 1.59; $I^2 = 76\%$; 4 studies, 786 participants), but the evidence is very uncertain. There is low-certainty evidence (downgraded for study limitations and imprecision, both by one level) that SMT may result in little to no difference in recovery compared to other conservative interventions at six months (RR 1.01, 95% CI 0.83 to 1.23; $I^2 = 0\%$; 2 studies, 413 participants). Lastly, there is very low-certainty evidence (downgraded for study limitations by one level and imprecision by two levels) that SMT may result in little to no difference in recovery compared to other conservative interventions at 12 months (RR 1.06, 95% CI 0.86 to 1.32; $I^2 = 0\%$; 2 studies, 410 participants), but the evidence is very uncertain.

Health-related quality of life

Ten trials (1788 participants) examined the effect of SMT versus other conservative interventions for health-related quality of life (HRQoL) (Bronfort 2011; Brønfort 1996; Castro-Sanchez 2021; Gudavalli 2006; Haas 2014; Petersen 2011; Rasmussen-Barr 2003; Schulz 2019; Ulger 2017; Zaproudina 2009) (Analysis 3.4). There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT likely results in little to no difference in health-related quality of life compared to other conservative interventions at one month (SMD -0.04, 95% CI -0.17 to 0.10; $I^2 = 0\%$; 6 studies, 925 participants) and six months (SMD -0.13, 95% CI -0.26 to 0.01; $I^2 = 0\%$; 4 studies, 925 participants). There is low-certainty evidence (downgraded for study limitations and inconsistency by one level each) that SMT results in little to no difference in health-related quality of life compared to other conservative interventions at

three months (SMD 0.02, 95% CI -0.15 to 0.19; $I^2 = 51\%$; 8 studies, 1261 participants). Lastly, there is very low-certainty evidence (downgraded for study limitations, inconsistency, and imprecision by one level each) that SMT results in little to no difference in health-related quality of life compared to other conservative interventions at 12 months (SMD -0.14, 95% CI -0.34 to 0.06; $I^2 = 50\%$; 5 studies, 916 participants), but the evidence is very uncertain.

Return to work

Five trials (1036 participants) examined the effects of SMT versus other conservative interventions for return to work (Brønfort 1996; Cecchi 2010; Gibson 1985; Gudavalli 2006; Hemmila 2002; Petersen 2011) (Analysis 3.5).

There is very low-certainty evidence (downgraded for study limitations, inconsistency, indirectness, and imprecision all by two levels) that SMT results in a small increase in return to work at one month (RR 0.73, 95% CI 0.13 to 4.12; 1 study, 14 participants), but the evidence is very uncertain. There is very low-certainty evidence (downgraded for study limitations and imprecision, both by two levels) that SMT results in little to no difference in return to work at three months (RR 1.15, 95% CI 0.92 to 1.45; $I^2 = 0\%$; 3 studies, 159 participants), but the evidence is very uncertain. There is low-certainty evidence (downgraded for study limitations and inconsistency, both by one level) that SMT results in little to no difference in return to work at six months (RR 1.00, 95% CI 0.82 to 1.21; $I^2 = 0\%$; 2 studies, 146 participants). There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT likely results in little to no difference in return to work at 12 months (RR 1.08, 95% CI 0.98 to 1.19; $I^2 = 0\%$; 5 studies, 386 participants).

Comparison 4. Effect of SMT as adjuvant therapy

Critical outcomes

Pain

In total, 16 trials examined the adjuvant effects of SMT for pain (2685 participants) (Blanco-Giménez 2024; Bronfort 2014; Dogan 2021; Grande-Alonso 2019; Hsieh 2002; Ibrahim 2023; Licciardone 2003; Nambi 2018; Nejati 2019; Rasmussen 2008; Saracoglu 2022; Schulz 2019; UK BEAM Trial 2004; Verma 2013; Vismara 2012; Wang 2019). There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT as adjuvant therapy likely results in a small reduction in pain at one month (MD -6.70, 95% CI -9.10 to -4.29; $I^2 = 53\%$; 15 studies, 1755 participants). There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT as adjuvant therapy likely results in little to no difference in pain at three months (MD -4.82, 95% CI -8.67 to -0.98; $I^2 = 73\%$; 8 studies, 1517 participants), and 12 months (MD -3.92, 95% CI -8.53 to 0.69; $I^2 = 77\%$; 5 studies, 1568 participants). Lastly, there is very low-certainty evidence (downgraded for study limitations, inconsistency, and imprecision) that SMT as adjuvant therapy may result in a small reduction in pain relief at six months (MD -5.40, 95% CI -12.12 to 1.33; $I^2 = 73\%$; 5 studies, 553 participants), but the evidence is very uncertain (Analysis 4.1).

Functional status

In total, 12 trials examined the adjuvant effects of SMT on functional status (2583 participants) (Blanco-Giménez 2024; Bronfort 2014; Grande-Alonso 2019; Hsieh 2002; Ibrahim 2023; Licciardone 2003;

Nambi 2018; Nejati 2019; Saracoglu 2022; Schulz 2019; UK BEAM Trial 2004; Vismara 2012). There is very low-certainty evidence (downgraded for study limitations by one level and inconsistency by two levels) that SMT as adjuvant therapy may result in a small improvement in functional status at one month (SMD -0.46, 95% CI -0.74 to -0.18; $I^2 = 82\%$; 11 studies, 1545 participants), but the evidence is very uncertain. There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT as adjuvant therapy likely results in little to no difference in functional status at three months (SMD -0.15, 95% CI -0.30 to 0.00; $I^2 = 39\%$; 8 studies, 1581 participants). There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT as adjuvant therapy likely results in a small improvement in functional status at six months (SMD -0.38, 95% CI -0.70 to -0.05; $I^2 = 67\%$; 5 studies, 552 participants) and low-certainty evidence (downgraded for study limitations and inconsistency, both by one level) that SMT as adjuvant therapy may result in a small improvement in functional status at 12 months (SMD -0.23, 95% CI -0.43 to -0.03; $I^2 = 73\%$; 4 studies, 1562 participants; Analysis 4.2). Converted to an MD on the Roland Morris Disability Questionnaire of 0 to 100, these effects amount to an MD of -7.81 (95% CI -12.56 to -3.05) at 1-month follow-up, MD of -2.56 (95% CI -5.12 to -0.0) at 3-month follow-up, MD of -6.44 (95% CI -11.87 to -0.85) at 6-month follow-up, and MD of -3.93 (95% CI -7.73 to -0.51) at 12-month follow-up (based on the SD of 16.97 at 1 month, SD of 17.07 at 3 months, SD of 16.96 at 6 months, and SD of 17.07 at 12 months from Schulz 2019, as this is a large, good-quality study that uses a common functional status scale).

Adverse events

In total, eight trials examined this outcome (Blanco-Giménez 2024, Bronfort 2014; Evans 1978; Hsieh 2002; Nambi 2018; Rasmussen 2008; Schulz 2019; UK BEAM Trial 2004) (Table 5). The evidence is very uncertain about the adverse effects of SMT (downgraded for study limitations by one level, and imprecision by two levels).

Serious adverse events

Six trials reported no serious adverse events with SMT (Blanco-Giménez 2024; Evans 1978; Hsieh 2002; Nambi 2018; Rasmussen 2008; UK BEAM Trial 2004). Bronfort 2014 reported five serious adverse events, and Schulz 2019 reported six serious adverse events that were unrelated to the study interventions.

Minor adverse events

Two trials examined minor adverse events (Bronfort 2014; Rasmussen 2008). The reported minor adverse events were transient in nature (flare-ups, muscle soreness and stiffness), required little or no change to activity levels, and were considered non-serious in both trials.

Important outcomes

Recovery

One study examined the adjuvant effects of SMT on perceived recovery (Evans 1978). There is very low-certainty evidence (downgraded for study limitations by two levels, inconsistency, imprecision, and indirectness by one level) that SMT may result in a strong increase in recovery at one month (RR 3.40, 95% CI 1.12 to 10.28; 32 participants; NNTB = 2) but the evidence is very uncertain (Analysis 4.3). There are no data for three, six, and 12 months.

Health-related quality of life

Four trials examined the adjuvant effects of SMT on health-related quality of life at varying time points (Bronfort 2014; Dogan 2021; Schulz 2019; UK BEAM Trial 2004). There is very low-certainty evidence (downgraded for study limitations, inconsistency, indirectness, and imprecision, all by one level) that SMT as adjuvant therapy may result in little to no difference in health-related quality of life at one month (MD 0.45, 95% CI -1.35 to 2.25; 2 studies, 223 participants) and six months (MD -0.30, 95% CI -2.21 to 1.61; 1 study, 156 participants), but the evidence is very uncertain. There is moderate-certainty evidence (downgraded for study limitations by one level) that SMT as adjuvant therapy likely results in little to no difference in health-related quality of life at three months (MD 1.78, 95% CI 0.19 to 3.36; $I^2 = 38\%$; 3 studies, 834 participants). There is low-certainty evidence (downgraded for study limitations and inconsistency, both by one level) that SMT as adjuvant therapy may result in little to no difference in health-related quality of life at 12 months (MD -0.49, 95% CI -2.67 to 1.69; $I^2 = 84\%$; 4 studies, 1041 participants) (Analysis 4.4).

Return to work

There are no data for return to work.

Secondary analysis: effect of SMT versus another kind of SMT

In total, nine trials (1227 participants) examined the effect of head-to-head comparisons of different types of SMT (Carrasco-Martinez 2019; Castro-Sanchez 2016; Cook 2013; De Oliveira 2020b; Hondras 2009; Lizis 2021; McCarthy 2019; Nim 2020; Xia 2016). Of these, treatment allocation appeared to be conducted properly in six trials (Carrasco-Martinez 2019; Castro-Sanchez 2016; De Oliveira 2020b; Hondras 2009; Lizis 2021; Xia 2016). Participants in three trials were blinded to the interventions (De Oliveira 2020b; Lizis 2021; Nim 2020).

Six trials directly compared manipulation with mobilisation. However, there was substantial heterogeneity in the techniques used across these studies, making direct comparisons challenging. Most trials did not demonstrate a significant difference in effectiveness at the one-month follow-up. Two studies reported a moderate improvement in pain outcomes favouring manipulation, while one study found a moderate benefit in functional outcomes for manipulation.

See for a detailed description, see [Supplementary material 11](#) and [Supplementary material 5](#).

Explanation of heterogeneity, sensitivity analysis, and meta-regression

We were unable to conduct any of the pre-planned subgroup and sensitivity analyses for the comparison of SMT with no treatment because there were too few trials and too little discriminatory ability among the prognostic factors, thus precluding any meaningful interpretation of the effects.

For the other comparisons, we used subgroup analyses to explore the impact of symptom duration, multimodal treatment, and country income status. Furthermore, for the SMT versus other conservative intervention comparison, we conducted subgroup analyses by dividing the control group into recommended versus non-recommended interventions, active versus passive interventions, and SMT versus exercise-based interventions.

We conducted subgroup analyses for the type of technique (mobilisation versus manipulation), radiating pain, or treatment type. In many cases, there was too little discriminatory ability among the prognostic factors or too much data were missing.

Subgroup analyses failed to explain the statistical heterogeneity; nor did they result in a difference in effects (for details, see [Supplementary material 11](#); [Supplementary material 5](#)).

Meta-regression

We conducted meta-regression for the outcomes of pain and functional status for the analyses that included more than 10 trials: (1) SMT versus sham/placebo at 1-month follow-up, (2) SMT versus other conservative interventions at one, three, six, and 12 months of follow-up, and (3) SMT as an adjuvant therapy at 1-month follow-up ([Table 1](#); [Table 2](#)). We assessed these variables: income of country, multimodal SMT, and duration of LBP. For all multivariable analyses, these variables did not explain the statistical heterogeneity. The moderator effects were typically small.

Sensitivity analysis

Exclusion of trials with a high risk of bias in the sensitivity analysis had no appreciable effect on pain and functional status for SMT versus sham/placebo, versus other conservative interventions, and SMT as adjuvant therapy (see [Supplementary material 11](#); [Supplementary material 5](#)).

Equity assessment

The results show minimal impact on the synthesised result for pain and functional status (see [Supplementary material 5](#); [Supplementary material 11](#)). There was no evidence of a difference in the effectiveness of SMT across countries with different income levels (high versus middle and low). Additionally, these analyses did not explain the statistical heterogeneity. Due to insufficient data, it was not feasible to explore differences between age groups (younger than 65 versus 65 and older).

Reporting biases

We provided our assessment of the risk of selective outcome reporting bias for each included study in [Risk of bias in included studies](#).

Publication bias

An examination of publication bias was possible only for the primary comparisons SMT versus sham/placebo, and SMT versus other conservative interventions, due to the scarcity of data for the other comparisons. Funnel plots were constructed for the outcomes of pain and functional status for these comparisons ([Figure 10](#); [Figure 11](#); [Figure 12](#); [Figure 13](#)), respectively. For the outcome of pain for SMT versus other conservative interventions, it might appear that small trials favouring SMT are missing in the comparison. This may indicate publication bias; in this case, an underestimation of the true effects, because some trials may have used SMT as a control group in a trial evaluating the effects of another intervention. However, it is difficult to come to a definitive conclusion.

Figure 10. Funnel plot of comparison: 1 SMT versus sham/placebo, outcome: pain

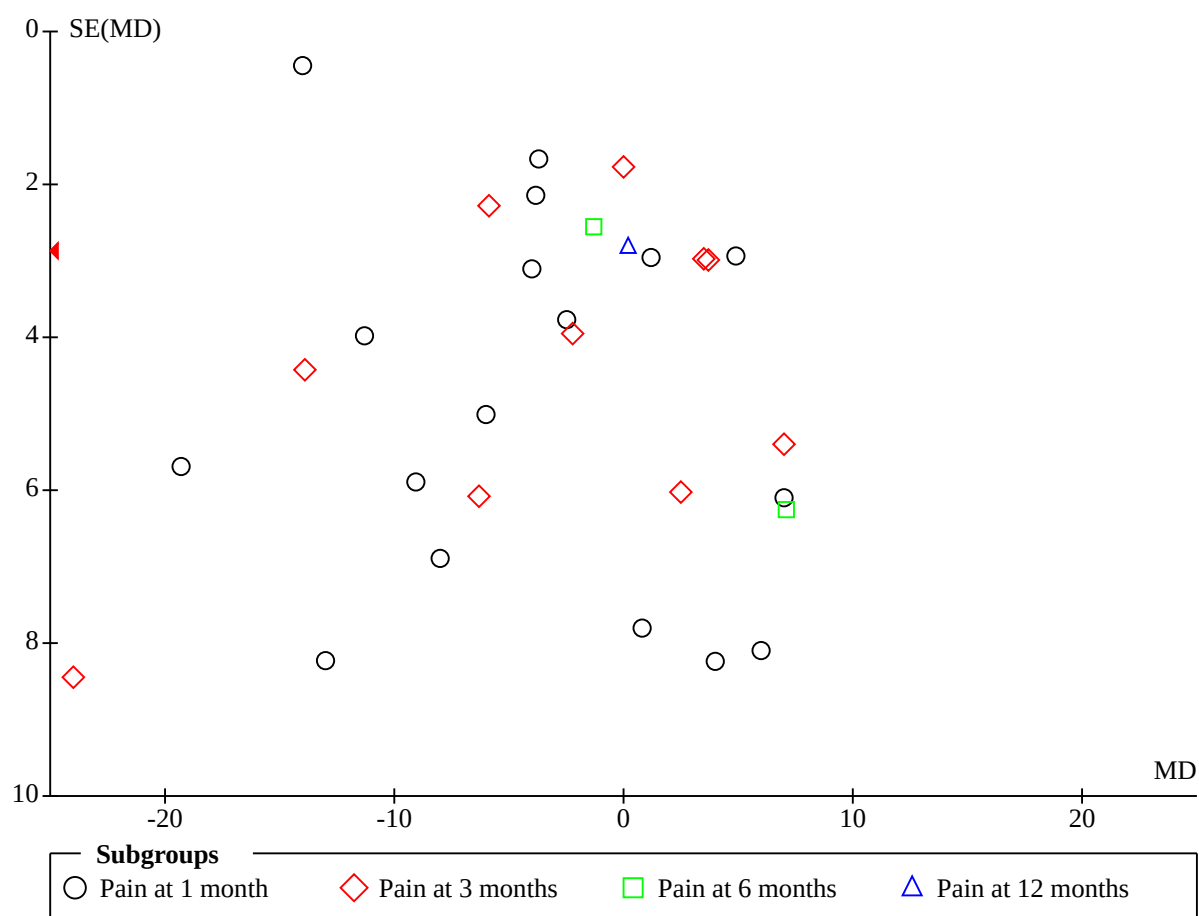


Figure 11. Funnel plot of comparison: 1 SMT versus sham/placebo, outcome: functional status

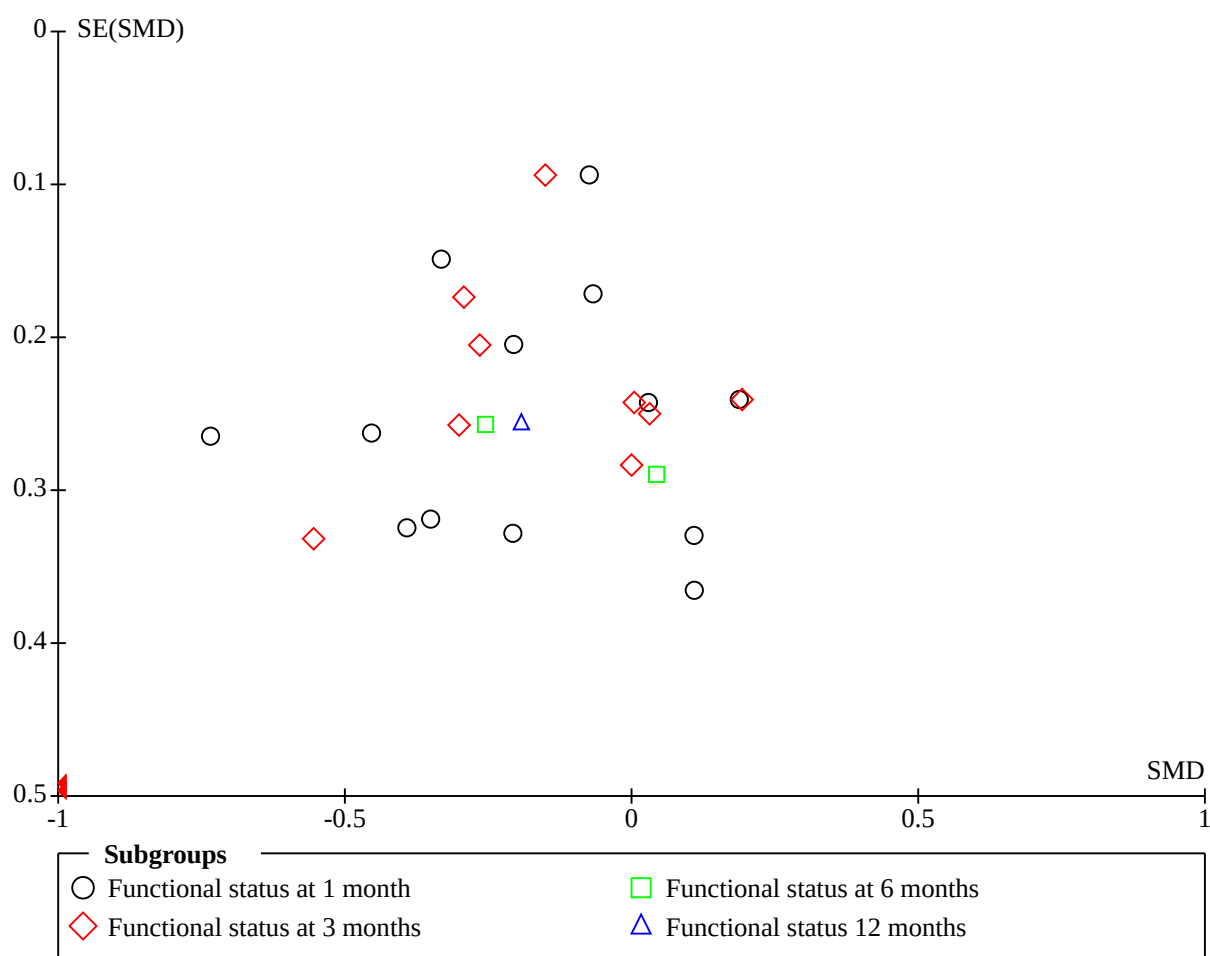


Figure 12. Funnel plot of comparison: SMT versus other conservative interventions, outcome: pain

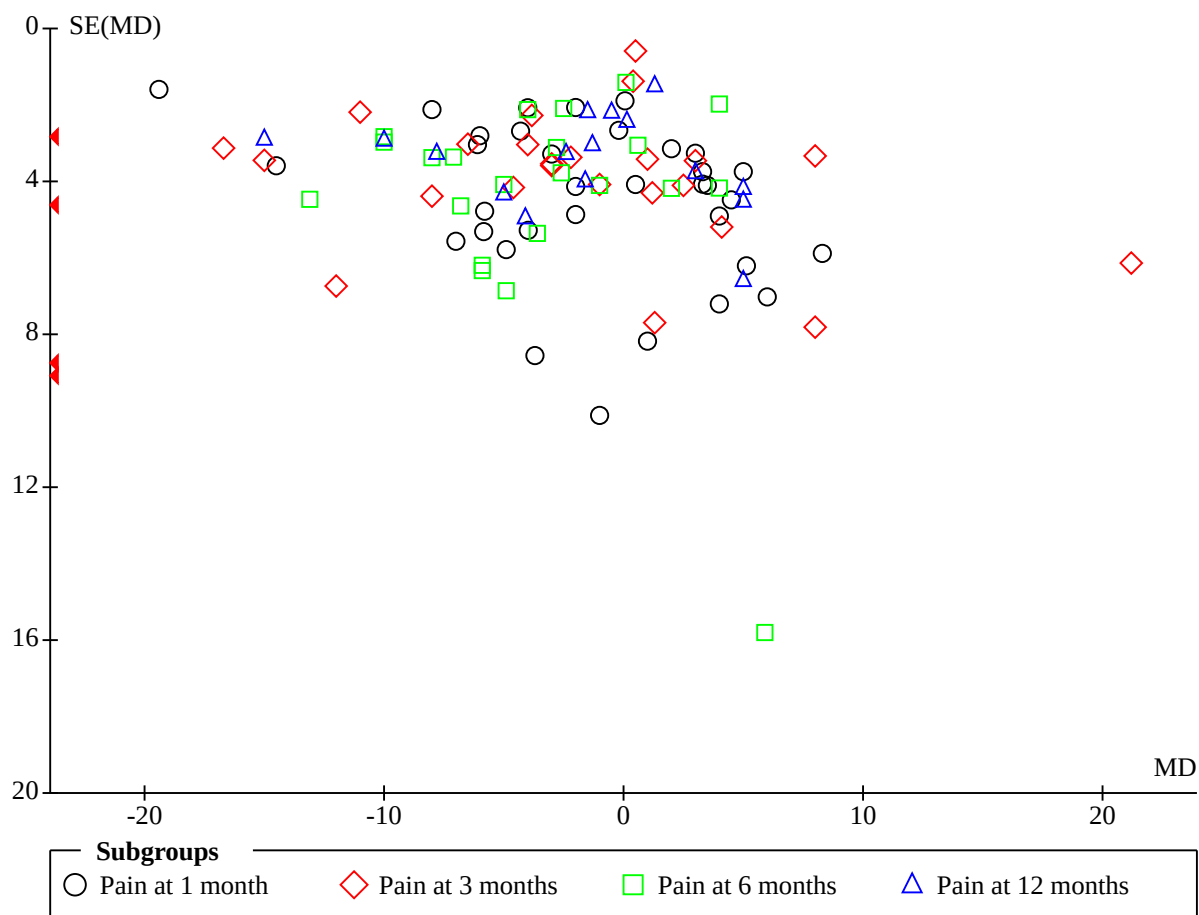
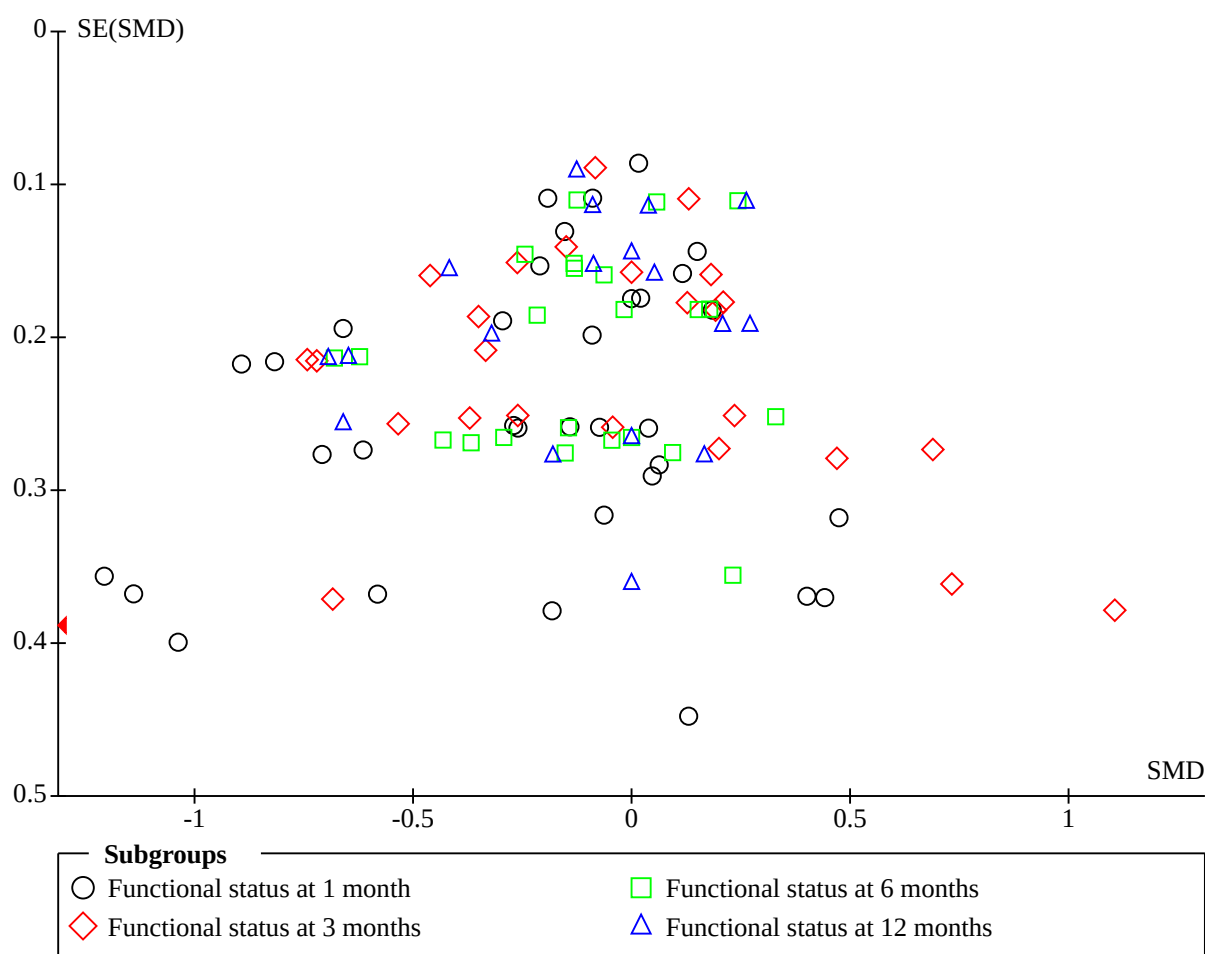


Figure 13. Funnel plot of comparison: SMT versus other conservative interventions, outcome: functional status

More important is to assess the number of trials that have been registered but have not been published or are ongoing (for details see [Supplementary material 4](#)). We identified two unpublished PhD theses (López Díaz 2016b [191]; Venegas-Rios 2009 [192]), and two conference abstracts of unpublished trials (Mandara 2008 [193]; Westrom 2010 [194, 195, 196]). Additionally, 11 trials were registered more than 10 years ago, but have not been published (ISRCTN47636118 2000; ISRCTN61808774 2000; NCT00163124 2005 [197]; NCT00410397 2006; NCT00394264 2006 [198]; NCT00964717 2012 [199]; NCT01861418 2013 [200]; NCT01765777 2013 [201]; NCT01842737 2013 [202]; NCT02034864 2014 [203]; NCT02123693 2014 [204]). In the last decade, 35 ongoing trials (identified in trial registers or as published protocols) have been identified, with 26 trials registered within the previous five years (CTRI 2022/08/044938 [205]; Fernandes 2017 [206]; Freitas 2022 [207]; Gedin 2017 [208]; IRCT20200511047391N1 2020 [209]; IRCT20230502058050N1 2023 [210]; KCT0008899 2023 [211]; NCT02982382 2016 [212]; NCT02983435 2017 [213]; NCT03049644 2017 [214]; NCT03181659 2017 [215]; NCT03190187 2017 [216]; NCT03794830 2019 [217]; NCT03836248 2019 [218]; NCT03880500 2019 [219]; NCT04082065 2019 [220]; NCT04646018 2020 [221]; NCT04744883 2021 [222]; NCT04843800 2021 [223]; NCT04924387 2021 [224]; NCT05042115 2021 [225]; NCT05057091 2021 [226]; NCT05162924 2021 [227]; NCT05214456 2022 [228]; NCT05282589

2022 [229]; ; NCT05460286 2022 [230]; NCT05784168 2023 [231]; NCT05986370 2023 [232]; NCT06150625 2023 [233]; NCT06277700 2024 [234]; NCT06291168 2024 [235]; NCT06294132 2024 [236]; NCT06404346 2024 [237]; PACTR201910655740780 2020 [238]; RBR-9zk7qx2 2023 [239]). Twenty-eight of these 35 ongoing RCTs involve fewer than 100 participants. The sample size in seven RCTs ranges from 108 to 600.

Further, of all 33 ongoing trials, 29 RCTs (2943 participants) investigate the effect of SMT compared to other conservative interventions. Only five trials (273 participants) examine SMT compared to no treatment, while 18 RCTs (1739 participants) compare SMT to sham SMT/placebo. If all the trials comparing SMT to sham SMT/placebo publish their results, it could potentially have an impact on our results. However, for the other comparisons, the impact on our results is likely to be minimal.

Finally, virtually all the trials appear to be non-commercially sponsored, so the impact of industry on these results would appear to be minimal. However, this does not rule out the possibility that small, clinician-driven trials may have impacted these results.

Conflicts of interest/industry-sponsored trials

Many of the older trials do not indicate the funding body or include declarations of interest, or both. Therefore, it is difficult to ascertain whether conflict of interest is an issue. The intervention, SMT, is not a marketable instrument and, therefore, this issue is likely to be less relevant than for pharmaceutical trials. However, many of the trials were conducted by clinicians/therapists who provide SMT and had full possession of the trial process from the beginning to the planned reporting, including the statistical analyses. Therefore, potential conflicts of interest cannot be ruled out.

DISCUSSION

Summary of main results

This updated review assessed the effectiveness of spinal manipulative therapy (SMT) for chronic low back pain (LBP) and now includes 76 RCTs with 11,866 participants, which suggests rather robust analyses. Although we provide new information and insight, challenges remain, particularly in addressing the substantial clinical and statistical heterogeneity. As a result, the certainty of evidence remains low to very low.

In summary, we found very low-certainty evidence (downgraded for inconsistency and study limitations) that SMT may result in a small reduction in pain compared to sham SMT/placebo at one month and very low-certainty evidence (downgraded for study limitations and inconsistency) that SMT may result in a medium improvement in functional status compared to sham SMT/placebo at one month ([Summary of findings 1](#)). For the comparison of SMT to no treatment, there is very low-certainty evidence that SMT results in a medium reduction in pain and a large improvement in functional status at one month ([Summary of findings 2](#)).

While the effect sizes for the outcomes pain and functional status met our prespecified threshold for medium to large benefit for the comparison between SMT and no treatment and met our prespecified threshold for small to medium benefit for the comparison between SMT and sham SMT/placebo, caution is warranted when interpreting these results due to the wide CIs and the low to very low certainty of the evidence.

Since the comparisons of SMT to sham SMT/placebo and no treatment demonstrated medium to large improvements in functional status, we also presented the results of SMT versus other conservative interventions ([Summary of findings 3](#)), as these results are particularly relevant for clinical practice and answer very different clinical questions. Specifically, the comparison of SMT versus placebo or no treatment answers the question of efficacy, while the comparison of the effect of SMT versus other conservative interventions answers a more pragmatic question; that is, how it works in clinical practice. Perhaps more importantly, most of the available data (41 trials including 8022 participants) compares SMT with other conservative interventions. In this comparison, a relatively higher number of studies – both in terms of quantity and participant numbers included – were of higher quality (i.e. had a lower risk of bias) compared to the other comparisons, making this comparison a more robust source of evidence. In summary, compared to other conservative interventions, SMT may result in little to no difference in pain and a small improvement in functional status at short-term follow-up. However, the effect sizes were small, and the certainty of the evidence was low. This low certainty is

primarily due to downgrading for inconsistency, which in turn was the result of including trials that investigated the effect of SMT in different settings and populations, with different types of SMT techniques, dosage, and frequency of treatment.

Only 47% of the RCTs reported on adverse events, which raises concerns about under-reporting. Importantly, of the included trials that examined adverse events, none reported serious complications as a result of SMT. Where adverse events were documented, most adverse events were transient, such as a temporary increase in pain, muscle soreness, or stiffness. Serious complications following SMT for LBP, which include cauda equina syndrome, fractures, and neurological or vascular compromise, are extremely rare and have primarily been documented in case reports [240, 241] and retrospective cohort studies [242]. Budtz and colleagues estimated that 0.1 in 1.6 million patients presenting to a physiotherapist were identified as having cauda equina syndrome [243]. Given the rarity of this condition, risk estimates after receiving SMT vary widely, ranging from seven cases in 100,000 participants after receiving chiropractic treatment [242], to one in 100,000 to 128,000 treatments [241]. Consequently, assessing serious adverse events in a review of RCTs provides limited information, and RCTs are not the study design of choice for evaluating these rare outcomes. Further, the absence of adverse events in the RCT data should not be interpreted as evidence of safety. Therefore, the true incidence of adverse events remains uncertain, and these results should also be interpreted with caution.

Lastly, many of the new trials on SMT were small and had a high risk of bias, weakening the evidence base, and prompting an urgent call for higher-quality trials to avoid wasting resources.

Limitations of the evidence included in the review

In this update, we identified approximately three times as many trials as in the previous edition.

In general, the certainty of evidence was low to very low across all comparisons. For the critical outcomes at most follow-up time points, we downgraded the evidence due to the diversity in results across the trials, which could not be explained by subgroup analysis, and study limitations, as most participants came from trials with a high risk of bias (mostly selection and performance bias). In addition to study limitations and inconsistency, we downgraded for imprecision due to the small number of participants for SMT compared to no treatment.

Interestingly, in the previous edition, we considered the evidence to be of high certainty for these critical outcomes for some comparisons. This is somewhat enigmatic because investigators would expect the evidence to become stronger, not weaker, as more, better-quality trials are published. This 'downgrading' of the evidence is a consequence of increased statistical heterogeneity caused by small, newly conducted, but poor-quality trials. This renders the interpretation of the true treatment effect size difficult. Also, the classification of the comparisons has been slightly adapted, and the GRADE approach has been refined, which may have led to a more rigorous GRADE assessment than in the previous review.

Further, we based our interpretation of the magnitude of effect on the American College of Physicians (ACP) guidelines [5]. We acknowledge that applying alternative frameworks, such

as the minimum clinically important improvement thresholds proposed by Tubach and colleagues [244], could have led to a more conservative interpretation, potentially suggesting that the observed effects fall below the threshold for a minimal clinically important difference. However, decisions to choose SMT are not solely based on the size of the benefit; they also consider factors such as cost, potential harms, availability of alternative treatments, and individual patient preferences. These broader considerations are reflected in the ACP guideline framework [5], which is why we selected it as the basis for our interpretation. Another limitation of this review is the exclusion of responder outcomes (e.g. proportion of participants achieving at least a 30% pain reduction), which may provide complementary insights beyond average effects on continuous scales.

As stated above, we identified an alarming development. Specifically, of the 22 trials published between 2018 and 2023, 16 trials had fewer than 100 participants, and nine trials had a high risk of bias (i.e. selection bias). This trend is concerning because these trials do not add anything to our understanding of the effectiveness of SMT, which has been confirmed in a recent trial sequential analysis (in preparation for publication) [245]. This is an interesting development because there appeared to be a trend toward better-quality trials examining the effects of SMT [246]. Furthermore, the small trials which we included quite typically had a high risk of bias and demonstrated a large effect, suggesting publication bias. Lastly, our observations represent an urgent call to trial authors and policymakers that these types of trials are a waste of resources [247].

Moreover, it is essential to acknowledge other important sources of heterogeneity when considering the effectiveness of SMT. These include the variances in SMT techniques employed by diverse healthcare providers, the degree of flexibility allowed in tailoring SMT to the individual participant by the healthcare providers, permissible co-interventions, the variation in the treatment duration and frequency, and the number of healthcare providers/clinics in each trial, as well as the diversity of treatments in the control group. We assessed heterogeneity through subgroup analyses (e.g. active versus passive control, radiating pain versus no radiation, manipulation versus mobilisation) and meta-regression (e.g. multimodal treatment) defined a priori, but these did not explain the observed heterogeneity. For other potential sources, such as treatment duration and provider variability, subgroup or meta-regression analyses were not defined a priori and not conducted due to insufficient data or inconsistent reporting across trials. We are aware that the presented trials are very heterogeneous. Future research can focus on better rationales for personalised treatment approaches, and treatment duration and frequency.

To further improve the certainty of the evidence, we need to be more specific with the inclusion criterion of chronic non-specific LBP. This criterion encompasses people with a wide range of symptoms, where the underlying mechanisms causing LBP can vary. This variability introduces heterogeneity in the trials and their results.

Regarding the reporting of the SMT intervention, nearly three-quarters of the trials provided a rationale for the conduct of the trial. Frequently, this rationale cited previous research indicating a modest short-term benefit of SMT, yet highlighted the absence of definitive evidence owing to a paucity of published trials. However,

only in a small number of trials did the reported rationale include a mechanism for how SMT might work or the intended goal of the SMT intervention. This is important because considering how an intervention works can enhance the alignment between inclusion/exclusion criteria and outcome measures. Such alignment may lead to a more homogenous group of participants, improved outcomes, and finally to better patient care.

In many of the trials, the list of authors included the clinicians/therapists providing SMT. Therefore, a potential conflict of interest cannot be ruled out even though the authors declared no potential risk of interest. Study authors may have been biased in their preference for a given intervention, resulting in an overestimation of its effectiveness. While the role of publication bias and the funding body do not appear to influence our results for those studies using a pragmatic and placebo design, we could not sufficiently determine the potential influence on the other comparisons and outcomes, so this issue remains unresolved.

The 76 RCTs included in this review were diverse in geographical location and predominantly carried out in high-income countries, with a few in middle-income countries and none in low-income countries. The trials were conducted in various settings, including the adult population with LBP with or without radiation into the leg(s), meaning the results are widely applicable to clinical practice in high- and middle-income countries. This is a significant limitation as approximately 80% of the global population lives in low-income or lower-middle countries, and the effectiveness of SMT in these settings remains uncertain.

A wide variety of SMT techniques were included, reflecting the diversity of practice. We classified the techniques by manipulation, mobilisation, or a mix of these. This broad inclusion likely contributed to the observed heterogeneity across trials. Additionally, the number and duration of treatments varied widely from one to 24 sessions over one week to six months, underscoring the lack of consensus on the optimal timing and frequency of applying SMT, thereby introducing further heterogeneity.

Further, the most available data pertain to the critical outcomes; far fewer trials measured important outcomes, leading to less certainty in drawing meaningful conclusions for these outcomes. Although this review focused on the clinical effects of SMT, there is a clear need to include economic evaluations. A systematic review on cost-effectiveness suggests that manual treatments, including SMT, may be cost-effective [248]. SMT is typically delivered over fewer treatment sessions, may result in fewer referrals to secondary care, and may potentially reduce recurrences. Consulting a chiropractor or physical therapist as the initial care provider has also been associated with lower healthcare costs. However, these conclusions are primarily based on observational studies, as cost-effectiveness is rarely evaluated in RCTs. Since only a few trials in this review included a cost-effectiveness analysis, and we did not assess cost-effectiveness in this review, it remains uncertain whether SMT is a cost-effective option for treating chronic LBP.

Adverse effects were reported in less than half of the trials, with an increasing trend in reporting observed in recent trials. The inconsistent and incomplete reporting of adverse events across studies is likely to introduce a risk of underestimation in the frequency and severity of adverse events. The quality and consistency of the reporting of adverse effects remain poor [249]. Consequently, drawing firm conclusions regarding the adverse

effects of SMT is challenging due to the scarcity of data, inadequate reporting, and small sample size.

Lastly, of the 22 trials published between 2018 and 2023, only six had a sample size exceeding 100 participants (ranging from 10 to 330 participants; median: 60, mean: 47). Conducting additional small trials is unlikely to improve our current evidence.

Limitations of the review processes

To minimise the potential biases in the review process, we adhered to the recommendations outlined by the Cochrane Musculoskeletal Group and the *Cochrane Handbook for Systematic Reviews of Interventions* [35]. Pairs of review authors independently assessed trials for inclusion, with a third review author resolving any discrepancies. Citations were tracked, and the trial list was cross-checked with trial authors and experts in the field of SMT to identify any missing trials. Additionally, the funnel plots did not suggest publication bias. Despite this structured approach, we may have missed potentially relevant trials. While we consider the possibility that we overlooked a large trial to be unlikely, it is important to note that the review's conclusion is not expected to change.

Further, most ongoing trials are small, do not address a main comparison and, therefore, are likely to have little impact on our results.

We may have introduced a systematic bias by using the Cochrane RoB 1 tool instead of the Cochrane RoB 2 tool (developed during the conduct of this review), and potentially misclassifying domains as high or low risk. We used selection bias (random sequence generation and allocation to treatment) as criteria for identifying trials with a high risk of bias. Assessment of these items does not differ in the RoB 2 tool. Consequently, this aspect may not have affected our conclusions. Additionally, sensitivity analyses to evaluate the impact of studies with a high risk of selection bias did not change our conclusions. Furthermore, a meta-epidemiological study that examined the association between the risk of bias and treatment effects in exercise RCTs for chronic LBP suggests there is no association between treatment effects and risk of bias, indicating that the risk of bias assessment may be less relevant than previously assumed [250]. Nevertheless, it remains a contentious issue.

In this update, we grouped the comparisons as recommended by the Cochrane Musculoskeletal Group, which did not entirely overlap with the comparisons analysed in the 2011 version. This entailed relocating some studies from one comparison to another (e.g. SMT versus 'all other interventions' to SMT versus other conservative interventions). These issues arose because of the ambiguous descriptions of the control arms in some studies. However, these reclassifications were few and do not appear to have influenced the results and conclusions.

Agreements and disagreements with other studies or reviews

Our results are, in short, consistent with our earlier review, notwithstanding the fact that we modified the choice of primary and secondary comparisons. Furthermore, these results are consistent with other recent systematic reviews [251, 252, 253, 254], and guidelines that recommend SMT as part of a broader treatment package [5, 6, 7, 8]. Furthermore, Gorrell and colleagues' review on reporting adverse events from SMT aligns with our conclusion,

indicating that serious events in SMT are rare [249]. RCTs are typically designed to detect intervention effects, not the risk of serious adverse events.

However, this review is not entirely in agreement with others. Lavazza and colleagues investigated the effect of SMT versus sham SMT and concluded there was no clinically relevant effect; however, they included a broader range of treatments (e.g. massage) than we did [255]. Coulter and colleagues concluded that high-velocity, low-amplitude (HVLA) spinal manipulation produces a larger effect than low-velocity, low-amplitude (LVLA) mobilisation techniques [256]. However, those conclusions were based upon indirect evidence, whereas our results are based upon direct, head-to-head comparisons of these two techniques.

Lastly, Sørensen and colleagues examined targeted versus non-targeted SMT and concluded that targeting a specific vertebral level did not result in improved outcomes for pain and functional status compared to a non-targeted approach [257]. Whereas they combined results from various studies, we chose not to conduct a meta-analysis for studies exploring this comparison due to the limited number of studies and a wide diversity of approaches.

AUTHORS' CONCLUSIONS

Implications for practice

Clinicians want information on the efficacy of spinal manipulative therapy for chronic low back pain; the results of this review will serve as guidance in various ways.

Spinal manipulative therapy may result in a small reduction in pain and medium improvement in functional status in the short term when compared to sham spinal manipulative therapy/placebo. Spinal manipulative therapy may result in a medium reduction in pain and a large improvement in functional status in the short term when compared to no treatment. Importantly, the certainty of the evidence ranged from low to very low. Improving the certainty of the evidence is unlikely, as the newly included randomised controlled trials in this updated review have neither significantly altered the evidence nor improved the certainty of the evidence. It is difficult to design a randomised controlled trial (RCT) with a convincing comparison group when studying the efficacy of spinal manipulative therapy. Consequently, the blinding is flawed, resulting in a lower certainty of evidence. The weak evidence does not diminish the importance of the question.

While spinal manipulative therapy can be delivered as a stand-alone therapy, it is frequently offered in a broader treatment package that includes advice and exercise. This approach aligns with the recommendations of the practice guidelines.

When chronic low back pain significantly interferes with quality of life, patients will seek assistance from a healthcare practitioner. In this context, understanding the effectiveness of spinal manipulative therapy is important. The results of our analysis yield the following insights: when comparing spinal manipulative therapy to other conservative interventions, it may demonstrate little to no effect in reducing pain and a small effect in improving functional status, indicating that it is as effective as other conservative interventions. This supports the inclusion of spinal manipulative therapy in international guidelines such as the World Health Organization guideline [8].

Similar results emerge when spinal manipulative therapy is utilised as adjuvant therapy: it may demonstrate a small reduction in pain and a small improvement in functional status. Furthermore, current evidence does not justify targeting spinal manipulative therapy based on specific patient characteristics [258]. Instead, the choice of treatment should be the result of a shared decision-making process, taking patient preferences and clinicians' experience and skills into account.

Adverse events were often not recorded and, when recorded, were measured differently across trials. Consequently, we were unable to pool these data, and unable to draw firm conclusions. Often, the reported adverse events were mild to moderate and temporary. Of those included studies which reported serious adverse events, none were deemed to be related to the intervention. According to a systematic review that included studies with multiple designs, but primarily case reports and case series, serious adverse events are rare [240]. Consequently, assessing serious adverse events in a review of RCTs may provide limited information, as this is not the design of choice for evaluating these rare outcomes. Also, the under-reporting of adverse events may introduce an underestimation in frequency and severity. Therefore, these results should also be interpreted with caution. Given this, clinicians must ensure that patients are fully informed of potential risks before undergoing treatment.

Equity-related implications for practice

The findings of this review should be interpreted with caution in relation to health equity. Most included studies were conducted in high-income countries and involved adults aged 18 to 65 years, often drawn from the general population or healthcare settings. As a result, the applicability of the findings to more diverse or underserved populations may be limited. Practitioners should consider the potential lack of representation of disadvantaged groups when applying these findings in a clinical setting.

Implications for research

Conducting additional randomised controlled trials in a similar way to those previously performed is unlikely to alter our conclusions. In particular, small studies should be avoided as they tend to have a high risk of bias. This high risk is caused by the absence of published protocols or registrations, substandard conduct, and inadequate reporting, largely due to limited financial resources. Consequently, this research does not contribute to evidence-based knowledge about spinal manipulative therapy.

Moving forward, any future randomised controlled trials investigating the effect of spinal manipulative therapy should adhere to rigorous standards. This entails a published protocol or registration in a trial registry, aiming for a large sample size, including the recommended core outcomes [39], as well as more participant and treatment characteristics, ensuring robust conduct of the trial, and adherence to the reporting guidelines. Additionally, there is a need for economic evaluations. Spinal manipulative therapy could prove to be more cost-effective than other conservative interventions because it is typically provided in fewer treatment sessions and may lead to fewer referrals to secondary care. Spinal manipulative therapy may also be more cost-effective as it may potentially reduce recurrences. However, this needs to be investigated in direct, head-to-head comparisons.

When planning future trials, researchers should focus on exploring the aetiology of chronic low back pain and how spinal manipulative therapy is thought to work. Inclusion and exclusion criteria should align with these priorities. This approach may refine the selection of participants for spinal manipulative therapy and thus provide more insight into the subgroups within the broad group of people suffering from non-specific low back pain. This will facilitate a more meaningful, effective comparison of interventions across trials.

Lastly, although a randomised controlled trial is not the optimal design for assessing the potential harms of spinal manipulative therapy, the reporting of adverse events should be improved. Future trials should systematically assess and report adverse events, and ensure balance in reporting of both benefits and harms of spinal manipulation in RCTs to provide a more comprehensive understanding of the safety of spinal manipulative therapy [249]

Equity-related implications for research

Health equity is an increasingly important consideration in research. In this review, country income level and older age were defined a priori as equity-relevant factors. Most included studies were conducted in high-income countries and primarily involved adults aged 18 to 65 years, with representation from both genders. Most studies recruited participants from the general population or healthcare setting, although some focused on specific subgroups such as women, university students, or company employees. Ethnicity, socioeconomic status, and educational background were reported in only some studies. Furthermore, equity analyses were limited to pain and function, excluding other important outcomes such as return to work and health-related quality of life due to the scant data. Future trials should aim to include more diverse and underserved populations, and systematically report health equity factors such as ethnicity, socioeconomic status, and education. Incorporating these elements can enhance the applicability and generalisability of our results to the global population.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD008112.pub3](https://doi.org/10.1002/14651858.CD008112.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 The GRADE approach to evidence synthesis

Supplementary material 8 Other conservative treatment categories and description

Supplementary material 9 Sources of risk of bias

Supplementary material 10 Criteria for a judgement of 'low risk of bias' for the sources of bias

Supplementary material 11 Detailed version of description of studies and synthesis of results

Supplementary material 12 Differences between the protocol and previous versions

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Musculoskeletal supported the authors in the development of this review.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Jordi Pardo Pardo, Campbell and Cochrane Health Equity Thematic Group;
- Managing Editor (provided editorial guidance to authors, edited the article): Jenny Bellorini, Cochrane Central Editorial Service;
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- Peer-reviewers (provided comments and recommended an editorial decision): Susana Tinoco Duarte, Escola Superior de Saúde - Instituto Politécnico de Setúbal (clinical/content review); Roger Chou, Oregon Health & Science University (clinical/content review); A/Prof Bruno Tirotti Saragiotto, Discipline of Physiotherapy, University of Technology Sydney, Australia (clinical/content review); Brian Duncan (patient and public review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

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Conception and design: SM Rubinstein, MW van Tulder, WJJ Assendelft

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Drafting of the review: A de Zoete, T Innocenti, MJ Petrozzi, SM Rubinstein

Critical revision of the article for important intellectual content: all authors

Final approval of the article: all authors

Statistical expertise: MR de Boer

Administrative, technical, or logistical support: SM Rubinstein

Collection and assembly of data: A de Zoete, T Innocenti, MJ Petrozzi, SM Rubinstein, M van Middelkoop (2011 publication), MR de Boer, WJJ Assendelft (2004 publication)

Declarations of interest

All authors comply with Cochrane's Conflict of Interest Policy.

A de Zoete, MJ Petrozzi, and SM Rubinstein all work in their private clinics as chiropractors.

T Innocenti, M van Middelkoop, WJJ Assendelft, MR de Boer, and MW van Tulder have declared that they have no conflicts of interest.

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Registration and protocol

Previous published versions of the review

Original review

Assendelft WJ, Morton SC, Yu EI, Suttorp MJ, Shekelle PG. Spinal manipulative therapy for low back pain. A meta-analysis of effectiveness relative to other therapies. *Annals of Internal Medicine* 2003;138:898-906.

Assendelft WJ, Morton SC, Yu EI, Suttorp MJ, Shekelle PG. Spinal manipulative therapy for low-back pain. *Cochrane Database of Systematic Reviews* 2004, Issue 1. Art. No.: CD000447. DOI: 10.1002/14651858.CD000447.pub2. [DOI: 10.1002/14651858.CD000447.pub2]

Review update

Protocol: Rubinstein SM, Van Middelkoop M, Assendelft WJ, De Boer M, Van Tulder MW. Spinal manipulative therapy for chronic low-back pain [Protocol]. *Cochrane Database of Systematic Reviews* 2009, Issue 4. Art. No.: doi.org/10.1002/14651858.CD008112

Rubinstein SM, Van Middelkoop M, Assendelft WJ, De Boer MR, Van Tulder MW. Spinal manipulative therapy for chronic low-back pain. *Cochrane Database of Systematic Reviews* 2011, Issue 2.

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Data, code and other materials

As part of the published Cochrane review, the following supplementary materials are made available for download for users of the Cochrane Library:

1. Search strategies ([Supplementary material 1](#))
2. Characteristics of included studies ([Supplementary material 2](#))
3. Characteristics of excluded studies ([Supplementary material 3](#))
4. Characteristics of ongoing studies ([Supplementary material 4](#))
5. Analyses ([Supplementary material 5](#))
6. Data package ([Supplementary material 6](#))
7. The GRADE approach to evidence synthesis ([Supplementary material 7](#))
8. Other conservative treatment categories and description ([Supplementary material 8](#))

What's new

Date	Event	Description
6 January 2026	New search has been performed	The search has been updated to 18 October 2024.
6 January 2026	New citation required but conclusions have not changed	This update includes 50 newly identified studies, bringing the total number of included studies to 76. While the additional data allowed for a more precise point estimate, the overall conclusions of the review remain unchanged. Notably, of the 22 trials identified between 2018 and 2023, 16 had fewer than 100 participants and nine were assessed as having a high risk of bias, primarily due to selection bias. The small sample sizes and methodological limitations of many of the newly included studies reduce the certainty of the evidence. See 'Differences between the protocol and previous versions' for methods changes (Supplementary material 12).

History

Protocol first published: Issue 4, 2009

Review first published: Issue 2, 2011

Date	Event	Description
10 December 2012	Amended	Updated author affiliations
1 August 2012	Amended	Updated search and added studies to 'Studies awaiting classification' list

9. Sources of risk of bias ([Supplementary material 9](#))
10. Criteria for a judgment of 'low risk of bias' for the sources of bias ([Supplementary material 10](#))
11. Detailed version of description of studies and synthesis of results ([Supplementary material 11](#))
12. Differences between the protocol and previous versions ([Supplementary material 12](#))

Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, Review Manager, using the inbuilt computation methods. Template data extraction forms are available from the authors by reasonable request.

Notes

Since the earliest publication (2004), this review has been split into two parts (acute and chronic low back pain). In the 2011 version of the review, 18 new studies were identified. In this update, we have identified 50 new studies.

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ADDITIONAL TABLES

Table 1. Meta-regression analysis. Comparisons: SMT versus sham SMT/placebo and SMT versus other conservative interventions for the outcome pain

SMT versus sham SMT/placebo					
Pain at 1 month			Initial model $I^2 = 89\%$		
Variable(s)	N = 17	B (95% CI)	P-value	$I^2 = 88$	$R^2 = 20$
Income country		-17.69 (-32.9 to 2.46)	0.03		
Duration LBP		2.4 (-12.65 to 17.45)	0.74		

Table 1. Meta-regression analysis. Comparisons: SMT versus sham SMT/placebo and SMT versus other conservative interventions for the outcome pain (Continued)

Multi-modal SMT		-0.14 (-27.02 to 26.74)	0.99		
SMT versus other conservative interventions					
Pain at 1 month			Initial model I ² = 90%		
Variable(s)	N = 38	B (95% CI)	P-value	I ² = 84	R ² = 8
Income country		-4.61 (-12.05 to 2.83)	0.22		
Duration LBP		-4.42 (-11.27 to 2.43)	0.20		
Pain at 3 months			Initial model I ² = 79%		
Variable(s)	N = 24	B (95% CI)	P-value	I ² = 76	R ² = 26
Income country		26.79 (9.27 to 44.32)	0.005		
Duration LBP		-2.0 (-3.83 to 8.01)	0.47		
Multi-modal SMT		3.38 (-2.55 to 9.31)	0.25		
Pain at 6 months			Initial model I ² = 54%		
Variable(s)	N= 22	B (95% CI)	P-value	I ² = 40	R ² = 43
Income country		9.91 (2.7 to 17.75)	0.016		
Duration LBP		-5.11 (-9.53 to -0.69)	0.07		
Multi-modal SMT		1.50 (-2.68 to 5.67)	0.46		
Pain at 12 months			Initial model I ² = 67%		
Variable(s)	N=16	B (95% CI)	P-value	I ² = 0	R ² =100
Duration LBP		-2.41 (-7.58 to 2.77)	0.33		
Multi-modal SMT		7.32 (-1.12 to 13.14)	0.018		
Income country		No middle to low income studies			

Reference: duration LBP (0 = subacute/chronic; 1 = chronic only); multi-modal SMT (0 = SMT alone; 1 = SMT in combination with other (minimal) modalities); Country (0 = high income countries; 1 = middle to low income countries).

Table 2. Meta-regression analysis. Comparisons: SMT versus sham SMT/placebo and SMT versus other conservative interventions for the outcome functional status

SMT versus sham SMT/placebo			Initial model $I^2 = 83\%$		
<i>Variable(s)</i>	<i>N = 15</i>	<i>B (95% CI)</i>	<i>P-value</i>	$I^2 = 89$	$R^2 = 38$
Income country		-1.39 (-2.40 to -.38)	0.01		
Duration LBP		-.06 (-1.33 to 1.21)	0.92		
Multi-modal SMT		-0.28 (-2.16 to 1.60)	0.75		
SMT versus other conservative interventions					
Functional status at 1 month			Initial model $I^2 = 73\%$		
<i>Variable(s)</i>	<i>N = 34</i>	<i>B (95% CI)</i>	<i>P-value</i>	$I^2 = 78$	$R^2 = 16$
Income country		-0.10 (-0.51 to 0.30)	0.61		
Duration LBP		-0.29 (-0.63 to 0.04)	0.08		
Multi-modal SMT		0.23 (-0.09 to 0.55)	0.16		
Functional status at 3 months			Initial model $I^2 = 70\%$		
<i>Variable(s)</i>	<i>N = 25</i>	<i>B (95% CI)</i>	<i>P-value</i>	$I^2 = 57$	$R^2 = 58$
Income country		1.40 (-2.32 to -0.48)	0.01		
Multi-modal SMT		0.37 (-0.62 to 0.12)	0.004		
Functional status at 12 months			Initial model $I^2 = 62\%$		
<i>Variable(s)</i>	<i>N = 18</i>	<i>B (95% CI)</i>	<i>P-value</i>	$I^2 = 53$	$R^2 = 43$
Duration LBP		-0.15 (-0.46 to 0.17)	0.33		
Multi-modal SMT		0.22 (-0.10 to 0.55)	0.17		
Income country		No middle- to low-income country studies			

Reference: duration LBP (0 = subacute/chronic; 1 = chronic only); multi-modal SMT (0 = SMT alone; 1 = SMT in combination with other (minimal) modalities); Country (0 = high income countries; 1 = middle to low income countries).

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically

Study ID	N randomised population category (age: mean; % female; other)	Comparator	Details of the intervention	Details of the control	Outcome do- main	Outcome measure	Time point of measure- ment	Methods of synthe- sis
Ali 2019	33 Healthcare population (35; not reported)	Other conserv- ative interven- tions	Maitland mo- bilisation	Mulligan mobil- isation	Pain	NPRS (0-10)	4 weeks	MA
					Functional status	ODI		
Balthaz- ard 2012	42 Healthcare population (43; 33%)	Sham SMT/ placebo	SMT plus ac- tive exercise	Detuned ultra- sound	Pain	VAS (0-10)	After 8th therapeu- tic session, 3- and 6- months af- ter the end of treatment	MA
					Functional status	ODI		
Bialosky 2014	110 General population (32; 70%)	Sham SMT/ placebo	SMT	Sham SMT	Pain	NPRS (0-100)	Immediate- ly following treatment and 2 weeks	MA
		No treatment	SMT	Sit quietly	Functional status	ODI		
Blan- co-Giménez 2024	55 General population (45; 56)	SMT as adju- vant therapy	SMT + exer- cise therapy	Exercise thera- py	Pain	VAS (0-10)	3, 6 and 12 weeks	MA
		Other conserv- ative interven- tions	SMT + exer- cise therapy	Kinesiotaping + exercise thera- py	Functional status	ODI		
					Adverse events			
Bond 2020	29 General population (24; 38%; university students)	Sham SMT/ placebo	SMT	Sham SMT	Pain	NPRS (0-10)	3 weeks	MA
					Functional status	ODI		
					Adverse events			
Bronfort 2011	301		SMT	Supervised ex- ercise therapy	Pain	NPRS (0-10)		MA

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically

	Study	Population (n; %)	Intervention	Comparison	Outcomes	Synthesis	Quality	Comments
		General population (45; 61%)	Other conservative interventions	Home exercise and advice	Functional status Recovery Adverse events	RMDQ Patient perceived global improvement (0-9)	4, 12, 26, and 52 weeks	
Bronfort 2014	192	General population (57; 64%)	SMT as adjunct therapy	SMT plus home exercise and advice	Home exercise and advice	Pain Functional status Recovery Adverse events	NPRS (0-10) RMDQ Patient perceived global improvement (0-9)	4, 12, 26, and 52 weeks MA
Brønfort 1996	174	General population (41; 47%)	Other conservative interventions	SMT + strengthening exercises	NSAIDs + strengthening exercises	Pain Functional status Adverse events	NPRS (0-10) RMDQ	5, 11 weeks and 1 year MA
Carrasco-Martinez 2019	171	Healthcare population (43; 53%)	SMT versus SMT	Flexion distraction HVLA SMT		Pain Functional status	VAS (0-100) ODI	4 weeks Summary
Castro-Sanchez 2016	62	Healthcare population (45; 64%)	SMT versus SMT	Functional technique SMT		Pain Functional status Adverse effects	NPRS (0-10) RMDQ, ODI	3 weeks and 7 weeks Summary

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically (Continued)

Castro-Sanchez 2021	64 Healthcare population (52; 61%)	Other conservative interventions	SMT	Myofascial release	Pain Functional status Adverse effects	NPRS (0-10) RMDQ	Post treatment (12 weeks) and 16 weeks	MA
Cecchi 2010	210 Healthcare population (59; 67%)	Other conservative interventions	SMT	Back school group Individual physiotherapy	Pain Functional status Adverse effects	Roland Morris Pain rating scale (0-6) RMDQ	On discharge, 3, 6 and 12 months	MA
Chown 2008	210 Healthcare population (43; 58%)	Other conservative interventions	Osteopathy	Physiotherapy Group exercise with physiotherapist	Functional status	ODI	6 weeks after discharge and 12 months	Fatal flaw
Cook 2013	154 Healthcare population (48; 53%)	SMT versus SMT	Thrust manipulation Non-thrust manipulation		Pain Functional status Adverse events	NPRS (0-10) ODI	After two visits, and at discharge from therapy services (up to 4 months)	Summary
De Oliveira 2020a	42 Healthcare population (48; 74%)	Other conservative interventions	Osteopathic manual therapy	Active control exercises	Pain Functional status	VAS (0-10) ODI	5 weeks	MA
De Oliveira 2020b	148 Healthcare population (45; 78%)	SMT versus SMT	Directed vertebral manipulation		Pain Functional status	NPRS (0-10) RMDQ	4, 12 and 26 weeks	Summary

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

			Generic verte- bral manipu- lation		Recovery	Global perceived effect scale		
Didehdar 2020	25 General population (37; 44%)	Sham SMT/ placebo	SMT	Sham SMT	Pain Functional status	NPRS (0-10) ODI	5 weeks post-treat- ment	MA
Dogan 2021	64 Not reported (37; 68)	SMT as adju- vant therapy	SMT + home exercise	Home exercise	Pain	VAS (0-10) at rest and with activity	24 hours, 1 and 4 weeks	MA
Dougherty 2014a	181 Healthcare population and general population (57; 31%)	Other conserv- ative interven- tions	SMT	Active exercise therapy	Pain Functional status Adverse events	VAS (0-100) ODI	4, 12 and 24 weeks	MA
Dougherty 2014b	136 Healthcare population? (77; 1%; Elderly)	Sham SMT/ placebo	SMT	Sham SMT	Pain Functional status Adverse events	VAS (0-10) ODI	5 and 12 weeks	MA
Evans 1978	36 Healthcare population (45; 53%)	SMT as adju- vant therapy	SMT with analgesics	Analgesics	Pain Recovery Adverse events	Pain scale (0-4) Global assessment (0-4)	Up to 6 weeks	MA
Fatima 2021	30 Healthcare population (31; 60%)	Other conserv- ative interven- tions	SMT	Muscle energy technique	Pain Functional status	NPRS (0-10) ODI	2 weeks?	MA
Ferreira 2007	240		SMT	General exer- cise	Pain	VAS (0-10)	2, 6 and 12 months	MA

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically

		Healthcare population (54; 69%)	Other conserv- ative interven- tions		Functional status	RMDQ, PSFS		
				Motor control exercise	Recovery	Global perceived effect (0-10)		
Gev- ers-Mon- toro 2024 ;	98	Healthcare and general population (48; 64%)	Sham SMT/ placebo	SMT	Placebo SMT	Pain	NPRS (0-100)	4, 8 and 16 weeks
						Functional Status	ODI	MA
						Adverse events		
Ghroubi 2007	64	Healthcare population (38; 80%)	Sham SMT/ placebo	SMT	Sham SMT	Pain	VAS (0-10)	1 and 2 months
						Functional status	ODI	MA
Gibson 1985	109	Healthcare population (36; 43%)	Sham SMT/ placebo	Osteopathic manipulation and mobilisa- tion	Detuned short- wave diathermy	Pain	VAS (0-100- daytime and nocturnal scores)	2, 4 and 12 weeks
			Other conserv- ative interven- tions		Active short- wave diathermy	Recovery	% patients pain free	MA
Goldby 2006	323	Healthcare population (42; 68%)	Other conserv- ative interven- tions	SMT	Spinal stabil- isation reha- bilitation pro- gramme	Pain	NPRS (0-100)	3.6.12 and 24 months
					Education min- imal interven- tion	Functional status	ODI	Dropout > 50%
Grande- Alonso 2019	52	General population (39; 56%; university popula- tion)	SMT as adju- vant therapy	Osteopathic manipulative therapy + be- havioural ed- ucation	Behavioural ed- ucation	Pain	VAS (0-10)	4 and 12 weeks
						Functional status	RMDQ	MA

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

					Recovery		Global perception of change		
					Pain				
Griswold 2019	65 Healthcare population (47; 47%)	Other conservative interventions	Non-thrust manipulation	Dry needling	Pain	NPRS (0-10)	Visit 2, 4 and 6 (treatment lasted 3 weeks)	MA	
					Functional status	Modified ODI			
					Recovery	% recovery			
Gudavalli 2006	235 Healthcare population (42; 38%)	Other conservative interventions	Flexion distraction	Exercise	Pain	VAS (0-10)	4 weeks, 3, 6 and 12 months	MA	
					Functional status	RMDQ			
					Adverse events				
Haas 2014	400 General population (41; 50%)	SMT as adjunct therapy	SMT plus massage	Massage	Pain	Modified Von Korff pain scales	6, 12, 18, 24, 39 and 52 weeks	MA	
					Functional status	Modified Von Korff disability scales			
		Other conservative interventions	SMT	Massage	Recovery	Perceived pain and disability improvement			
					Adverse events				
Hemmila 2002	132 Healthcare population and general population (42; 43%)	Other conservative interventions	Bone setting	Physiotherapy	Pain	VAS (0-10)	6 weeks, 3 and 6 months	MA	
				Home exercise with individual instruction by physical therapist	Functional status	ODI			
Hondras 2009	240 General population (63; 44%; Elderly over 55)	Other conservative interventions	HVLA SMT	Medical care	Pain	VAS (0-10)	3, 6, 12, and 24 weeks	MA	
					Functional status	RMDQ			

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

		SMT versus SMT	LVVA SMT		Recovery	Perceived recovery (0-10)		
					Adverse events			
Hsieh 2002	206 General population (48; 35%)	SMT as adjuvant therapy	SMT and Myofascial therapy	Myofascial therapy	Pain	VAS (0-10)	3 weeks and 6 months	MA
					Functional status	RMDQ		
		Other conservative interventions	SMT	Myofascial therapy Back school	Adverse events			
Hurwitz 2002	681 Healthcare population (51; 52%)	Other conservative interventions	Chiropractic care only	Medical care only	Pain	NPRS (0-10)	2 and 6 weeks	MA
			Chiropractic care with physical modalities	Medical care with physical modalities	Functional status	RMDQ		
					Recovery	Perceived recovery (0-3)		
Ibrahim 2023	54 Healthcare population (22; 19%)	SMT as adjuvant therapy	Maitland mobilisation + selected exercise programme	Selected exercise programme	Pain	VAS (0-100)	4 weeks	MA
					Functional status	ODI		
Kamali 2019	40 Healthcare population (41; 70%)	Other conservative interventions	Manipulation	Stabilising exercise	Pain	VAS (0-10)	After the treatment period (2 weeks for SMT group and 4 weeks for SE group)	MA
					Functional status	ODI		
					Adverse events			
Koes 1992	256	Sham SMT/ placebo	SMT	Detuned short-wave diathermy and detuned ultrasound	Pain	WHYMPI (0-6)	3, 6, 12, 26 and 52 weeks	Summary

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically

	Healthcare population and general population (43; 48%)	Other conservative interventions		Physiotherapy	Functional status	Sickness impact profile		
				Standard GP care	Recovery	Global perceived effect (0-6)		
Krek-oukias 2017	75 Healthcare population (47; 44%)	Sham SMT/ placebo	Manual therapy	Sham manual therapy	Pain	NPRS (0-10)		5 weeks
		Other conservative interventions		Conventional physiotherapy	Functional status	RMDQ, ODI		MA
Licciar-done 2003	91 Healthcare population and general population (50; 64%)	Sham SMT/ placebo	Osteopathic therapy	Sham SMT	Pain	VAS (0-10)		1, 3, and 6 months
		No treatment		No intervention control	Functional status	RMDQ		MA
Licciar-done 2013	455 Healthcare population and general population (41 (median); 59%)	Sham SMT/ placebo	Osteopathic manual treatment plus ultrasound	Sham osteopathic manual treatment plus ultrasound	Pain	VAS (0-100)		4, 8 and 12 weeks
					Functional status	RMDQ		
			Osteopathic manual treatment plus sham ultrasound	Sham osteopathic manual treatment plus sham ultrasound	Adverse events			
Lizis 2021	68 Healthcare population and general population (50; 34%)	SMT versus SMT	Osteopathic manual treatment		Pain	NPRS (0-10)		6 weeks
					Functional status	ODI		Summary
			Kaltenborn-Evjenth orthopaedic manual therapy		Adverse events			

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically (Continued)

McCarthy 2019	60 Healthcare population (37; not reported)	SMT versus SMT	Targeted manipulative thrust		Pain	VAS (0-100)	4 and 7-9 days	Summary
			Generalised manipulative thrust		Functional status	RMDQ		
Mohseni-Bandpei 2006	120 Healthcare population (36; 59%)	Other conservative interventions	SMT + exercise	Ultrasound + exercise	Pain	VAS (0-100)	6 weeks; 6 months	MA
					Functional status	ODI		
Muller 2005	155 Healthcare population (39; 47%)	Other conservative interventions	SMT	Acupuncture	Pain	VAS (0-10)	4 and 9 weeks, and 12 months	Fatal flaw
					Functional status	ODI		
				Medication	Adverse events			
Nambi 2018	330 Healthcare population and general population (21-60 range; 61%)	SMT as adjunct therapy	SMT plus laser therapy and conventional exercise	Laser therapy and conventional exercise	Pain	VAS (0-10)	4 weeks, 6 and 12 months	MA
					Functional status	RMDQ		
					Adverse events			
Nejati 2019	56 Healthcare population (47; 58%)	SMT as adjunct therapy	Manual therapy plus exercise	Exercise	Pain	VAS (0-10)	6, 12, and 24 weeks	MA
		Other conservative interventions	Manual therapy	Exercise	Functional status	ODI		
Nim 2020	132 Healthcare population (45; 46%)	SMT versus SMT	Stiffest segment SMT		Pain	NPRS (0-10)	14 days	Summary
			Lowest pain threshold SMT		Adverse events			

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically (Continued)

Ozcan 2022	40 Healthcare workers (29; 70%)	SMT versus SMT	Chiropractic manipulation	Mulligan mobilisation	Pain Function	VAS (0-100) ODI	4 weeks	MA
Paatelma 2008	134 Healthcare population (44; 35%)	Other conservative interventions	Orthopaedic manual therapy	McKenzie Advice-only	Pain Functional status	VAS (0-100) RMDQ	3, 6 and 12 months	MA
Petersen 2011	350 Healthcare population (38; 56%)	Other conservative interventions	SMT	McKenzie	Pain Functional status Recovery	Modified Von Korf pain scale RMDQ Global perceived effect	Post-treatment, 2- and 12 months post-treatment	MA
Pope 1994	164 General population (32; 38%; including university students)	Other conservative interventions	SMT	Soft-tissue massage Corset Transcutaneous muscle stimulation	Pain	VAS (0-10)	1, 2 and 3 weeks	Summary
Postacchini 1988	459 Healthcare population (39; 50%)	Other conservative interventions	SMT	Diclofenac 'full dose' Physiotherapy Bed rest Back school Placebo gel	Pain Functional status	Scale (0-4) Scale (0-4)	3 weeks, 2, and 6 months	Summary
Rasmussen 2008	72 Healthcare population (40; 56%)	SMT as adjunct therapy	SMT plus exercise	Exercise	Pain	NPRS (0-10)	2 and 4 weeks, and 1 year	MA

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

					Adverse events					
Rasmussen-Barr 2003	47 Healthcare population (38; 75%)	Other conservative interventions	SMT	Stabilising training	Pain	VAS (0-10)	6 weeks (post-treatment), 3 and 12 months	MA		
					Functional status	ODI				
Samir 2016	30 Healthcare population (42; not reported)	Other conservative interventions	Maitland mobilisation with conventional physical exercise therapy	Mulligan with movement with conventional physical exercise therapy	Pain	VAS (0-10)	4 weeks (post-treatment)	MA		
Saracoglu 2022	69 Healthcare population (40; 56%)	SMT as adjunct therapy	Manual therapy and home exercise programme	Home exercise programme	Pain	NPRS (0-10)	4 weeks (post treatment) and 12 weeks	MA		
					Functional status	ODI and back performance scale (0-15)				
Sarker 2017	90 Healthcare population (25; not reported)	Other conservative interventions	SMT	Core stability exercise	Pain	NPRS (0-10)	2 weeks	MA		
				Supervised exercise and ergonomic advice						
Schulz 2019	241 Healthcare population (75; 52%; Elderly over 65)	SMT as adjunct therapy	SMT + home exercise programme	Home exercise programme	Pain	NPRS (0-10)	4, 12, 26, and 52 weeks	MA		
					Functional status	RMDQ				
		Other conservative interventions		Supervised exercise programme + home exercise programme	Recovery	Global improvement (-4 - 4)				
					Adverse events					
Senna 2011	93	Sham SMT/ placebo	Non-maintained SMT	Sham SMT	Pain	VAS (0-10)	1, 4, 7, and 12 months	MA		

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically

Study	N	Healthcare population (n; %)	Intervention	Comparison	Outcomes	Synthesis	Follow-up	MA
Skillgate 2007	409	General population (47; 71%; employees at 2 large public companies)	Other conservative interventions	Naprapathy	Standard care or 'evidence-based' care	Pain Functional status Recovery Adverse events	Modified von Korff pain scale Modified von Korff disability scale (0-10)	3, 7 and 12 weeks MA
Sturion 2020	10	General population (44; not reported)	Other conservative interventions	SMT	Muscle energy technique	Pain Functional status	NPRS (0-10) and McGill Pain Questionnaire RMDQ	15 days MA
Tavares 2017	60	Not reported (37; 85%)	Sham SMT/ placebo No treatment	Mobilisation	Sham mobilisation Control, no intervention	Pain Functional status	NPRS (0-10) ODI	Following the 10th therapeutic session (5 weeks) MA
Teixiera 2016	14	Not reported (51; 50%)	Other conservative interventions	SMT	Stretching and exercise therapy	Pain Functional status	Modified von Korff pain scale RMDQ	2 weeks of manual therapy followed by 2 weeks of physio with one week of wash-out Summary
Thomas 2020	162	General population (25; 58%)	Sham SMT/ placebo	Spinal manipulation	Sham laser	Pain Functional status	NPRS (0-10) RMDQ	After completing 3 weeks of treatment, and 4 weeks after treatment MA

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

		SMT versus SMT	Spinal mobilisation		Adverse events		ment completion	
UK BEAM Trial 2004	1334 Healthcare population (43; 56%)	SMT as adjunct therapy	Best care plus private manipulation alone	Best care in general practice	Pain	Modified von Korff pain scale	3 and 12 months	MA
			Best care plus NHS manipulation		Functional status	RMDQ; Modified von Korff disability scale		
			st					
			Best care plus private manipulation plus exercise	Best care plus exercise alone				
		Other conservative interventions	Best care plus private manipulation alone	Best care plus exercise alone	Adverse events			
			Best care plus NHS manipulation					
			st					
Ulger 2017	113 Healthcare population (43; 59%)	Other conservative interventions	Manual therapy	Spinal stabilisation	Pain	VAS (0-10)	After treatment period (6 weeks)	MA
					Functional status	ODI		
Verma 2013	30 Healthcare population (34; 44%)	SMT as adjunct therapy	SMT + short wave diathermy + exercise	Short wave diathermy + exercise	Pain	VAS (0-100)	2 and 4 weeks	MA
Vismara 2012	21 Healthcare population (44; 100%; BMI > 30)	SMT as adjunct therapy	Osteopathic manipulative treatment	Specific exercise	Pain	VAS (0-10)	After 10 sessions (18-20 days)	MA
					Functional status	RMDQ; ODI		

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

Waagen 1986	29 Healthcare population (25; 54%)	Sham SMT/ placebo	Manipulation and specific exercise	Sham manipu- lation	Pain	VAS (0-10)	2 weeks	MA
Walker 2013	183 General population (55; 37%)	Sham SMT/ placebo	Usual chiro- practic care	Sham treat- ment including detuned ultra- sound	Pain	NPRS (0-10)	2 weeks	MA
					Functional status	Functional rating index		
					Adverse events			
Wang 2019	60 Not reported (31; 35%)	SMT as adju- vant therapy	SMT plus sling-exer- ciset therapy	Sling-exer- ciset therapy	Pain	VAS (0-10)	2, 3, and 4 weeks	MA
					Functional status	ODI		
Wilkey 2008	30 Healthcare population (44; 57%)	Other conserv- ative interven- tions	Chiropractic treatment	Hospital pain clinic	Pain	NPRS (0-100)	2, 4, 6 and 8 weeks	MA
					Functional status	RMDQ		
Xia 2016	192 General population (40; 54%)	No treatment	Thrust	Control	Pain	VAS (0-10)	3 weeks	MA
					Functional status	RMDQ		
		SMT versus SMT	Non thrust		Adverse events			
Zaidi 2020	60 Not reported (37; not re- ported)	Other conserv- ative interven- tions	Maitland mo- bilisation	Muscle energy techniques	Pain	VAS (0-10)	4 weeks	MA
					Functional status	Modified ODI		
Zaproudi- na 2009	131 General population (41; 51%)	Other conserv- ative interven- tions	Traditional bone setting	Physical thera- py	Pain	VAS (0-10)	3, 9, and 15 months post-base- line	MA
					Functional status	ODI		

Table 3. Overview of included studies and synthesis table illustrating key study characteristics, outcomes, and synthesis – sorted alphabetically *(Continued)*

Recovery
Perceived recovery
(0-10)

BMI: body mass index; **HVLA:** high-velocity low-amplitude; **LVLA:** low-velocity low-amplitude; **MA:** meta-analysis; **NHS:** National Health Service; **NPRS:** numerical pain rating scale; **ODI:** Oswestry Disability Index; **PSFS:** Patient Specific Functional Scale; **RMDQ:** Roland Morris Disability Questionnaire; **SMT:** spinal manipulative therapy; **VAS:** visual analogue scale; **WHYMPI:** West Haven-Yale Multidimensional Pain Inventory

Table 4. Specific clinical and treatment characteristics of the included studies

Author	Type radiat- ing pain	Duration LBP: ac- cording to inclusion criteria	Duration LBP: current episode for the popula- tion	Type of manipula- tor (N = number of manipulators)	Type of manip- ulation	Max. no. SMT treatments allowed and duration
Ali 2019	Unclear	> 12 weeks	Not reported	Not reported	Maitland and Mulligan mobil- isation	16 over 4 weeks
Balthazard 2012	With or with- out symptoms in the lower extremity	> 12 and < 26 weeks	Not stated, but > 12 weeks for the population and < 26 weeks	Physiotherapist (N = 1)	Manipulation and mobilisa- tion	8 over 4–8 weeks
Bialosky 2014	Unclear	No restric- tion	Median: 16.03 weeks	Physical therapist (N = 2)	Manipulation	6 over 2 weeks
Blan- co-Giménez 2024	Unclear	Chronic LBP, not further specified	Unclear	Physical specialist (n = 1) and Physio- therapist (n = 1)	Manipulation	24 over 12 weeks
Bond 2020	Without radia- tion	> 12 weeks	Mean 44.00 (SD 28.07) months	Physiotherapist and/or chiropractor (N = 2)	Manipulation	6 over 2 weeks
Bronfort 2011	With or with- out symptoms in the lower extremity	At least 6 weeks du- ration	Median: 5.0 years	Chiropractor (N = 9)	Manipulation	Participants were dis- charged from care if the treating clini- cian felt that maximum clinical ben- efit was ob- tained; 12 weeks of care
Bronfort 2014	With pain ra- diating to the leg	> 4 weeks	104 weeks	Chiropractor (N = 11)	Manipulation	20 sessions over 12 weeks
Brønfort 1996	With or with- out radiation to one or both legs to the knee	> 6 weeks	Median: 2.5 years	Chiropractor (N = 5)	Manipulation	10 over 5 weeks
Carras- co-Martinez 2019	Without radia- tion	> 3 months	Not stated	Physiotherapist (N = 1)	Manipulation and flexion dis- traction	4 over 2 weeks
Cas- tro-Sanchez 2016	Unclear	> 3 months	Mean 7, 6–8 months	Physical therapist (N = 2)	Manipulation and mobilisa- tion	3 over 3 weeks

Table 4. Specific clinical and treatment characteristics of the included studies (Continued)

Cas-tro-Sanchez 2021	Unclear	> 3 months	Time with pain (months (SD)) SMT = 10.91 (4.28); MR = 11.91 (4.55)	Physical therapist (N = 2)	Manipulation	12 over 12 weeks
Cecchi 2010	Unclear	> 6 months	Not stated per group, but > 6 months for the population	Physicians (N = 2)	Manipulation and mobilisation	4–6 over 4–6 weeks
Chown 2008	Without radiation	> 3 months	Unclear	Osteopathy & Manipulative therapy (N = ?)	Manipulation or mobilisation (depending upon group assignment)	5 over 3 months
Cook 2013	Unclear	No restriction	Range: 30.2 to 37.7 weeks	Physiotherapist (N = 17)	Manipulation or mobilisation (depending upon group assignment)	1st 2 visits only; afterwards, clinician was allowed to choose technique they felt most beneficial for the patient
De Oliveira 2020a	Unclear	> 3 months	Mean (SD) = 59.8 (56.2) months	Osteopath (N = 1) and Physiotherapy students (N = 2)	Manipulation and mobilisation	5 over 5 weeks
De Oliveira 2020b	Unclear	> 3 months	Median: 6 months (range 3–24 months)	Physiotherapist with osteopathic certification (N = 1)	Manipulation	10 over 4 weeks
Didehdar 2020	Unclear	> 3 months	Unclear	Not reported	Manipulation	3 over 1 week
Dogan 2021	Unclear	Chronic LBP confirmed by the authors	Unclear	Physiatrist (n = 1)	Mobilisation	3 over 3 weeks
Dougherty 2014a	Without radiation	> 3 months	Range: 138–261 months	Chiropractor (N = 3)	Manipulation and mobilisation	8 over 4 weeks
Dougherty 2014b	Without radiation	> 3 months	Unclear	Chiropractor (N = 2)	Manipulation	8 over 2 weeks
Evans 1978	With or without femoral or sciatic radiation	> 3 weeks	Median: 10 months	Medical manipulator (N = 1)	Manipulation	3 over 3 weeks
Fatima 2021	Unclear	Between 3 and 6 months	Unclear	Physical therapists (N = ?)	Manipulation	6 over 2 weeks

Table 4. Specific clinical and treatment characteristics of the included studies (Continued)

Ferreira 2007	With or without radiation	> 3 months	75% > 1 year	Physical therapists (N = ?)	Mobilisation or manipulation; Maitland	12 over 8 weeks
Gev-ers-Mon-toro 2024	With or without radiation	> 3 months	SMT group: mean, 110 months Placebo SMT group: 118 months	Chiropractor (N = 1)	Manipulation	12 over 4 weeks
Ghroubi 2007	Without radiation	> 6 months	Range: 16 to 19 months	Manual or physical therapist? (N = 1)	Unclear; presumably manipulation?	4 over 4 weeks
Gibson 1985	Unclear	> 2 months to < 12 months	Range: 4 to 4 ½ months	Osteopath (N = 1)	Manipulation and mobilisation	4 over 4 weeks
Goldby 2006	Unclear	> 3 months	Mean: 11.7 years	Manual therapist (N = ?)	Unclear	10 over 10 weeks?
Grande-Alonso 2019	Without radiation	> 3 months	Mean: 71.46 months in intervention group, 82.75 in control group	Unclear	Mobilisation	8 over 4 weeks
Griswold 2019	Without radiation	At least six weeks	90% were chronic	Physical therapist (N = ?)	Manipulation	6 over 3 weeks
Gudavalli 2006	With or without radiculopathy	> 3 months	Unclear	Chiropractor (N = ?)	MOB (flexion-distraction)	16 over 4 weeks
Haas 2014	With or without radiation to the knee	> 3 months	Range: 11.2-12.5 years	Chiropractor (N = 12)	Manipulation or MOB	18 over 6 weeks
Hemmila 2002	With or without radiation below the knee	> 7 weeks	Range: 6.8 to 7.5 years	Bone-setter (N = 4)	Primarily mobilisation? No Manipulation	10 over 6 weeks
Hondras 2009	Primarily (85%) with or without radiation to the knee	> 4 weeks	Range: 9.6 to 15.1 years	Chiropractor (N = 4)	Manipulation or mobilisation (flexion-distraction) (depending upon group assignment)	12 over 6 weeks
Hsieh 2002	With or without leg pain, but no neurological signs	> 3 weeks to < 6 months	Range: 10.7 to 11.8 weeks	Chiropractor (N = ?)	Manipulation	9 over 3 weeks
Hurwitz 2002	With or without leg pain	No restriction	58% > 3 months	Chiropractor (N = 4)	Manipulation	? - at discretion of therapist

Table 4. Specific clinical and treatment characteristics of the included studies *(Continued)*

Ibrahim 2023	Unclear	> 3 months	Not stated per group but > 3 months for the population	Unclear	Mobilisation	12 over 4 weeks
Kamali 2019	With or without radiation to the knee	At least 4 weeks	Manual therapy group: mean (SD) 4.73 (3.07) years; Exercise group: 4.63 years (2.21)	Physiotherapist (N = 2)	Manipulation	6 over 2 weeks (SMT group) and 12 over 4 weeks (control group)
Koes 1992	With or without radiation to the knee	> 6 weeks	Median: 1 year	Manual therapist (N = 7)	Manipulation and mobilisation	Avg. 5 over 9 weeks
Krekoukias 2017	Unclear	> 3 months	Not stated per group but > 3 months for the population	Physiotherapist (N = 1)	Mobilisation	5 over 5 weeks
Licciardone 2003	With or without sciatica, but no neurological signs	> 3 months	Range: 39% to 63% > 1 year	Osteopath (N = ?)	Manipulation or mobilisation	7 over 5 months
Licciardone 2013	Unclear	> 3 months	51% to 66% with LBP > 1 year	Osteopath (N = 15)	Manipulation or mobilisation	6 over 8 weeks
Lizis 2021	Unclear	> 3 months	Not reported	Physiotherapist (N = 1)	Manipulation or mobilisation (depending upon group assignment)	10 over 5 weeks
McCarthy 2019	Unclear	At least 3 weeks	Not reported	Osteopath (N = 1)	Manipulation	3 over max. 9 days
Mohseni-Bandpei 2006	Unclear	> 3 months	Range: 31 to 56 months	Manual therapist (N = 1)	Manipulation (Maitland)	7 over 4 weeks?
Muller 2005	Without	> 3 months	Range: 4 mo to 45 years	Chiropractor (N = 1?)	Manipulation	? - but equal per therapy group
Nambi 2018	With or without sciatica	> 3 months	Not reported	Physical therapist (N = ?)	Manipulation	12 over 4 weeks.
Nejati 2019	Without pain radiating below to the knee	At least 3 months	Not reported	Sports medicine specialist (N = ?)	Manipulation and mobilisation	1 session over 12 weeks.
Nim 2020	Unclear	> 3 months	Stiffest segment SMT mean 14.10 (67.10) months	Chiropractor (N = 2)	Manipulation	4 sessions over 14 days

Table 4. Specific clinical and treatment characteristics of the included studies (Continued)

			Lowest pain threshold SMT mean 17.50 (49.35) months			
Özcan 2022	Unclear	Chronic LBP not further specified	Chronic LBP not further specified	Chiropractor who was also a qualified physiotherapist (n = 1)	Manipulation or mobilisation	8 over 4 weeks
Paatelma 2008	With or without sciatica	No restriction	> 50% symptoms > 3 months	Orthopaedic manual therapist (N = 1)	Manipulation or mobilisation	7 over ? weeks Mean: 6 treatments/group
Petersen 2011	With or without leg pain	> 6 weeks	Range 94–97 weeks	Chiropractor (N = 3)	Manipulation or mobilisation	Max. 15 over 12 weeks
Pope 1994	Without sciatica	3 weeks to 6 months, preceded by 3-week pain-free episode	29% < 6 months; 35% between 6 months to 2 years; 36% > 2 years	Chiropractor (N = 5)	Manipulation	3 or more sessions/week for 3 weeks
Postacchini 1988	2 grps. = with and without radiation to knee	Grp.C = > 9 weeks	Grp.C range: 9 to 11 months	Chiropractor (N = ?)	Manipulation?	12 over 6 weeks
Rasmussen 2008	With or without radiation to the knee	> 3 months	Range: 8 to 17 months	Medical manipulator (N = 1?)	Manipulation	3 over 4 weeks
Rasmussen-Barr 2003	With or without radiation to the knee	> 6 weeks	90% > 3 months	Manual therapist (N = ?)	Mobilisation	6 over 6 weeks
Samir 2016	Unclear	> 3 months	Not stated per group but > 3 months for the population	Physiotherapist (N = ?)	Mobilisation	12 over 1 month
Saracoglu 2022	Unclear	> 6 months	SMT + exercise 35 months (SD 15), Exercise group: 32 (SD 14)	Physiotherapist (N = 1?)	Mobilisation	8 over 4 weeks
Sarker 2017	Unclear	> 3 months	Not stated per group but > 3 months for the population	Physiotherapist? (N = ?)	Manipulation	? over 15 days
Schulz 2019	With leg pain	≥ 6 weeks	78% chronic	Physiotherapist (N = 2) and Chiropractor (N = 1)	Manipulation and mobilisation	20 over 12 weeks
Senna 2011	Unclear	> 6 months	Range: 18.4–18.8 months	Physician (N = ?)	Manipulation	12 over 1 month
Skillgate 2007	Unclear	> 2 weeks	Range: 72% to 78% > 3 months	Naprapath (N = 8)	Manipulation or mobilisation	6 over 6 weeks

Table 4. Specific clinical and treatment characteristics of the included studies *(Continued)*

Sturion 2020	Without proximal radicular pain	> 3 months	50% of participants had pain duration of 24–36 months	Physiotherapist (N = 2)	Manipulation	3 over 3 weeks (RCT cross-over)
Tavares 2017	Unclear	> 3 months	Unclear	Unclear	Mobilisation	10 over 5 weeks
Teixiera 2016	Unclear	> 3 months	Unclear	Unclear	Manipulation	8 over 5 weeks (RCT cross-over)
Thomas 2020	Unclear	> 3 months	Mean (SD) 6.2 (5.1) years	Licenced clinicians (either a doctor of osteopathic medicine or physical therapist) (N = ?)	Manipulation or mobilisation	6 over 5 weeks
UK BEAM Trial 2004	(Primarily) with or without radiation to the knee	(Essentially) > 3 weeks	59% > 3 months	Chiropractor, osteopath or physiotherapist (N = 84)	Manipulation or mobilisation	8 over 12 weeks
Ulger 2017	Unclear	> 3 months	Not stated per group, but > 3 months for the population	Physiotherapist (N = ?)	Manipulation or mobilisation	18 over 6 weeks
Verma 2013	Unclear	> 3 months	Not stated per group, but > 3 months for the population	Physiotherapist (N = ?)	Mobilisation	8 over 4 weeks
Vismara 2012	Unclear	> 6 months	Not stated per group, but > 6 months for the population	Osteopath (N = 1)	Manipulation or mobilisation	10 over 10 weeks?
Waagen 1986	With or without the knee	> 3 weeks	Range: 2.5 to 2.8 years	Chiropractor (N = ?)	Manipulation	6 over 2 weeks
Walker 2013	Unclear	> 1 week	98% had experienced spinal pain for more than 3 months	Chiropractor (N = 8)	Manipulation or mobilisation	2 over 2 weeks
Wang 2019	Without radiation to the legs	> = 3 months	Not reported	Physician (N = ?)	Manipulation and mobilisation	4 over 4 weeks
Wilkey 2008	With or without radiation to the legs	> 3 months	Range: 0.5 to 20 years	Chiropractor (N = ?)	Manipulation	16 over 8 weeks
Xia 2016	With or without radiation to the legs	> 4 weeks	71% > 3 months	Chiropractor (N = 4)	Manipulation or mobilisation	4 over 2 weeks
Zaidi 2020	Without radiation to the legs	Unclear	Unclear	Unclear	Mobilisation	12 over 4 weeks

Table 4. Specific clinical and treatment characteristics of the included studies (Continued)

Zaproudina 2009	With or without radiation to the legs	> 3 months	Unclear	Bone-setters (N = ?)	Mobilisation	5 over 10 weeks
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grp(s): group(s); **LBP:** low back pain; **MOB:** mobilisation; **MR:** myofascial release; **SD:** standard deviation; **?** = unclear

Table 5. Summary of adverse event assessments among trials included in this systematic review of SMT for chronic low back pain

Study, sample size	Methods used to assess adverse events	Adverse events assessed	Adverse events reported (for SMT or control group)
Balthazard 2012, n = 42	Not reported	Any adverse event	No adverse events were reported, but one participant dropped out.
Blanco-Giménez 2024, n = 48	Not reported	Any unfavourable outcomes that participants may encounter during or after the treatment	No adverse events or unintended effects were reported.
Bond 2020, n = 29	A written log of injuries/adverse events	Any adverse events	A single adverse event of transient (< 48 h) local, mild joint discomfort in the SMT group; no adverse events in the sham SMT.
Bronfort 2011, n = 301	Self reported throughout follow-up	Any adverse event	"All adverse events were transient in nature, required little or no change to activity levels, and were considered non-serious." Six (2%) participants were treated with rescue pain medication during treatment period: severe back pain, acute flare-up of low back and buttock pain, neck pain, and inability to sleep because of pain. Four (1%) participants reported similar adverse events but declined rescue medication.
Bronfort 2014, n = 192	Not reported	Any adverse event	Five serious adverse events occurred during the trial, all unrelated to study interventions (1 due to bowel obstruction in the home exercise and advice (HEA) group and 4 due to anaphylaxis, sports-related trauma, heart condition, and menorrhagia in the SMT plus HEA group). The adverse events were mild to moderate, self-limiting, and reported by 30% of participants in the SMT as adjuvant group and 42% in the control group.
Brønfort 1996, n = 174	Not reported	Any adverse event	Non-steroidal anti-inflammatory drug group: 2 (4%) participants developed severe nausea and vomiting and subsequently discontinued the study, 8 (16%) developed substantial nausea and dyspepsia, and 1 (2%) developed severe tinnitus; SMT + exercise groups: 1 (2%) participant discontinued exercise because she did not tolerate it well and 7 (14%) developed muscle soreness and stiffness, including neck pain after exercise — these symptoms gradually abated and did not prevent completion of the study; 1 (1%) developed symptoms of a myocardial infarction unrelated to exercise. "Overall, both strengthening and stretching exercise and SMT were well tolerated."
Carrasco-Martinez 2019	Not reported but in communication	Not reported	No adverse events reported

Table 5. Summary of adverse event assessments among trials included in this systematic review of SMT for chronic low back pain (Continued)

	with the author, it was reported that no adverse events were observed.		
Castro-Sanchez 2016, n = 62	Self-reported after treatment and follow-up	Any adverse event	No adverse events reported
Castro-Sanchez 2021	No reported	Any adverse events	No adverse events were reported by any participant during the course of the study or at the time of the 1-month follow-up.
Cecchi 2010, n = 210	Not reported	Any adverse event	No adverse events reported
Cook 2013, n = 154	Physiotherapists queried at end of study	Any adverse event	No adverse events reported
Dougherty 2014a, n = 181	Assessed at each treatment visit and via phone calls during follow-up period	Any adverse event	243 adverse events were reported during the study: 55% in exercise group and 45% in SMT group. Of 110 events reported in the SMT group, the Data Safety Monitoring Board (DSMB) judged 14 as definitely or probably associated with SMT. Most adverse events consisted of musculoskeletal soreness and resolved within study period. During the study period, 10 serious adverse events were reported (5 control group, 5 SMT group); DSMB judged none of the serious adverse events to be associated with the study intervention.
Dougherty 2014b, n = 136	Adverse events were collected at each visit and follow-up. Data Safety and Management Board evaluated the reported adverse events.	Adverse event and serious adverse event	250 adverse events were reported: 56% in the SMT group and 44% in the sham intervention group. Of the 141 adverse events in the SMT group, the DSMB judged 35 as definitely or probably associated with the intervention. Of the 109 adverse events in the sham group, the DSMB judged 10 as definitely or probably associated with the intervention. Most adverse were mild to moderate and were related to musculoskeletal soreness. There were no differences in the severity nor in frequency between the two groups of the study. Only 10% of the adverse events were judged to be definitely related to the study. Pre-existing conditions accounted for 42% of the adverse events, and new events accounted for 58% of the events. Six serious adverse events were reported (5 in the SMT group and 1 in the sham intervention group). Two episodes of syncope, 1 chest pain, 1 paraesthesia, 1 myocardial infarction, and 1 fall and injury to neck. The DSMB judged that none of them were associated with the interventions.
Evans 1978, n = 36	Not reported	Any adverse event	One (3%) participant reported constipation after consumption of 24 codeine phosphate capsules in first 4 days; no serious adverse events reported
Ferreira 2007, n = 240	Not reported	Any adverse event	No adverse events were reported. One participant died, and one was admitted to hospital, in control group.
Gevers-Montoro 2024, n = 102	Reported by participants using an online questionnaire or when the partici-	The type of adverse event, time of onset, duration, and severity of the ad-	A total of 377 adverse events were reported in the study, 198 in the placebo group and 179 in the SMT group, representing 34.1% and 30.7% of treatment sessions, respectively. The most commonly reported events were an increase in back pain

Table 5. Summary of adverse event assessments among trials included in this systematic review of SMT for chronic low back pain (Continued)

	participant reported a 30-point increase in pain intensity.	verse events. Adverse events were categorised into 4 main categories: muscle stiffness, increased pain, radiating discomfort, and others.	(36.3% in the SMT group, 41.9% in the placebo group), muscle soreness/stiffness (31.3% in the SMT group, 31.8% in the placebo group), pain irradiating to the lower extremity (22.4% in the SMT group, 14.2% in the placebo group), changes in the usual pain pattern or location (7.8% in the SMT group, 9.6% in the placebo group). "Other" (2.2% in the SMT group and 2.5% in the placebo group).
Griswold 2019, n = 235	Not reported	Any adverse event	No adverse events reported
Gudavalli 2006	Not reported	Any adverse event	No adverse events or side effects were reported from participants in either intervention group.
Haas 2014, n = 400	Not reported	Any adverse event	Three (1%) participants reported seeking care for symptomatic relief of low back pain exacerbation related to study; one (1%) lost several days of work followed by complete resolution during treatment phase; and one (1%) dropped out after an exacerbation associated with lifting a child; no serious adverse events reported
Hondras 2009, n = 240	Not reported	Any adverse event	20 (8%) participants reported an adverse event, all resolved within 6 days, and none required referral for outside care. Adverse events in SMT groups consisted of soreness or stiffness. One participant reported a skin rash in drug group; no serious adverse events reported.
Hsieh 2002, n = 206	Not reported	Any adverse event	23 (12%) participants reported adverse events: 17 (11%) in control groups (combined), 6 (12%) in SMT group; adverse events were limited to transient exacerbations of symptoms, except for one case of constant tinnitus in a control group; 2 (4%) participants claimed SMT had aggravated their condition; no serious adverse events reported
Kamali 2019, n = 40	Physiotherapist monitored adverse events during each visit	Mild and moderate musculoskeletal adverse events	There were no adverse events or side effects in either group.
Licciardone 2013, n = 455	Not reported	Any adverse event	27 (6%) participants reported an adverse event; 9 (2%) reported a serious adverse event ("none was definitely or probably related to a study intervention" according to DSMB); no significant differences between groups in frequency of (serious) adverse events; 6 participants who received SMT developed a contraindication to continued study participation (SMT was adjudicated by DSMB to possibly be related to development in only one of these).
Lizis 2021	Not reported	Any adverse event	No adverse events were reported
Muller 2005, n = 115	Not reported	Any adverse event	3 (6%) participants in drug group experienced an adverse event; no serious adverse events reported
Nambi 2018, n = 330	Self-reported during the conduct of the intervention	Any adverse event	No serious or severe adverse events reported

Table 5. Summary of adverse event assessments among trials included in this systematic review of SMT for chronic low back pain (Continued)

Nim 2020, n = 132	Any adverse events that occurred were recorded for each SMT session.	Any adverse event	69% reported minor side effects. This included reporting of increased local muscle pain by 63%, 33% reported increased lumbar stiffness, 10% reported headaches, and 8% reported worsening of leg pain. Only 5% had other minor side effects such as nausea, dizziness, etc. One participant reported continued anterior chest pain after the 4th session at follow-up.
Rasmussen 2008, n = 72	Not reported	Any adverse event	Four (11%) participants in SMT group reported worsening of low back pain versus 3 (8%) in control group; no serious adverse events reported
Schulz 2019, n = 241	Serious adverse events were recorded and classified according to relationship with treatment	Serious adverse events	Six serious adverse events occurred during the trial; all were determined to be unrelated to trial interventions. These include: one hospitalisation for cholecystectomy and one death due to lung cancer in the intervention alone group; one hospitalisation for acute cardiac symptoms, one new diagnosis of prostate cancer, one injury attending a hockey game, and one transient ischaemic attack during follow-up phase (not actively in intervention) in the SMT + intervention group.
Senna 2011, n = 93	Not reported	Any adverse event	Most common were local discomfort and tiredness, which were transient and began within 24 hours after treatment, and were of mild to moderate severity; no serious adverse events
Skillgate 2007, n = 409	Self-reported events at a follow-up visit	Any adverse event	Minor short-term events limited to muscle soreness, tiredness, and increased pain, most commonly after first and second treatments; no serious adverse events reported
Thomas 2020, 162	Serious adverse events were measured, but not reported how	Serious adverse events	Not reported
UK BEAM Trial 2004, n = 1334	Monitoring by research team; not elucidated further	Serious adverse events only, defined as admission to hospital or death within one week of treatment	No serious adverse events reported
Walker 2013, n = 183	Self-reported events at each follow-up visit	Any adverse event	30 (33%) of participants in sham group and 39 (42%) in SMT group reported at least 1 adverse event; common adverse events were increased pain (sham 29%; SMT 36%), muscle stiffness (sham 29%; SMT 37%), and headache (sham 17%; SMT 9%). The relative risk was not significant for adverse event occurrence (1.24, 95% confidence interval 0.85 to 1.81), occurrence of severe adverse events (1.9, 0.98 to 3.99), adverse event onset (0.16, 0.02 to 1.34), or duration of adverse events (1.13, 0.59 to 2.18); no serious adverse events
Xia 2016, n = 192	Not reported	Not reported	No serious adverse events reported
Zaproudina 2009, n = 131	Not reported	Not reported	One (2%) participant in SMT group and two (3%) in control group discontinued treatment owing to worsening of low back pain; no serious adverse events reported

